8 NGAP Procedures

8.1 List of NGAP Elementary Procedures

In the following tables, all EPs are divided into Class 1 and Class 2 EPs (see subclause 3.1 for explanation of the different classes):

Table 8.1-1: Class 1 procedures

Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
		Response message	Response message
AMF Configuration Update	AMF CONFIGURATION UPDATE	AMF CONFIGURATION UPDATE ACKNOWLEDGE	AMF CONFIGURATION UPDATE FAILURE
RAN Configuration Update	RAN CONFIGURATION UPDATE	RAN CONFIGURATION UPDATE ACKNOWLEDGE	RAN CONFIGURATION UPDATE FAILURE
Handover Cancellation	HANDOVER CANCEL	HANDOVER CANCEL ACKNOWLEDGE	
Handover Preparation	HANDOVER REQUIRED	HANDOVER COMMAND	HANDOVER PREPARATION FAILURE
Handover Resource Allocation	HANDOVER REQUEST	HANDOVER REQUEST ACKNOWLEDGE	HANDOVER FAILURE
Initial Context Setup	INITIAL CONTEXT SETUP REQUEST	INITIAL CONTEXT SETUP RESPONSE	INITIAL CONTEXT SETUP FAILURE
NG Reset	NG RESET	NG RESET ACKNOWLEDGE	
NG Setup	NG SETUP REQUEST	NG SETUP RESPONSE	NG SETUP FAILURE
Path Switch Request	PATH SWITCH REQUEST	PATH SWITCH REQUEST ACKNOWLEDGE	PATH SWITCH REQUEST FAILURE
PDU Session Resource Modify	PDU SESSION RESOURCE MODIFY REQUEST	PDU SESSION RESOURCE MODIFY RESPONSE	
PDU Session Resource Modify Indication	PDU SESSION RESOURCE MODIFY INDICATION	PDU SESSION RESOURCE MODIFY CONFIRM	
PDU Session Resource Release	PDU SESSION RESOURCE RELEASE COMMAND	PDU SESSION RESOURCE RELEASE RESPONSE	
PDU Session Resource Setup	PDU SESSION RESOURCE SETUP REQUEST	PDU SESSION RESOURCE SETUP RESPONSE	
UE Context Modification	UE CONTEXT MODIFICATION REQUEST	UE CONTEXT MODIFICATION RESPONSE	UE CONTEXT MODIFICATION FAILURE
UE Context Release	UE CONTEXT RELEASE COMMAND	UE CONTEXT RELEASE COMPLETE	
Write-Replace Warning	WRITE-REPLACE WARNING REQUEST	WRITE-REPLACE WARNING RESPONSE	
PWS Cancel	PWS CANCEL REQUEST	PWS CANCEL RESPONSE	
UE Radio Capability Check	UE RADIO CAPABILITY CHECK REQUEST	UE RADIO CAPABILITY CHECK RESPONSE	
UE Context Suspend	UE CONTEXT SUSPEND REQUEST	UE CONTEXT SUSPEND RESPONSE	UE CONTEXT SUSPEND FAILURE
UE Context Resume	UE CONTEXT RESUME REQUEST	UE CONTEXT RESUME RESPONSE	UE CONTEXT RESUME FAILURE

Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
		Response message	Response message
UE Radio Capability ID Mapping	UE RADIO CAPABILITY ID MAPPING REQUEST	UE RADIO CAPABILITY ID MAPPING RESPONSE	
Broadcast Session Setup	BROADCAST SESSION SETUP REQUEST	BROADCAST SESSION SETUP RESPONSE	BROADCAST SESSION SETUP FAILURE
Broadcast Session Modification	BROADCAST SESSION MODIFICATION REQUEST	BROADCAST SESSION MODIFICATION RESPONSE	BROADCAST SESSION MODIFICATION FAILURE
Broadcast Session Release	BROADCAST SESSION RELEASE REQUEST	BROADCAST SESSION RELEASE RESPONSE	
Distribution Setup	DISTRIBUTION SETUP REQUEST	DISTRIBUTION SETUP RESPONSE	DISTRIBUTION SETUP FAILURE
Distribution Release	DISTRIBUTION RELEASE REQUEST	DISTRIBUTION RELEASE RESPONSE	
Multicast Session Activation	MULTICAST SESSION ACTIVATION REQUEST	MULTICAST SESSION ACTIVATION RESPONSE	MULTICAST SESSION ACTIVATION FAILURE
Multicast Session Deactivation	MULTICAST SESSION DEACTIVATION REQUEST	MULTICAST SESSION DEACTIVATION RESPONSE	
Multicast Session Update	MULTICAST SESSION UPDATE REQUEST	MULTICAST SESSION UPDATE RESPONSE	MULTICAST SESSION UPDATE FAILURE

Table 8.1-2: Class 2 procedures

Elementary Procedure	Message
Downlink RAN Configuration Transfer	DOWNLINK RAN CONFIGURATION TRANSFER
Downlink RAN Status Transfer	DOWNLINK RAN STATUS TRANSFER
Downlink NAS Transport	DOWNLINK NAS TRANSPORT
Error Indication	ERROR INDICATION
Uplink RAN Configuration Transfer	UPLINK RAN CONFIGURATION TRANSFER
Uplink RAN Status Transfer	UPLINK RAN STATUS TRANSFER
Handover Notification	HANDOVER NOTIFY
Initial UE Message	INITIAL UE MESSAGE
NAS Non Delivery Indication	NAS NON DELIVERY INDICATION
Paging	PAGING
PDU Session Resource Notify	PDU SESSION RESOURCE NOTIFY
Reroute NAS Request	REROUTE NAS REQUEST
UE Context Release Request	UE CONTEXT RELEASE REQUEST
Uplink NAS Transport	UPLINK NAS TRANSPORT
AMF Status Indication	AMF STATUS INDICATION
PWS Restart Indication	PWS RESTART INDICATION
PWS Failure Indication	PWS FAILURE INDICATION
Downlink UE Associated NRPPa Transport	DOWNLINK UE ASSOCIATED NRPPA TRANSPORT
Uplink UE Associated NRPPa Transport	UPLINK UE ASSOCIATED NRPPA TRANSPORT
Downlink Non UE Associated NRPPa Transport	DOWNLINK NON UE ASSOCIATED NRPPA TRANSPORT
Uplink Non UE Associated NRPPa Transport	UPLINK NON UE ASSOCIATED NRPPA TRANSPORT
Trace Start	TRACE START
Trace Failure Indication	TRACE FAILURE INDICATION
Deactivate Trace	DEACTIVATE TRACE
Cell Traffic Trace	CELL TRAFFIC TRACE
Location Reporting Control	LOCATION REPORTING CONTROL
Location Reporting Failure Indication	LOCATION REPORTING FAILURE INDICATION
Location Report	LOCATION REPORT
UE TNLA Binding Release	UE TNLA BINDING RELEASE REQUEST
UE Radio Capability Info Indication	UE RADIO CAPABILITY INFO INDICATION
RRC Inactive Transition Report	RRC INACTIVE TRANSITION REPORT
Overload Start	OVERLOAD START
Overload Stop	OVERLOAD STOP
Secondary RAT Data Usage Report	SECONDARY RAT DATA USAGE REPORT
Uplink RIM Information Transfer	UPLINK RIM INFORMATION TRANSFER
Downlink RIM Information Transfer	DOWNLINK RIM INFORMATION TRANSFER
Retrieve UE Information	RETRIEVE UE INFORMATION
UE Information Transfer	UE INFORMATION TRANSFER
RAN CP Relocation Indication	RAN CP RELOCATION INDICATION
Connection Establishment Indication	CONNECTION ESTABLISHMENT INDICATION
AMF CP Relocation Indication	AMF CP RELOCATION INDICATION
Handover Success	HANDOVER SUCCESS
Uplink RAN Early Status Transfer	UPLINK RAN EARLY STATUS TRANSFER
Downlink RAN Early Status Transfer	DOWNLINK RAN EARLY STATUS TRANSFER
Multicast Group Paging	MULTICAST GROUP PAGING
Broadcast Session Release Required	BROADCAST SESSION RELEASE REQUIRED

9 Elements for NGAP Communication

9.0 General

Subclauses 9.2 and 9.3 present the NGAP message and IE definitions in tabular format. The corresponding ASN.1 definition is presented in subclause 9.4. In case there is contradiction between the tabular format and the ASN.1

definition, the ASN.1 shall take precedence, except for the definition of conditions for the presence of conditional IEs, where the tabular format shall take precedence.

The messages have been defined in accordance to the guidelines specified in TR 25.921 [7].

When specifying IEs which are to be represented by bitstrings, if not otherwise specifically stated in the semantics description of the concerned IE or elsewhere, the following principle applies with regards to the ordering of bits:

- The first bit (leftmost bit) contains the most significant bit (MSB);
- The last bit (rightmost bit) contains the least significant bit (LSB);
- When importing bitstrings from other specifications, the first bit of the bitstring contains the first bit of the concerned information;

9.1 Tabular Format Contents

9.1.1 Presence

All IEs are marked mandatory, optional or conditional according to table 9.1.1-1.

Table 9.1.1-1: Meaning of content within "Presence" column

Abbreviation	Meaning
M	IEs marked as Mandatory (M) shall always be included in the message.
O	IEs marked as Optional (O) may or may not be included in the message.
C	IEs marked as Conditional (C) shall be included in a message only if the condition is satisfied. Otherwise the IE shall not be included.

9.1.2 Criticality

Each IE or group of IEs may have criticality information applied to it according to table 9.1.2-1.

Table 9.1.2-1: Meaning of content within "Criticality" column

Abbreviation	Meaning
–	No criticality information is applied explicitly.
YES	Criticality information is applied. This is usable only for non-repeatable IEs
GLOBAL	The IE and all its repetitions together have one common criticality information. This is usable only for repeatable IEs.
EACH	Each repetition of the IE has its own criticality information. It is not allowed to assign different criticality values to the repetitions. This is usable only for repeatable IEs.

9.1.3 Range

The Range column indicates the allowed number of copies of repetitive IEs/IE groups.

9.1.4 Assigned Criticality

The Assigned Criticality column provides the actual criticality information as defined in subclause 10.3.2, if applicable.

9.2 Message Functional Definition and Content

9.2.1 PDU Session Management Messages

9.2.1.1 PDU SESSION RESOURCE SETUP REQUEST

This message is sent by the AMF and is used to request the NG-RAN node to assign resources on Uu and NG-U for one or several PDU session resources.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
NAS-PDU	O		9.3.3.4		YES	reject
PDU Session Resource Setup Request List		*1*			YES	reject
>PDU Session Resource Setup Request Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session NAS-PDU	O		NAS-PDU 9.3.3.4		-	
>>S-NSSAI	M		9.3.1.24		-	
>>PDU Session Resource Setup Request Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Request Transfer* IE specified in subclause 9.3.4.1.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
UE Aggregate Maximum Bit Rate	O		9.3.1.58		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.2 PDU SESSION RESOURCE SETUP RESPONSE

This message is sent by the NG-RAN node as a response to the request to assign resources on Uu and NG-U for one or several PDU session resources.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Setup Response List		*0..1*			YES	ignore
>PDU Session Resource Setup Response Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Response Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Response Transfer* IE specified in subclause 9.3.4.2.	-	
PDU Session Resource Failed to Setup List		*0..1*			YES	ignore
>PDU Session Resource Failed to Setup Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Unsuccessful Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Unsuccessful Transfer* IE specified in subclause 9.3.4.16.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.3 PDU SESSION RESOURCE RELEASE COMMAND

This message is sent by the AMF and is used to request the NG-RAN node to release already established PDU session resources for a given UE.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
NAS-PDU	O		9.3.3.4		YES	ignore
PDU Session Resource to Release List		*1*			YES	reject
>PDU Session Resource to Release Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Release Command Transfer	M		OCTET STRING	Containing the *PDU Session Resource Release Command Transfer* IE specified in subclause 9.3.4.12.	-	

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.4 PDU SESSION RESOURCE RELEASE RESPONSE

This message is sent by the NG-RAN node as a response to the request to release already established PDU session resources for a given UE.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Released List		*1*			YES	ignore
>PDU Session Resource Released Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Release Response Transfer	M		OCTET STRING	Containing the *PDU Session Resource Release Response Transfer* IE specified in subclause 9.3.4.21.	-	
User Location Information	O		9.3.1.16		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.1.5 PDU SESSION RESOURCE MODIFY REQUEST

This message is sent by the AMF and is used to request the NG-RAN node to enable modifications of already established PDU session resources for a given UE.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
PDU Session Resource Modify Request List		*1*			YES	reject
>PDU Session Resource Modify Request Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>NAS-PDU	O		9.3.3.4		-	
>>PDU Session Resource Modify Request Transfer	M		OCTET STRING	Containing the *PDU Session Resource Modify Request Transfer* IE specified in subclause 9.3.4.3.	-	
>>S-NSSAI	O		9.3.1.24		YES	reject
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.6 PDU SESSION RESOURCE MODIFY RESPONSE

This message is sent by the NG-RAN node and is used to report the outcome of the request from the PDU SESSION RESOURCE MODIFY REQUEST message.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Modify Response List		*0..1*			YES	ignore
>PDU Session Resource Modify Response Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Response Transfer	M		OCTET STRING	Containing the *PDU Session Resource Modify Response Transfer* IE specified in subclause 9.3.4.4.	-	
PDU Session Resource Failed to Modify List		*0..1*			YES	ignore
>PDU Session Resource Failed to Modify Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Unsuccessful Transfer	M		OCTET STRING	Containing the *PDU Session Resource Modify Unsuccessful Transfer* IE specified in subclause 9.3.4.17.	-	
User Location Information	O		9.3.1.16		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.7 PDU SESSION RESOURCE NOTIFY

This message is sent by the NG-RAN node to notify that the QoS requirements of already established GBR QoS flow(s) for which notification control has been requested are either not fulfilled anymore or fulfilled again by the NG-RAN node. This message can also be sent by the NG-RAN node to notify that PDU session resource(s) for a given UE are released.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
PDU Session Resource Notify List		*0..1*			YES	reject
>PDU Session Resource Notify Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Notify Transfer	M		OCTET STRING	Containing the *PDU Session Resource Notify Transfer* IE specified in subclause 9.3.4.5.	-	
PDU Session Resource Released List		*0..1*			YES	ignore
>PDU Session Resource Released Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Notify Released Transfer	M		OCTET STRING	Containing the *PDU Session Resource Notify Released Transfer* IE specified in subclause 9.3.4.13.	-	
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.8 PDU SESSION RESOURCE MODIFY INDICATION

This message is sent by the NG-RAN node and is used to request the AMF to enable modifications of already established PDU session resources for a given UE.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
PDU Session Resource Modify Indication List		*1*			YES	reject
>PDU Session Resource Modify Indication Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Indication Transfer	M		OCTET STRING	Containing the *PDU Session Resource Modify Indication Transfer* IE specified in subclause 9.3.4.6.	-	
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.1.9 PDU SESSION RESOURCE MODIFY CONFIRM

This message is sent by the AMF and is used to confirm the outcome of the request from the PDU SESSION RESOURCE MODIFY INDICATION message.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Modify Confirm List		*0..1*			YES	ignore
>PDU Session Resource Modify Confirm Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Confirm Transfer	M		OCTET STRING	Containing the *PDU Session Resource Modify Confirm Transfer* IE specified in subclause 9.3.4.7.	-	
PDU Session Resource Failed to Modify List		*0..1*			YES	ignore
>PDU Session Resource Failed to Modify Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Modify Indication Unsuccessful Transfer	M		OCTET STRING	Containing the *PDU Session Resource Modify Indication Unsuccessful Transfer* IE specified in subclause 9.3.4.22.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2 UE Context Management Messages

9.2.2.1 INITIAL CONTEXT SETUP REQUEST

This message is sent by the AMF to request the setup of a UE context.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Old AMF	O		AMF Name 9.3.3.21		YES	reject
UE Aggregate Maximum Bit Rate	C-ifPDUsessionResourceSetup		9.3.1.58		YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
GUAMI	M		9.3.3.3		YES	reject
PDU Session Resource Setup Request List		*0..1*			YES	reject
>PDU Session Resource Setup Request Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session NAS-PDU	O		NAS-PDU 9.3.3.4		-	
>>S-NSSAI	M		9.3.1.24		-	
>>PDU Session Resource Setup Request Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Request Transfer* IE specified in subclause 9.3.4.1.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
Allowed NSSAI	M		9.3.1.31	Indicates the S-NSSAIs permitted by the network	YES	reject
UE Security Capabilities	M		9.3.1.86		YES	reject
Security Key	M		9.3.1.87		YES	reject
Trace Activation	O		9.3.1.14		YES	ignore
Mobility Restriction List	O		9.3.1.85		YES	ignore
UE Radio Capability	O		9.3.1.74		YES	ignore
Index to RAT/Frequency Selection Priority	O		9.3.1.61		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore
NAS-PDU	O		9.3.3.4		YES	ignore
Emergency Fallback Indicator	O		9.3.1.26		YES	reject
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
Redirection for Voice EPS Fallback	O		9.3.1.116		YES	ignore
Location Reporting Request Type	O		9.3.1.65		YES	ignore
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
IAB Authorized	O		9.3.1.129		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore

Extended Connected Time	O		9.3.3.31		YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services Authorized	O		9.3.1.147		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE User Plane CIoT Support Indicator	O		9.3.1.160		YES	ignore
RG Level Wireline Access Characteristics	O		OCTET STRING	Specified in TS 23.316 [34]. Indicates the wireline access technology specific QoS information corresponding to a specific wireline access subscription.	YES	ignore
Management Based MDT PLMN List	O		MDT PLMN List 9.3.1.168		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
QMC Configuration Information	O		9.3.1.223		YES	ignore
Target NSSAI Information	O		9.3.1.229		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore
5G ProSe Authorized	O		9.3.1.233		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

Condition	Explanation
ifPDUsessionResourceSetup	This IE shall be present if the *PDU Session Resource Setup List* IE is present.

9.2.2.2 INITIAL CONTEXT SETUP RESPONSE

This message is sent by the NG-RAN node to confirm the setup of a UE context.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Setup Response List		*0..1*			YES	ignore
>PDU Session Resource Setup Response Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Response Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Response Transfer* IE specified in subclause 9.3.4.2.	-	
PDU Session Resource Failed to Setup List		*0..1*			YES	ignore
>PDU Session Resource Failed to Setup Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Unsuccessful Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Unsuccessful Transfer* IE specified in subclause 9.3.4.16.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.3 INITIAL CONTEXT SETUP FAILURE

This message is sent by the NG-RAN node to indicate that the setup of the UE context was unsuccessful.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Failed to Setup List		*0..1*			YES	ignore
>PDU Session Resource Failed to Setup Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Setup Unsuccessful Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Unsuccessful Transfer* IE specified in subclause 9.3.4.16.	-	
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.4 UE CONTEXT RELEASE REQUEST

This message is sent by the NG-RAN node to request the release of the UE-associated logical NG-connection over the NG interface.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
PDU Session Resource List		*0..1*			YES	reject
>PDU Session Resource Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
Cause	M		9.3.1.2		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.5 UE CONTEXT RELEASE COMMAND

This message is sent by the AMF to request the release of the UE-associated logical NG-connection over the NG interface.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
CHOICE *UE NGAP IDs*	M				YES	reject
>UE NGAP ID pair						
>>AMF UE NGAP ID	M		9.3.3.1		-	
>>RAN UE NGAP ID	M		9.3.3.2		-	
>AMF UE NGAP ID						
>>AMF UE NGAP ID	M		9.3.3.1		-	
Cause	M		9.3.1.2		YES	ignore

9.2.2.6 UE CONTEXT RELEASE COMPLETE

This message is sent by the NG-RAN node to confirm the release of the UE-associated logical NG-connection over the NG interface.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore
Information on Recommended Cells and RAN Nodes for Paging	O		9.3.1.100		YES	ignore
PDU Session Resource List		*0..1*			YES	reject
>PDU Session Resource Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>PDU Session Resource Release Response Transfer	O		OCTET STRING	Containing the *PDU Session Resource Release Response Transfer* IE specified in subclause 9.3.4.21.	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.7 UE CONTEXT MODIFICATION REQUEST

This message is sent by the AMF to provide UE Context information changes to the NG-RAN node.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
Security Key	O		9.3.1.87		YES	reject
Index to RAT/Frequency Selection Priority	O		9.3.1.61		YES	ignore
UE Aggregate Maximum Bit Rate	O		9.3.1.58		YES	ignore
UE Security Capabilities	O		9.3.1.86		YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
Emergency Fallback Indicator	O		9.3.1.26		YES	reject
New AMF UE NGAP ID	O		AMF UE NGAP ID 9.3.3.1		YES	reject
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
New GUAMI	O		GUAMI 9.3.3.3		YES	reject
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
IAB Authorized	O		9.3.1.129		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services Authorized	O		9.3.1.147		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
RG Level Wireline Access Characteristics	O		OCTET STRING	Specified in TS 23.316 [34]. Indicates the wireline access technology specific QoS information corresponding to a specific wireline access subscription.	YES	ignore
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
QMC Configuration Information	O		9.3.1.223		YES	ignore
QMC Deactivation	O		9.3.1.222		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore
Management Based MDT PLMN Modification	O		MDT PLMN Modification List		YES	ignore

List			9.3.1.243			
5G ProSe Authorized	O		9.3.1.233		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore

9.2.2.8 UE CONTEXT MODIFICATION RESPONSE

This message is sent by the NG-RAN node to confirm the performed UE context updates.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
RRC State	O		9.3.1.92		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.9 UE CONTEXT MODIFICATION FAILURE

This message is sent by the NG-RAN node in case the performed UE context update is not successful.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.10 RRC INACTIVE TRANSITION REPORT

This message is sent by the NG-RAN node to notify the 5GC the UE enters or leaves RRC_INACTIVE state.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RRC State	M		9.3.1.92		YES	ignore
User Location Information	M		9.3.1.16		YES	ignore

9.2.2.11 CONNECTION ESTABLISHMENT INDICATION

This message is sent by the AMF to complete the establishment of the UE-associated logical NG-connection.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
UE Radio Capability	O		9.3.1.74		YES	ignore
End Indication	O		9.3.3.32		YES	ignore
S-NSSAI	O		9.3.1.24		YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network	YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
DL CP Security Information	O		9.3.3.49		YES	ignore
NB-IoT UE Priority	O		9.3.1.145		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Masked IMEISV	O		9.3.1.54		YES	ignore
Old AMF	O		AMF Name 9.3.3.21		YES	reject

9.2.2.12 AMF CP RELOCATION INDICATION

This message is sent by the AMF to inform the NG-RAN node that the UE is to be relocated as described in TS. 38.300 [8].

Direction: AMF ☑☰ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
S-NSSAI	O		9.3.1.24		YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network	YES	ignore

9.2.2.13 RAN CP RELOCATION INDICATION

This message is sent by the NG-RAN node to initiate the establishment of a UE-associated logical NG-connection, following the reception of re-establishment request.

Direction: NG-RAN node ☑☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
5G-S-TMSI	M		9.3.3.20		YES	reject
E-UTRA CGI	M		9.3.1.9		YES	ignore
TAI	M		9.3.3.11		YES	ignore
UL CP Security Information	M		9.3.3.48		YES	reject

9.2.2.14 RETRIEVE UE INFORMATION

The message is sent by the NG-RAN node to request UE information over the NG interface.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
5G-S-TMSI	M		9.3.3.20		YES	reject

9.2.2.15 UE INFORMATION TRANSFER

The message is sent by the AMF to transfer UE information over the NG interface.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
5G-S-TMSI	M		9.3.3.20		YES	reject
NB-IoT UE Priority	O		9.3.1.145		YES	ignore
UE Radio Capability	O		9.3.1.74		YES	ignore
S-NSSAI	O		9.3.1.24		YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network	YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore

9.2.2.16 UE CONTEXT SUSPEND REQUEST

This message is sent by the NG-RAN node to request the AMF to suspend the UE context and the related PDU session contexts.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Information on Recommended Cells and RAN Nodes for Paging	O		9.3.1.100		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore
PDU Session Resource Suspend List		*0..1*			YES	reject
>PDU Session Resource Suspend Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>UE Context Suspend Request Transfer	M		Containing the *UE Context Suspend Request Transfer* IE specified in subclause 9.3.4.26.		-	

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.17 UE CONTEXT SUSPEND RESPONSE

This message is sent by the AMF to indicate to the NG-RAN node the UE context and the related PDU session contexts have been suspended.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Security Context	O		9.3.1.88		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.18 UE CONTEXT SUSPEND FAILURE

This message is sent by the AMF to indicate to the NG-RAN node that suspension of the UE context has failed in the 5GC.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.2.19 UE CONTEXT RESUME REQUEST

This message is sent by the NG-RAN node to request the AMF to resume the UE-associated logical NG-connection and UE context.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RRC Resume Cause	M		RRC Establishment Cause 9.3.1.111		YES	ignore
PDU Session Resource Resume List		*0..1*			YES	reject
>PDU Session Resource Resume Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>UE Context Resume Request Transfer	M		OCTET STRING	Containing the *UE Context Resume Request Transfer* IE specified in subclause 9.3.4.24	-	
PDU Session Resource Failed to Resume List		*0..1*			YES	reject
>PDU Session Resource Failed to Resume Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Cause	M		9.3.1.2		-	
Suspend Request Indication	O		9.3.1.158		YES	ignore
Information on Recommended Cells and RAN Nodes for Paging	O		9.3.1.100		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.20 UE CONTEXT RESUME RESPONSE

This message is sent by the AMF to indicate to the NG-RAN node that the UE context and the related PDU session contexts have been resumed in the 5GC.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Resume List		*0..1*			YES	reject
>PDU Session Resource Resume Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>UE Context Resume Response Transfer	M		OCTET STRING	Containing the *UE Context Resume Response Transfer* IE specified in subclause 9.3.4.25	-	
PDU Session Resource Failed to Resume List		*0..1*			YES	reject
>PDU Session Resource Failed to Resume Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Cause	M		9.3.1.2		-	
Security Context	O		9.3.1.88		YES	reject
Suspend Response Indication	O		9.3.1.159		YES	ignore
Extended Connected Time	O		9.3.3.31		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.2.21 UE CONTEXT RESUME FAILURE

This message is sent by the AMF to indicate to the NG-RAN node that resumption of the UE context and the related PDU session contexts has failed in the 5GC.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.3 UE Mobility Management Messages

9.2.3.1 HANDOVER REQUIRED

This message is sent by the source NG-RAN node to the AMF to request the preparation of resources at the target.

Direction: NG-RAN node ☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Handover Type	M		9.3.1.22		YES	reject
Cause	M		9.3.1.2		YES	ignore
Target ID	M		9.3.1.25		YES	reject
Direct Forwarding Path Availability	O		9.3.1.64		YES	ignore
PDU Session Resource List		*1*			YES	reject
>PDU Session Resource Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Required Transfer	M		OCTET STRING	Containing the *Handover Required Transfer* IE specified in subclause 9.3.4.14.	-	
Source to Target Transparent Container	M		9.3.1.20		YES	reject

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.2 HANDOVER COMMAND

This message is sent by the AMF to inform the source NG-RAN node that resources for the handover have been prepared at the target side.

Direction: AMF☰ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Handover Type	M		9.3.1.22		YES	reject
NAS Security Parameters from NG-RAN	C-iftoEPSUTRA		9.3.3.26		YES	reject
PDU Session Resource Handover List		*0..1*			YES	ignore
>PDU Session Resource Handover Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Command Transfer	M		OCTET STRING	Containing the *Handover Command Transfer* IE specified in subclause 9.3.4.10.	-	
PDU Session Resource to Release List		*0..1*			YES	ignore
>PDU Session Resource to Release Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Preparation Unsuccessful Transfer	M		OCTET STRING	Containing the *Handover Preparation Unsuccessful Transfer* IE specified in subclause 9.3.4.18.	-	
Target to Source Transparent Container	M		9.3.1.21		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

Condition	Explanation
iftoEPSUTRA	This IE shall be present if the *Handover Type* IE is set to the value "5GStoEPS" or "5GtoUTRA".

9.2.3.3 HANDOVER PREPARATION FAILURE

This message is sent by the AMF to inform the source NG-RAN node that the Handover Preparation has failed.

Direction: AMF ☰ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Target to Source Failure Transparent Container	O		9.3.1.186		YES	ignore

9.2.3.4 HANDOVER REQUEST

This message is sent by the AMF to the target NG-RAN node to request the preparation of resources.

Direction: AMF ☰ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
Handover Type	M		9.3.1.22		YES	reject
Cause	M		9.3.1.2		YES	ignore
UE Aggregate Maximum Bit Rate	M		9.3.1.58		YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
UE Security Capabilities	M		9.3.1.86		YES	reject
Security Context	M		9.3.1.88		YES	reject
New Security Context Indicator	O		9.3.1.55		YES	reject
NASC	O		NAS-PDU 9.3.3.4	Refers to either the “Intra N1 mode NAS transparent container” or the “S1 mode to N1 mode NAS transparent container”, the details of the IE definition and the encoding arespecified in TS 24.501 [26].	YES	reject
PDU Session Resource Setup List		*1*			YES	reject
>PDU Session Resource Setup Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>S-NSSAI	M		9.3.1.24		-	
>>Handover Request Transfer	M		OCTET STRING	Containing the *PDU Session Resource Setup Request Transfer* IE specified in subclause 9.3.4.1.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
Allowed NSSAI	M		9.3.1.31	Indicates the S-NSSAIs permitted by the network.	YES	reject
Trace Activation	O		9.3.1.14		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore
Source to Target Transparent Container	M		9.3.1.20		YES	reject
Mobility Restriction List	O		9.3.1.85		YES	ignore
Location Reporting Request Type	O		9.3.1.65		YES	ignore
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
GUAMI	M		9.3.3.3		YES	reject
Redirection for Voice EPS Fallback	O		9.3.1.116		YES	ignore
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore

IAB Authorized	O		9.3.1.129		YES	reject
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services Authorized	O		9.3.1.147		YES	ignore
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE User Plane CIoT Support Indicator	O		9.3.1.160		YES	ignore
Management Based MDT PLMN List	O		MDT PLMN List 9.3.1.168		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Extended Connected Time	O		9.3.3.31		YES	ignore
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
UE Slice Maximum Bit Rate List	O		9.3.1.231		YES	ignore
5G ProSe Authorized	O		9.3.1.233		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.5 HANDOVER REQUEST ACKNOWLEDGE

This message is sent by the target NG-RAN node to inform the AMF about the prepared resources at the target.

Direction: NG-RAN node ☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2	Allocated at the target NG-RAN node.	YES	ignore
PDU Session Resource Admitted List		*1*			YES	ignore
>PDU Session Resource Admitted Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Request Acknowledge Transfer	M		OCTET STRING	Containing the *Handover Request Acknowledge Transfer* IE specified in subclause 9.3.4.11.	-	
PDU Session Resource Failed to Setup List		*0..1*			YES	ignore
>PDU Session Resource Failed to Setup Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Handover Resource Allocation Unsuccessful Transfer	M		OCTET STRING	Containing the *Handover Resource Allocation Unsuccessful Transfer* IE specified in subclause 9.3.4.19.	-	
Target to Source Transparent Container	M		9.3.1.21		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore
NPN Access Information	O		9.3.3.46		YES	reject
RedCap Indication	O		9.3.1.228		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.6 HANDOVER FAILURE

This message is sent by the target NG-RAN node to inform the AMF that the preparation of resources has failed.

Direction: NG-RAN node ☑☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Target to Source Failure Transparent Container	O		9.3.1.186		YES	ignore

9.2.3.7 HANDOVER NOTIFY

This message is sent by the target NG-RAN node to inform the AMF that the UE has been identified in the target cell and the handover has been completed.

Direction: NG-RAN node ☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
Notify Source NG-RAN Node	O		ENUMERATED (NotifySource, …)		YES	ignore

9.2.3.8 PATH SWITCH REQUEST

This message is sent by the NG-RAN node to inform the AMF of the new serving NG-RAN node and to transfer some NG-U DL tunnel termination point(s) to the SMF via the AMF for one or multiple PDU session resources.

Direction: NG-RAN node ☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Source AMF UE NGAP ID	M		AMF UE NGAP ID 9.3.3.1		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
UE Security Capabilities	M		9.3.1.86		YES	ignore
PDU Session Resource to be Switched in Downlink List		*1*			YES	reject
>PDU Session Resource to be Switched in Downlink Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Transfer	M		OCTET STRING	Containing the *Path Switch Request Transfer* IE specified in subclause 9.3.4.8.	-	
PDU Session Resource Failed to Setup List		*0..1*			YES	ignore
>PDU Session Resource Failed to Setup Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Setup Failed Transfer	M		OCTET STRING	Containing the *Path Switch Request Setup Failed Transfer* IE specified in subclause 9.3.4.15.	-	
RRC Resume Cause	O		RRC Establishment Cause 9.3.1.111		YES	ignore
RedCap Indication	O		9.3.1.228		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.9 PATH SWITCH REQUEST ACKNOWLEDGE

This message is sent by the AMF to inform the NG-RAN node that the path switch has been successfully completed in the 5GC.

Direction: AMF ⇉ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
UE Security Capabilities	O		9.3.1.86		YES	reject
Security Context	M		9.3.1.88		YES	reject
New Security Context Indicator	O		9.3.1.55		YES	reject
PDU Session Resource Switched List		*1*			YES	ignore
>PDU Session Resource Switched Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Acknowledge Transfer	M		OCTET STRING	Containing the *Path Switch Request Acknowledge Transfer* IE specified in subclause 9.3.4.9.	-	
>>PDU Session Expected UE Activity Behaviour	O		Expected UE Activity Behaviour 9.3.1.94	Expected UE Activity Behaviour for the PDU Session.	YES	ignore
PDU Session Resource Released List		*0..1*			YES	ignore
>PDU Session Resource Released Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Unsuccessful Transfer	M		OCTET STRING	Containing the *Path Switch Request Unsuccessful Transfer* IE specified in subclause 9.3.4.20.	-	
Allowed NSSAI	M		9.3.1.31	Indicates the S-NSSAIs permitted by the network.	YES	reject
Core Network Assistance Information for RRC INACTIVE	O		9.3.1.15		YES	ignore
RRC Inactive Transition Report Request	O		9.3.1.91		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
Redirection for Voice EPS Fallback	O		9.3.1.116		YES	ignore
CN Assisted RAN Parameters Tuning	O		9.3.1.119		YES	ignore
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
Extended Connected Time	O		9.3.3.31		YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
NR V2X Services Authorized	O		9.3.1.146		YES	ignore
LTE V2X Services	O		9.3.1.147		YES	ignore

Authorized						
NR UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.148	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
LTE UE Sidelink Aggregate Maximum Bit Rate	O		9.3.1.149	This IE applies only if the UE is authorized for LTE V2X services.	YES	ignore
PC5 QoS Parameters	O		9.3.1.150	This IE applies only if the UE is authorized for NR V2X services.	YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE User Plane CIoT Support Indicator	O		9.3.1.160		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Management Based MDT PLMN List	O		MDT PLMN List 9.3.1.168	This IE is ignored if the *Management Based MDT PLMN Modification List* IE is present.	YES	ignore
Time Synchronisation Assistance Information	O		9.3.1.220		YES	ignore
5G ProSe Authorized	O		9.3.1.233		YES	ignore
5G ProSe UE PC5 Aggregate Maximum Bit Rate	O		NR UE Sidelink Aggregate Maximum Bit Rate 9.3.1.148	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
5G ProSe PC5 QoS Parameters	O		9.3.1.234	This IE applies only if the UE is authorized for 5G ProSe services.	YES	ignore
Management Based MDT PLMN Modification List	O		MDT PLMN Modification List 9.3.1.243		YES	ignore
IAB Authorized	O		9.3.1.129		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.10 PATH SWITCH REQUEST FAILURE

This message is sent by the AMF to inform the NG-RAN node that a failure has occurred in the 5GC during the Path Switch Request procedure.

Direction: AMF ☰ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Released List		*1*			YES	ignore
>PDU Session Resource Released Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Path Switch Request Unsuccessful Transfer	M		OCTET STRING	Containing the PDU session *Path Switch Request Unsuccessful Transfer* IE specified in subclause 9.3.4.20.	-	
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.3.11 HANDOVER CANCEL

This message is sent by the source NG-RAN node to the AMF to request the cancellation of an ongoing handover.

Direction: NG-RAN node ☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.3.12 HANDOVER CANCEL ACKNOWLEDGE

This message is sent by the AMF to the source NG-RAN node to confirm that the ongoing handover was cancelled.

Direction: AMF ☰ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.3.13 UPLINK RAN STATUS TRANSFER

This message is sent by the source NG-RAN node to transfer the uplink PDCP-SN and HFN receiver status and the downlink PDCP SN and HFN transmitter status during intra 5GC NG-based handover.

Direction: NG-RAN node ☑☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Status Transfer Transparent Container	M		9.3.1.108		YES	reject

9.2.3.14 DOWNLINK RAN STATUS TRANSFER

This message is sent by the AMF to the target NG-RAN node to transfer the uplink PDCP-SN and HFN receiver status and the downlink PDCP SN and HFN transmitter status during intra 5GC NG-based handover.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
RAN Status Transfer Transparent Container	M		9.3.1.108		YES	reject

9.2.3.15 HANDOVER SUCCESS

This message is sent by the AMF to the source NG-RAN node to indicate the successful access of the UE toward the target NG-RAN node.

Direction: AMF ☑☰ NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject

9.2.3.16 UPLINK RAN EARLY STATUS TRANSFER

This message is sent by the source NG-RAN node to transfer the COUNT value(s) of the first forwarded downlink SDU(s) during NG DAPS Handover.

Direction: NG-RAN node ☑☰ AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Early Status Transfer Transparent Container	M		9.3.1.190		YES	reject

9.2.3.17 DOWNLINK RAN EARLY STATUS TRANSFER

This message is sent by the AMF to transfer the COUNT value(s) of the first forwarded downlink SDU(s) during NG DAPS Handover.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Early Status Transfer Transparent Container	M		9.3.1.190		YES	reject

9.2.4 Paging Messages

9.2.4.1 PAGING

This message is sent by the AMF and is used to page a UE in one or several tracking areas.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
UE Paging Identity	M		9.3.3.18		YES	ignore
Paging DRX	O		9.3.1.90		YES	ignore
TAI List for Paging		*1*			YES	ignore
>TAI List for Paging Item		*1..<maxnoofTAIforPaging>*			-	
>>TAI	M		9.3.3.11		-	
Paging Priority	O		9.3.1.78		YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
Paging Origin	O		9.3.3.22		YES	ignore
Assistance Data for Paging	O		9.3.1.69		YES	ignore
NB-IoT Paging eDRX Information	O		9.3.1.138		YES	ignore
NB-IoT Paging DRX	O		9.3.1.139	If this IE is present, the *Paging DRX* IE is ignored.	YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
WUS Assistance Information	O		9.3.1.143		YES	ignore
E-UTRA Paging eDRX Information	O		9.3.1.154		YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
NR Paging eDRX Information	O		9.3.1.227		YES	ignore
Paging Cause	O		ENUMERATED (voice, …)		YES	ignore
PEIPS Assistance Information	O		9.3.1.232		YES	ignore

Range bound	Explanation
maxnoofTAIforPaging	Maximum no. of TAIs for paging. Value is 16.

9.2.4.2 MULTICAST GROUP PAGING

This message is sent by the AMF and is used to notify involved UEs about the activation of a multicast MBS session.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
MBS Session ID	M		9.3.1.206		YES	ignore
MBS Service Area	O		9.3.1.208		YES	ignore
Multicast Group Paging Area List		*1*			YES	ignore
>Multicast Group Paging Area Item		*1..<maxnoofPagingAreas>*			-	
>>Multicast Group Paging Area	M		9.3.1.216		-	
>>UE Paging List		*0..1*			-	
>>>UE Paging Item		*1..<maxnoofUEsforPaging>*			-	
>>>>UE Identity Index Value	M		9.3.3.23		-	
>>>>Paging DRX	O		9.3.1.90		-	

Range bound	Explanation
maxnoofPagingAreas	Maximum no. of paging areas for multicast group paging. Value is 64.
maxnoofUEsforPaging	Maximum no. of UEs allowed within one paging area for multicast group paging. Value is 4096.

9.2.5 NAS Transport Messages

9.2.5.1 INITIAL UE MESSAGE

This message is sent by the NG-RAN node to transfer the initial layer 3 message to the AMF over the NG interface.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NAS-PDU	M		9.3.3.4		YES	reject
User Location Information	M		9.3.1.16		YES	reject
RRC Establishment Cause	M		9.3.1.111		YES	ignore
5G-S-TMSI	O		9.3.3.20		YES	reject
AMF Set ID	O		9.3.3.12		YES	ignore
UE Context Request	O		ENUMERATED (requested, ...)		YES	ignore
Allowed NSSAI	O		9.3.1.31		YES	reject
Source to Target AMF Information Reroute	O		9.3.3.27		YES	ignore
Selected PLMN Identity	O		PLMN Identity 9.3.3.5	Indicates the selected PLMN id for the non-3GPP access.	YES	ignore
IAB Node Indication	O		ENUMERATED (true, ...)	Indication of an IAB node	YES	reject
CE-mode-B Support Indicator	O		9.3.1.156		YES	reject
LTE-M Indication	O		9.3.1.157		YES	ignore
EDT Session	O		ENUMERATED (true, …)		YES	ignore
Authenticated Indication	O		ENUMERATED (true, …)	Indicates the FN-RG has been authenticated by the access network.	YES	ignore
NPN Access Information	O		9.3.3.46		YES	reject
RedCap Indication	O		9.3.1.228		YES	ignore

9.2.5.2 DOWNLINK NAS TRANSPORT

This message is sent by the AMF and is used for carrying NAS information over the NG interface.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Old AMF	O		AMF Name 9.3.3.21		YES	reject
RAN Paging Priority	O		9.3.3.15		YES	ignore
NAS-PDU	M		9.3.3.4		YES	reject
Mobility Restriction List	O		9.3.1.85		YES	ignore
Index to RAT/Frequency Selection Priority	O		9.3.1.61		YES	ignore
UE Aggregate Maximum Bit Rate	O		9.3.1.58		YES	ignore
Allowed NSSAI	O		9.3.1.31	Indicates the S-NSSAIs permitted by the network.	YES	reject
SRVCC Operation Possible	O		9.3.1.128		YES	ignore
Enhanced Coverage Restriction	O		9.3.1.140		YES	ignore
Extended Connected Time	O		9.3.3.31		YES	ignore
UE Differentiation Information	O		9.3.1.144		YES	ignore
CE-mode-B Restricted	O		9.3.1.155		YES	ignore
UE Radio Capability	O		9.3.1.74		YES	ignore
UE Capability Info Request	O		9.3.1.192		YES	ignore
End Indication	O		9.3.3.32		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject
Target NSSAI Information	O		9.3.1.229		YES	ignore
Masked IMEISV	O		9.3.1.54		YES	ignore

9.2.5.3 UPLINK NAS TRANSPORT

This message is sent by the NG-RAN node and is used for carrying NAS information over the NG interface.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NAS-PDU	M		9.3.3.4		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
W-AGF Identity Information	O		OCTET STRING	Containing the *WAgfInfo* IE specified in TS 29.510 [36].	YES	reject
TNGF Identity Information	O		OCTET STRING	Containing the *TngfInfo* IE specified in TS 29.510 [36].	YES	reject
TWIF Identity Information	O		OCTET STRING	Containing the *TwifInfo* IE specified in TS 29.510 [36].	YES	reject

9.2.5.4 NAS NON DELIVERY INDICATION

This message is sent by the NG-RAN node and is used for reporting the non-delivery of a NAS PDU previously received within a DOWNLINK NAS TRANSPORT message or the *NAS-PDU* IE previously received within the PDU SESSION RESOURCE SETUP REQUEST message over the NG interface.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NAS-PDU	M		9.3.3.4		YES	ignore
Cause	M		9.3.1.2		YES	ignore

9.2.5.5 REROUTE NAS REQUEST

This message is sent by the AMF in order to request for a rerouting of the INITIAL UE MESSAGE to another AMF.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
AMF UE NGAP ID	O		9.3.3.1		YES	ignore
NGAP Message	M		OCTET STRING	Contains the INITIAL UE MESSAGE	YES	reject
AMF Set ID	M		9.3.3.12		YES	reject
Allowed NSSAI	O		9.3.1.31		YES	reject
Source to Target AMF Information Reroute	O		9.3.3.27		YES	ignore

9.2.6 Interface Management Messages

9.2.6.1 NG SETUP REQUEST

This message is sent by the NG-RAN node to transfer application layer information for an NG-C interface instance.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Global RAN Node ID	M		9.3.1.5		YES	reject
RAN Node Name	O		PrintableString (SIZE(1..150, …))		YES	ignore
Supported TA List		*1*		Supported TAs in the NG-RAN node.	YES	reject
>Supported TA Item		*1..<maxnoofTACs>*			-	
>>TAC	M		9.3.3.10	Broadcast TAC	-	
>>Broadcast PLMN List		*1*			-	
>>>Broadcast PLMN Item		*1..<maxnoofBPLMNs>*			-	
>>>>PLMN Identity	M		9.3.3.5	Broadcast PLMN	-	
>>>>TAI Slice Support List	M		Slice Support List 9.3.1.17	Supported S-NSSAIs per TAC, per PLMN or per SNPN.	-	
>>>>NPN Support	O		9.3.3.44	If the *NID* IE is included, it identifies a SNPN together with the *PLMN Identity* IE.	YES	reject
>>>>Extended TAI Slice Support List	O		Extended Slice Support List 9.3.1.191	Additional Supported S-NSSAIs per TAC, per PLMN or per SNPN.	YES	reject
>>>>TAI NSAG Support List	O		9.3.1.238	NSAG information associated with the slices per TAC, per PLMN or per SNPN.	YES	ignore
>>Configured TAC Indication	O		9.3.3.50		YES	ignore
>>RAT Information	O		9.3.1.125	RAT information associated with the TAC of the indicated PLMN(s).	YES	reject
Default Paging DRX	M		Paging DRX 9.3.1.90		YES	ignore
UE Retention Information	O		9.3.1.117		YES	ignore
NB-IoT Default Paging DRX	O		9.3.1.137		YES	ignore
Extended RAN Node Name	O		9.3.1.193		YES	ignore

Range bound	Explanation
maxnoofTACs	Maximum no. of TACs. Value is 256.
maxnoofBPLMNs	Maximum no. of Broadcast PLMNs. Value is 12.

9.2.6.2 NG SETUP RESPONSE

This message is sent by the AMF to transfer application layer information for an NG-C interface instance.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF Name	M		9.3.3.21		YES	reject
Served GUAMI List		*1*			YES	reject
>Served GUAMI Item		*1..<maxnoofServed GUAMIs>*			-	
>>GUAMI	M		9.3.3.3		-	
>>Backup AMF Name	O		AMF Name 9.3.3.21		-	
>>GUAMI Type	O		ENUMERATED (native, mapped, …)		YES	ignore
Relative AMF Capacity	M		9.3.1.32		YES	ignore
PLMN Support List		*1*			YES	reject
>PLMN Support Item		*1..<maxnoofPLMNs>*			-	
>>PLMN Identity	M		9.3.3.5		-	
>>Slice Support List	M		9.3.1.17	Supported S-NSSAIs per PLMN or per SNPN.	-	
>>NPN Support	O		9.3.3.44	If *NID* IE is included, it identifies a SNPN together with the *PLMN Identity* IE.	YES	reject
>>Extended Slice Support List	O		9.3.1.191	Additional Supported S-NSSAIs per PLMN or per SNPN.	YES	reject
>>Onboarding Support	O		ENUMERATED (true, ...)	Indication of onboarding support.	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
UE Retention Information	O		9.3.1.117		YES	ignore
IAB Supported	O		ENUMERATED (true, ...)	Indication of support for IAB.	YES	ignore
Extended AMF Name	O		9.3.3.51		YES	ignore

Range bound	Explanation
maxnoofServedGUAMIs	Maximum no. of GUAMIs served by an AMF. Value is 256.
maxnoofPLMNs	Maximum no. of PLMNs per message. Value is 12.

9.2.6.3 NG SETUP FAILURE

This message is sent by the AMF to indicate NG setup failure.

Direction: AMF NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to Wait	O		9.3.1.56		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.4 RAN CONFIGURATION UPDATE

This message is sent by the NG-RAN node to transfer updated application layer information for an NG-C interface instance.

Direction: NG-RAN node AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
RAN Node Name	O		PrintableString (SIZE(1..150, …))		YES	ignore
Supported TA List		*0..1*		Supported TAs in the NG-RAN node.	YES	reject
>Supported TA Item		*1..<maxnoofTACs>*			-	
>>TAC	M		9.3.3.10	Broadcast TAC	-	
>>Broadcast PLMN List		*1*			-	
>>>Broadcast PLMN Item		*1..<maxnoofBPLMNs>*			-	
>>>>PLMN Identity	M		9.3.3.5	Broadcast PLMN	-	
>>>>TAI Slice Support List	M		Slice Support List 9.3.1.17	Supported S-NSSAIs per TAC, per PLMN or per SNPN.	-	
>>>>NPN Support	O		9.3.3.44	If the *NID* IE is included, it identifies a SNPN together with the *PLMN Identity* IE.	YES	reject
>>>>Extended TAI Slice Support List	O		Extended Slice Support List 9.3.1.191	Additional Supported S-NSSAIs per TAC, per PLMN or per SNPN.	YES	reject
>>>>TAI NSAG Support List	O		9.3.1.238	NSAG information associated with the slices per TAC, per PLMN or per SNPN.	YES	ignore
>>Configured TAC Indication	O		9.3.3.50		YES	ignore
>>RAT Information	O		9.3.1.125	RAT information associated with the TAC of the indicated PLMN(s).	YES	reject
Default Paging DRX	O		Paging DRX 9.3.1.90		YES	ignore
Global RAN Node ID	O		9.3.1.5		YES	ignore
NG-RAN TNL Association to Remove List		*0..1*			YES	reject
>NG-RAN TNL Association to Remove Item		*1..<maxnoofTNLAssociations>*			-	
>>TNL Association Transport Layer Address	M		CP Transport Layer Information 9.3.2.6	Transport layer address of the NG-RAN node.	-	
>>TNL Association Transport Layer Address at AMF	O		CP Transport Layer Information 9.3.2.6	Transport layer address of the AMF.	-	
NB-IoT Default Paging DRX	O		9.3.1.137		YES	ignore
Extended RAN Node Name	O		9.3.1.193		YES	ignore

Range bound	Explanation
maxnoofTACs	Maximum no. of TACs. Value is 256.
maxnoofBPLMNs	Maximum no. of Broadcast PLMNs. Value is 12.
maxnoofTNLAssociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.2.6.5 RAN CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by the AMF to acknowledge the NG-RAN node transfer of updated information for an NG-C interface instance.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.6 RAN CONFIGURATION UPDATE FAILURE

This message is sent by the AMF to indicate RAN configuration update failure.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to Wait	O		9.3.1.56		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.7 AMF CONFIGURATION UPDATE

This message is sent by the AMF to transfer updated information for an NG-C interface instance.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF Name	O		9.3.3.21		YES	reject
Served GUAMI List		*0..1*			YES	reject
>Served GUAMI Item		*1..<maxnoofServed GUAMIs>*			-	
>>GUAMI	M		9.3.3.3		-	
>>Backup AMF Name	O		AMF Name 9.3.3.21		-	
>>GUAMI Type	O		ENUMERATED (native, mapped, …)		YES	ignore
Relative AMF Capacity	O		9.3.1.32		YES	ignore
PLMN Support List		*0..1*			YES	reject
>PLMN Support Item		*1..<maxnoofPLMNs>*			-	
>>PLMN Identity	M		9.3.3.5		-	
>>Slice Support List	M		9.3.1.17	Supported S-NSSAIs per PLMN or per SNPN.	-	
>>NPN Support	O		9.3.3.44	If the *NID* IE is included, it identifies a SNPN together with the *PLMN Identity* IE.	YES	reject
>>Extended Slice Support List	O		9.3.1.191	Additional Supported S-NSSAIs per PLMN or per SNPN.	YES	reject
>>Onboarding Support	O		ENUMERATED (true, ...)	Indication of onboarding support.	YES	ignore
AMF TNL Association to Add List		*0..1*			YES	ignore
>AMF TNL Association to Add Item		*1..<maxnoofTNLAssociations>*			-	
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	AMF Transport Layer information used to set up the new TNL association.	-	
>>TNL Association Usage	O		9.3.2.9		-	
>>TNL Address Weight Factor	M		9.3.2.10		-	
AMF TNL Association to Remove List		*0..1*			YES	ignore
>AMF TNL Association to Remove Item		*1..<maxnoofTNLAssociations>*			-	
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	Transport Layer Address of the AMF.	-	
>>TNL Association Transport Layer Address NG-RAN	O		CP Transport Layer Address 9.3.2.6	Transport Layer Address of the NG-RAN node.	YES	reject
AMF TNL Association to Update List		*0..1*			YES	ignore
>AMF TNL Association to Update		*1..<maxnoofTNLAss*			-	

Item		*ociations>*				
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	AMF Transport Layer information used to identify the TNL association to be updated.	-	
>>TNL Association Usage	O		9.3.2.9		-	
>>TNL Address Weight Factor	O		9.3.2.10		-	
Extended AMF Name	O		9.3.3.51		YES	ignore

Range bound	Explanation
maxnoofServedGUAMIs	Maximum no. of GUAMIs served by an AMF. Value is 256.
maxnoofPLMNs	Maximum no. of PLMNs per message. Value is 12.
maxnoofTNLAssociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.2.6.8 AMF CONFIGURATION UPDATE ACKNOWLEDGE

This message is sent by the NG-RAN node to acknowledge the AMF transfer of updated information for an NG-C interface instance.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF TNL Association Setup List		*0..1*			YES	ignore
>AMF TNL Association Setup Item		*1..<maxnoofTNLAssociations>*			-	
>>AMF TNL Association Address	M		CP Transport Layer Information 9.3.2.6	Previously received AMF Transport Layer information for the TNL association.	-	
AMF TNL Association Failed to Setup List	O		TNL Association List 9.3.2.7		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

Range bound	Explanation
maxnoofTNLAssociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.2.6.9 AMF CONFIGURATION UPDATE FAILURE

This message is sent by the NG-RAN node to indicate AMF configuration update failure.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
Time to Wait	O		9.3.1.56		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.10 AMF STATUS INDICATION

This message is sent by the AMF to support AMF management functions.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Unavailable GUAMI List		*1*		Indicates the GUAMIs configured to be unavailable at the AMF	YES	reject
>Unavailable GUAMI Item		*1..<maxnoofServed GUAMIs>*			-	
>>GUAMI	M		9.3.3.3		-	
>>Timer Approach for GUAMI Removal	O		ENUMERATED (apply timer, ...)		-	
>>Backup AMF Name	O		AMF Name 9.3.3.21		-	

Range bound	Explanation
maxnoofServedGUAMIs	Maximum no. of GUAMIs served by an AMF. Value is 256.

9.2.6.11 NG RESET

This message is sent by both the NG-RAN node and the AMF to request that the NG interface, or parts of the NG interface, be reset.

Direction: NG-RAN node ☰ AMF and AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Cause	M		9.3.1.2		YES	ignore
CHOICE *Reset Type*	M				YES	reject
>NG interface						
>>Reset All	M		ENUMERATED (Reset all, …)		-	
>Part of NG interface						
>>UE-associated Logical NG-connection List	M		9.3.3.25		-	

9.2.6.12 NG RESET ACKNOWLEDGE

This message is sent by both the NG-RAN node and the AMF as a response to an NG RESET message.

Direction: NG-RAN node ☰ AMF and AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
UE-associated Logical NG-connection List	O		9.3.3.25		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.6.13 ERROR INDICATION

This message is sent by both the NG-RAN node and the AMF to indicate that some error has been detected in the node.

Direction: NG-RAN node ☑☰ AMF and AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	O		9.3.3.1		YES	ignore
RAN UE NGAP ID	O		9.3.3.2		YES	ignore
Cause	O		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore
5G-S-TMSI	O		9.3.3.20		YES	ignore

9.2.6.14 OVERLOAD START

This message is sent by the AMF and is used to indicate to the NG-RAN node that the AMF is overloaded.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF Overload Response	O		Overload Response 9.3.1.104		YES	reject
AMF Traffic Load Reduction Indication	O		Traffic Load Reduction Indication 9.3.1.106		YES	ignore
Overload Start NSSAI List		*0..1*			YES	ignore
>Overload Start NSSAI Item		*1..<maxnoofSliceItems>*			-	
>>Slice Overload List	M		9.3.1.107		-	
>>Slice Overload Response	O		Overload Response 9.3.1.104		-	
>>Slice Traffic Load Reduction Indication	O		Traffic Load Reduction Indication 9.3.1.106		-	

Range bound	Explanation
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.

9.2.6.15 OVERLOAD STOP

This message is sent by the AMF and is used to indicate that the AMF is no longer overloaded.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject

9.2.7 Configuration Transfer Messages

9.2.7.1 UPLINK RAN CONFIGURATION TRANSFER

This message is sent by the NG-RAN node in order to transfer RAN configuration information.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
SON Configuration Transfer	O		9.3.3.6		YES	ignore
EN-DC SON Configuration Transfer	O		OCTET STRING	Contains the *EN-DC SON Configuration Transfer* IE as defined in TS 36.413 [16].	YES	ignore
Inter-system SON Configuration Transfer	O		9.3.3.33		YES	ignore

9.2.7.2 DOWNLINK RAN CONFIGURATION TRANSFER

This message is sent by the AMF in order to transfer RAN configuration information.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
SON Configuration Transfer	O		9.3.3.6		YES	ignore
EN-DC SON Configuration Transfer	O		OCTET STRING	Contains the *EN-DC SON Configuration Transfer* IE as defined in TS 36.413 [16].	YES	ignore
Inter-system SON Configuration Transfer	O		9.3.3.33		YES	ignore

9.2.8 Warning Message Transmission Messages

9.2.8.1 WRITE-REPLACE WARNING REQUEST

This message is sent by the AMF to request the start or overwrite of the broadcast of a warning message.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Warning Area List	O		9.3.1.37		YES	ignore
Repetition Period	M		9.3.1.49		YES	reject
Number of Broadcasts Requested	M		9.3.1.38		YES	reject
Warning Type	O		9.3.1.39		YES	ignore
Warning Security Information	O		OCTET STRING (SIZE(50))	This IE is not used in the specification. If received, the IE is ignored.	YES	ignore
Data Coding Scheme	O		9.3.1.41		YES	ignore
Warning Message Contents	O		9.3.1.42		YES	ignore
Concurrent Warning Message Indicator	O		9.3.1.46		YES	reject
Warning Area Coordinates	O		9.3.1.112		YES	ignore

9.2.8.2 WRITE-REPLACE WARNING RESPONSE

This message is sent by the NG-RAN node to acknowledge the AMF on the start or overwrite request of a warning message.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Broadcast Completed Area List	O		9.3.1.43		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.8.3 PWS CANCEL REQUEST

This message is forwarded by the AMF to the NG-RAN node to cancel an already ongoing broadcast of a warning message.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Warning Area List	O		9.3.1.37		YES	ignore
Cancel-All Warning Messages Indicator	O		9.3.1.47		YES	reject

9.2.8.4 PWS CANCEL RESPONSE

This message is sent by the NG-RAN node to indicate the list of warning areas where cancellation of the broadcast of the identified message was successful and unsuccessful.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
Message Identifier	M		9.3.1.35		YES	reject
Serial Number	M		9.3.1.36		YES	reject
Broadcast Cancelled Area List	O		9.3.1.44		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.8.5 PWS RESTART INDICATION

This message is sent by the NG-RAN node to inform the AMF that PWS information for some or all cells of the NG-RAN node are available for reloading from the CBC if needed.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
CHOICE *Cell List for Restart*	M				YES	reject
>E-UTRA						
>>E-UTRA Cell List for Restart		*1..<maxnoofCellsingeNB>*			-	
>>>E-UTRA CGI	M		9.3.1.9		-	
>NR						
>>NR Cell List for Restart		*1..<maxnoofCellsingNB>*			-	
>>>NR CGI	M		9.3.1.7		-	
Global RAN Node ID	M		9.3.1.5		YES	reject
TAI List for Restart		*1..<maxnoofTAIforRestart>*			YES	reject
>TAI	M		9.3.3.11		-	
Emergency Area ID List for Restart		*0..<maxnoofEAIforRestart>*			YES	reject
>Emergency Area ID	M		9.3.1.48		-	

Range bound	Explanation
maxnoofCellsingeNB	Maximum no. of cells that can be served by an ng-eNB. Value is 256.
maxnoofCellsingNB	Maximum no. of cells that can be served by a gNB. Value is 16384.
maxnoofTAIforRestart	Maximum no. of TAIs subject for reloading warning message broadcast. Value is 2048.
maxnoofEAIforRestart	Maximum no. of Emergency Area IDs subject for reloading warning message broadcast. Value is 256.

9.2.8.6 PWS FAILURE INDICATION

This message is sent by the NG-RAN node to inform the AMF that ongoing PWS operation for one or more cells of the NG-RAN node has failed.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
CHOICE *PWS Failed Cell List*	M				YES	reject
>E-UTRA						
>>PWS Failed E-UTRA Cell List		*1..<maxnoofCellsinngeNB>*			-	
>>>E-UTRA CGI	M		9.3.1.9		-	
>NR						
>>PWS Failed NR Cell List		*1..<maxnoofCellsingNB>*			-	
>>>NR CGI	M		9.3.1.7		-	
Global RAN Node ID	M		9.3.1.5		YES	reject

Range bound	Explanation
maxnoofCellsinngeNB	Maximum no. of cells that can be served by an ng-eNB. Value is 256.
maxnoofCellsingNB	Maximum no. of cells that can be served by a gNB. Value is 16384.

9.2.9 NRPPa Transport Messages

9.2.9.1 DOWNLINK UE ASSOCIATED NRPPA TRANSPORT

This message is sent by the AMF and is used for carrying NRPPa message over the NG interface.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.9.2 UPLINK UE ASSOCIATED NRPPA TRANSPORT

This message is sent by the NG-RAN node and is used for carrying NRPPa message over the NG interface.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.9.3 DOWNLINK NON UE ASSOCIATED NRPPA TRANSPORT

This message is sent by the AMF and is used for carrying NRPPa message over the NG interface.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.9.4 UPLINK NON UE ASSOCIATED NRPPA TRANSPORT

This message is sent by the NG-RAN node and is used for carrying NRPPa message over the NG interface.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
Routing ID	M		9.3.3.13		YES	reject
NRPPa-PDU	M		9.3.3.14		YES	reject

9.2.10 Trace Messages

9.2.10.1 TRACE START

This message is sent by the AMF to initiate a trace session for a UE.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Trace Activation	M		9.3.1.14		YES	ignore

9.2.10.2 TRACE FAILURE INDICATION

This message is sent by the NG-RAN node to indicate that a Trace Start procedure or a Deactivate Trace procedure has failed for a UE.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	As per NG-RAN Trace ID in *Trace Activation* IE	YES	ignore
Cause	M		9.3.1.2		YES	ignore

9.2.10.3 DEACTIVATE TRACE

This message is sent by the AMF to deactivate a trace session.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	As per NG-RAN Trace ID in *Trace Activation* IE	YES	ignore

9.2.10.4 CELL TRAFFIC TRACE

This message is sent by the NG-RAN node to transfer trace specific information.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [11] (leftmost 6 octets, with PLMN information encoded as in 9.3.3.5), and Trace Recording Session Reference defined in TS 32.422 [11] (last 2 octets).	YES	ignore
NG-RAN CGI	M		9.3.1.73		YES	ignore
Trace Collection Entity IP Address	M		Transport Layer Address 9.3.2.4	For File based Reporting. Defined in TS 32.422 [11]. This IE is ignored if the *Trace Collection Entity URI* IE is present	YES	ignore
Privacy Indicator	O		ENUMERATED (Immediate MDT, Logged MDT, ...)		YES	ignore
Trace Collection Entity URI	O		URI 9.3.2.14	For Streaming based Reporting. Defined in TS 32.422 [11].	YES	ignore

9.2.11 Location Reporting Messages

9.2.11.1 LOCATION REPORTING CONTROL

This message is used by the AMF to request the NG-RAN node to report the location of the UE.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Location Reporting Request Type	M		9.3.1.65		YES	ignore

9.2.11.2 LOCATION REPORTING FAILURE INDICATION

This message is sent by the NG-RAN node and is used to indicate the failure of location reporting.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.11.3 LOCATION REPORT

This message is used to provide the UE's location.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
UE Presence in Area of Interest List	O		9.3.1.67		YES	ignore
Location Reporting Request Type	M		9.3.1.65	Contains the Location Reporting Request Type to which the Location Report refers.	YES	ignore

9.2.12 UE TNLA Binding Messages

9.2.12.1 UE TNLA BINDING RELEASE REQUEST

This message is sent by the AMF to request the NG-RAN node to release the TNLA binding for the respective UE.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject

9.2.13 UE Radio Capability Management Messages

9.2.13.1 UE RADIO CAPABILITY INFO INDICATION

This message is sent by the NG-RAN node to provide UE radio capability related information to the AMF.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
UE Radio Capability	M		9.3.1.74		YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
UE Radio Capability – E-UTRA Format	O		9.3.1.74a		YES	ignore

9.2.13.2 UE RADIO CAPABILITY CHECK REQUEST

This message is sent by the AMF to request the NG-RAN node to check the compatibility between the UE radio capabilities and network configuration on IMS voice.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
UE Radio Capability	O		9.3.1.74		YES	ignore
UE Radio Capability ID	O		9.3.1.142		YES	reject

9.2.13.3 UE RADIO CAPABILITY CHECK RESPONSE

This message is sent by the NG-RAN node to report IMS voice compatibility between the UE radio capabilities and network configuration.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
IMS Voice Support Indicator	M		9.3.1.89		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.13.4 UE RADIO CAPABILITY ID MAPPING REQUEST

This message is sent by the NG-RAN node to request the AMF to provide mapping information for the indicated UE Radio Capability ID.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
UE Radio Capability ID	M		9.3.1.142		YES	reject

9.2.13.5 UE RADIO CAPABILITY ID MAPPING RESPONSE

This message is sent by the AMF to provide UE Radio Capability information which is mapped to the UE Radio Capability ID indicated by the NG-RAN node in the UE RADIO CAPABILITY ID MAPPING REQUEST message.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
UE Radio Capability ID	M		9.3.1.142		YES	reject
UE Radio Capability	M		9.3.1.74		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.14 Data Usage Reporting Messages

9.2.14.1 SECONDARY RAT DATA USAGE REPORT

This message is sent by the NG-RAN node to report Secondary RAT data usage.

Direction: NG-RAN ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	ignore
RAN UE NGAP ID	M		9.3.3.2		YES	ignore
PDU Session Resource Secondary RAT Usage List		*1*			YES	ignore
>PDU Session Resource Secondary RAT Usage Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>Secondary RAT Data Usage Report Transfer	M		OCTET STRING	Containing the *Secondary RAT Data Usage Report Transfer* IE specified in subclause 9.3.4.23	-	
Handover Flag	O		ENUMERATED (handover_preparation, …)		YES	ignore
User Location Information	O		9.3.1.16		YES	ignore

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.

9.2.15 RIM Information Transfer Messages

9.2.15.1 UPLINK RIM INFORMATION TRANSFER

This message is sent by the NG-RAN node to the AMF to transfer the RIM Information.

Direction: NG-RAN ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
RIM Information Transfer	O		9.3.3.28		YES	ignore

9.2.15.2 DOWNLINK RIM INFORMATION TRANSFER

This message is sent by the AMF to the NG-RAN node to transfer the RIM Information.

Direction: AMF ☑☰ NG-RAN

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
RIM Information Transfer	O		9.3.3.28		YES	ignore

9.2.16 Broadcast Session Management Messages

9.2.16.1 BROADCAST SESSION SETUP REQUEST

This message is sent by the AMF to establish MBS session resources for a broadcast MBS session.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
S-NSSAI	M		9.3.1.24		YES	reject
MBS Service Area	M		9.3.1.208		YES	reject
MBS Session Setup Request Transfer	M		OCTET STRING	Containing the *MBS Session Setup or Modification Request Transfer* IE specified in subclause 9.3.5.3	YES	reject

9.2.16.2 BROADCAST SESSION SETUP RESPONSE

This message is sent by the NG-RAN node to report the successful outcome of the request from the BROADCAST SESSION SETUP REQUEST message.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Setup Response Transfer	O		OCTET STRING	Containing the *MBS Session Setup or Modification Response Transfer* IE specified in subclause 9.3.5.5	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.3 BROADCAST SESSION SETUP FAILURE

This message is sent by the NG-RAN node to report the unsuccessful outcome of the request from the BROADCAST SESSION SETUP REQUEST message.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Setup Failure Transfer	O		OCTET STRING	Containing the *MBS Session Setup or Modification Failure Transfer* IE specified in subclause 9.3.5.6	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.4 BROADCAST SESSION MODIFICATION REQUEST

This message is sent by the AMF to modify MBS session resources or the MBS session broadcast area of a previously established broadcast MBS session.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Service Area	O		9.3.1.208		YES	reject
MBS Session Modification Request Transfer	O		OCTET STRING	Containing the *MBS Session Setup or Modification Request Transfer* IE specified in subclause 9.3.5.3	YES	reject

9.2.16.5 BROADCAST SESSION MODIFICATION RESPONSE

This message is sent by the NG-RAN node to report the successful outcome of the request from the BROADCAST SESSION MODIFICATION REQUEST message.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Modification Response Transfer	O		OCTET STRING	Containing the *MBS Session Setup or Modification Response Transfer* IE specified in subclause 9.3.5.5	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.6 BROADCAST SESSION MODIFICATION FAILURE

This message is sent by the NG-RAN node to report the unsuccessful outcome of the request from the BROADCAST SESSION MODIFICATION REQUEST message.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Modification Failure Transfer	O		OCTET STRING	Containing the *MBS Session Setup or Modification Failure Transfer* IE specified in subclause 9.3.5.6	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.7 BROADCAST SESSION RELEASE REQUEST

This message is sent by the AMF to release the MBS session resources of a previously established broadcast MBS session.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.16.8 BROADCAST SESSION RELEASE RESPONSE

This message is sent by the NG-RAN node to acknowledge the BROADCAST SESSION RELEASE REQUEST message.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Session Release Response Transfer	O		OCTET STRING	Containing the *MBS Session Release Response Transfer* IE specified in subclause 9.3.5.14	YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.16.9 BROADCAST SESSION RELEASE REQUIRED

This message is sent by the NG-RAN node to trigger the AMF to initiate the release of corresponding MBS session resources.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.17 Multicast Session Management Messages

9.2.17.1 DISTRIBUTION SETUP REQUEST

This message is sent by the NG-RAN node to request the setup of the NG-U transport for a multicast MBS session, or for one area session of a location dependent multicast MBS session.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Setup Request Transfer	M		OCTET STRING	Containing the *MBS Distribution Setup Request Transfer* IE specified in subclause 9.3.5.7.	YES	reject

9.2.17.2 DISTRIBUTION SETUP RESPONSE

This message is sent by the AMF to confirm the setup of the NG-U transport.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Setup Response Transfer	M		OCTET STRING	Containing the *MBS Distribution Setup Response Transfer* IE specified in subclause 9.3.5.8.	YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.3 DISTRIBUTION SETUP FAILURE

This message is sent by the AMF to indicate that the setup of the NG-U transport was unsuccessful.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Setup Unsuccessful Transfer	M		OCTET STRING	Containing the *MBS Distribution Setup Unsuccessful Transfer* IE specified in subclause 9.3.5.9.	YES	ignore
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.4 DISTRIBUTION RELEASE REQUEST

This message is sent by the NG-RAN node to request the release of the NG-U transport.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
MBS Distribution Release Request Transfer	M		OCTET STRING	Containing the *MBS Distribution Release Request Transfer* IE specified in subclause 9.3.5.10.	YES	reject
Cause	M		9.3.1.2		YES	ignore

9.2.17.5 DISTRIBUTION RELEASE RESPONSE

This message is sent by the AMF to confirm the release of the NG-U transport.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.6 MULTICAST SESSION ACTIVATION REQUEST

This message is sent by the AMF to a NG-RAN node to request for activating MBS session resources for a multicast MBS session.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Multicast Session Activation Request Transfer	M		OCTET STRING	Containing the *Multicast Session Activation Request Transfer* IE specified in subclause 9.3.5.11.	YES	reject

9.2.17.7 MULTICAST SESSION ACTIVATION RESPONSE

This message is sent by the NG-RAN node to the AMF to indicate that the MBS session resources have been activated.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.8 MULTICAST SESSION ACTIVATION FAILURE

This message is sent by the NG-RAN node to the AMF to indicate that the requested activation of the MBS session resources has failed.

Direction: NG-RAN node ☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.2.13		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.9 MULTICAST SESSION DEACTIVATION REQUEST

This message is sent by the AMF to a NG-RAN node to request to deactivate the MBS session resources of a multicast MBS session.

Direction: AMF ☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Multicast Session Deactivation Request Transfer	M		OCTET STRING	Containing the *Multicast Session Deactivation Request Transfer* IE specified in subclause 9.3.5.12.	YES	reject

9.2.17.10 MULTICAST SESSION DEACTIVATION RESPONSE

This message is sent by the NG-RAN node to the AMF to indicate that the MBS session resources have been deactivated.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.11 MULTICAST SESSION UPDATE REQUEST

This message is sent by the AMF to a NG-RAN node to update the MBS session resources information.

Direction: AMF ☑☰ NG-RAN node

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Multicast Session Update Request Transfer	M		OCTET STRING	Containing the *Multicast Session Update Request Transfer* IE specified in subclause 9.3.5.13.	YES	reject

9.2.17.12 MULTICAST SESSION UPDATE RESPONSE

This message is sent by the NG-RAN node to the AMF to confirm the update of MBS information.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.2.17.13 MULTICAST SESSION UPDATE FAILURE

This message is sent by the NG-RAN node to the AMF to indicate multicast session update failure.

Direction: NG-RAN node ☑☰ AMF

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	reject
MBS Session ID	M		9.3.1.206		YES	reject
MBS Area Session ID	O		9.3.1.207		YES	reject
Cause	M		9.3.1.2		YES	ignore
Criticality Diagnostics	O		9.3.1.3		YES	ignore

9.3 Information Element Definitions

9.3.1 Radio Network Layer Related IEs

9.3.1.1 Message Type

The *Message Type* IE uniquely identifies the message being sent. It is mandatory for all messages.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Procedure Code	M		INTEGER (0..255)	
Type of Message	M		CHOICE (Initiating Message, Successful Outcome, Unsuccessful Outcome, …)	

9.3.1.2 Cause

The purpose of the *Cause* IE is to indicate the reason for a particular event for the NGAP protocol.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Cause Group*	M			
>Radio Network Layer				
>>Radio Network Layer Cause	M		ENUMERATED (Unspecified, TXnRELOCOverall expiry, Successful handover, Release due to NG-RAN generated reason, Release due to 5GC generated reason, Handover cancelled, Partial handover, Handover failure in target 5GC/NG-RAN node or target system, Handover target not allowed, TNGRELOCoverall expiry, TNGRELOCprep expiry, Cell not available, Unknown target ID, No radio resources available in target cell, Unknown local UE NGAP ID, Inconsistent remote UE NGAP ID, Handover desirable for radio reasons, Time critical handover, Resource optimisation handover, Reduce load in serving cell, User inactivity, Radio connection with UE lost, Radio resources not available, Invalid QoS combination, Failure in the radio interface procedure, Interaction with other procedure, Unknown PDU Session ID, Unknown QoS Flow ID, Multiple PDU Session ID Instances, Multiple QoS Flow ID Instances, Encryption and/or integrity protection algorithms not supported, NG intra-system handover triggered, NG inter-system handover triggered, Xn handover triggered, Not supported 5QI value, UE context transfer, IMS voice EPS fallback or RAT fallback triggered, UP integrity protection not possible, UP confidentiality protection not possible, Slice(s) not supported, UE in RRC_INACTIVE state not reachable, Redirection, Resources not available for the slice(s), UE maximum integrity protected data rate reason, Release due to CN-detected mobility, …, N26 interface not available,	

			Release due to pre-emption, Multiple Location Reporting Reference ID Instances, RSN not available for the UP, NPN access denied, CAG only access denied, Insufficient UE Capabilities, RedCap UE not supported, Unknown MBS Session ID, Indicated MBS Session Area Information not served by the gNB, Inconsistent slice info for the session, Misaligned association for the multicast and unicast sessions or flows)	
>Transport Layer				
>>Transport Layer Cause	M		ENUMERATED (Transport resource unavailable, Unspecified, …)	
>NAS				
>>NAS Cause	M		ENUMERATED (Normal release, Authentication failure, Deregister, Unspecified, …, UE not in PLMN serving area)	
>Protocol				
>>Protocol Cause	M		ENUMERATED (Transfer syntax error, Abstract syntax error (reject), Abstract syntax error (ignore and notify), Message not compatible with receiver state, Semantic error, Abstract syntax error (falsely constructed message), Unspecified, …)	
>Miscellaneous				
>>Miscellaneous Cause	M		ENUMERATED (Control processing overload, Not enough user plane processing resources, Hardware failure, O&M intervention, Unknown PLMN or SNPN, Unspecified, …)	

The meaning of the different cause values is described in the following tables. In general, "not supported" cause values indicate that the related capability is missing. On the other hand, "not available" cause values indicate that the related capability is present, but insufficient resources were available to perform the requested action.

Radio Network Layer cause	Meaning
Unspecified	Sent for radio network layer cause when none of the specified cause values applies.
TXnRELOCOverall expiry	The timer guarding the handover that takes place over Xn has abnormally expired.
Successful handover	Successful handover.
Release due to NG-RAN generated reason	Release is initiated due to NG-RAN generated reason.
Release due to 5GC generated reason	Release is initiated due to 5GC generated reason.
Handover cancelled	The reason for the action is cancellation of Handover.
Partial handover	Provides a reason for the handover cancellation. The HANDOVER COMMAND message from AMF contained *PDU Session Resource to Release List* IE or *QoS flow to Release List* and the source NG-RAN node estimated service continuity for the UE would be better by not proceeding with handover towards this particular target NG-RAN node.
Handover failure in target 5GC/ NG-RAN node or target system	The handover failed due to a failure in target 5GC/NG-RAN node or target system.
Handover target not allowed	Handover to the indicated target cell is not allowed for the UE in question.
$TNG_{RELOCoverall}$ expiry	The reason for the action is expiry of timer $TNG_{RELOCoverall}$.
$TNG_{RELOCprep}$ expiry	Handover Preparation procedure is cancelled when timer $TNG_{RELOCprep}$ expires.
Cell not available	The concerned cell is not available.
Unknown target ID	Handover rejected because the target ID is not known to the AMF.
No radio resources available in target cell	Load on target cell is too high.
Unknown local UE NGAP ID	The action failed because the receiving node does not recognise the local UE NGAP ID.
Inconsistent remote UE NGAP ID	The action failed because the receiving node considers that the received remote UE NGAP ID is inconsistent.
Handover desirable for radio reasons	The reason for requesting handover is radio related.
Time critical handover	Handover is requested for time critical reason i.e., this cause value is reserved to represent all critical cases where the connection is likely to be dropped if handover is not performed.
Resource optimisation handover	The reason for requesting handover is to improve the load distribution with the neighbour cells.
Reduce load in serving cell	Load on serving cell needs to be reduced. When applied to handover preparation, it indicates the handover is triggered due to load balancing.
User inactivity	The action is requested due to inactivity on all user data radio bearers (i.e., DRBs and, if applicable, MRBs as per section 16.10.5.2 in TS 38.300 [8]), e.g., NG is requested to be released in order to optimise the radio resources. For L2 U2N Relay UE, this action is requested due to user inactivity on all PDU sessions of L2 U2N Relay UE and its served remote UE(s).
Radio connection with UE lost	The action is requested due to losing the radio connection to the UE.
Radio resources not available	No requested radio resources are available.
Invalid QoS combination	The action was failed because of invalid QoS combination.
Failure in the radio interface procedure	Radio interface procedure has failed.
Interaction with other procedure	The action is due to an ongoing interaction with another procedure.
Unknown PDU Session ID	The action failed because the PDU Session ID is unknown in the NG-RAN node.
Unknown QoS Flow ID	The action failed because the QoS Flow ID is unknown in the NG-RAN node.
Multiple PDU Session ID instances	The action failed because multiple instance of the same PDU Session had been provided to/from the NG-RAN node.
Multiple QoS Flow ID instances	The action failed because multiple instances of the same QoS flow had been provided to the NG-RAN node.
Encryption and/or integrity protection algorithms not supported	The NG-RAN node is unable to support any of the encryption and/or integrity protection algorithms supported by the UE.
NG intra-system handover triggered	The action is due to a NG intra-system handover that has been triggered.
NG inter-system handover triggered	The action is due to a NG inter-system handover that has been triggered.
Xn handover triggered	The action is due to an Xn handover that has been triggered.
Not supported 5QI value	The QoS flow setup failed because the requested 5QI is not supported.
UE context transfer	The action is due to a UE resumes from the NG-RAN node different from the one which sent the UE into RRC_INACTIVE state.
IMS voice EPS fallback or RAT	The setup of QoS flow is failed due to EPS fallback or RAT fallback for IMS voice

fallback triggered	using handover or redirection.
UP integrity protection not possible	The PDU session cannot be accepted according to the required user plane integrity protection policy.
UP confidentiality protection not possible	The PDU session cannot be accepted according to the required user plane confidentiality protection policy.
Slice(s) not supported	Slice(s) not supported.
UE in RRC_INACTIVE state not reachable	The action is requested due to RAN paging failure.
Redirection	The release is requested due to inter-system redirection or intra-system redirection.
Resources not available for the slice(s)	The requested resources are not available for the slice(s).
UE maximum integrity protected data rate reason	The request is not accepted in order to comply with the maximum data rate for integrity protection supported by the UE.
Release due to CN-detected mobility	The context release is requested by the AMF because the UE is already served by another CN node (same or different system), or another NG interface of the same CN node.
N26 interface not available	The action failed due to a temporary failure of the N26 interface.
Release due to pre-emption	Release is initiated due to pre-emption.
Multiple Location Reporting Reference ID Instances	The action failed because multiple areas of interest are set with the same Location Reporting Reference ID.
RSN not available for the UP	The redundant user plane resources indicated by RSN are not available.
NPN access denied	Access was denied, or release is requested, for NPN reasons.
CAG only access denied	Access was denied because the cell is a non-CAG cell and UE is only allowed to access CAG cells.
Insufficient UE Capabilities	The procedure can’t proceed due to insufficient UE capabilities.
RedCap UE not supported	The action failed because target NG-RAN node does not support RedCap UE.
Unknown MBS Session ID	The action failed because the MBS Session ID is unknown.
Indicated MBS Session Area Information not served by the gNB	The action failed because the none of the cells in indicacted MBS Session Area Infomration served by the NG-RAN node.
Inconsistent slice info for the session	The action failed because the slice info of the multicast session is inconsistent.
Misaligned association for the multicast and unicast sessions or flows	The action failed because the Associated Unicast QoS Flow ID has already been used, or the Associated Unicast QoS Flow ID is not defined, or the Associated Unicast QoS Flow ID is not released, or multiple MBS QoS flows associated to the same unicast QoS flow, or same multicast session associated to multiple PDU Sessions.

Transport Layer cause	Meaning
Transport resource unavailable	The required transport resources are not available.
Unspecified	Sent when none of the above cause values applies but still the cause is Transport Network Layer related.

NAS cause	Meaning
Normal release	The release is normal.
Authentication failure	The action is due to authentication failure.
Deregister	The action is due to deregister.
Unspecified	Sent when none of the above other cause values applies but still the cause is NAS related.
UE not in PLMN serving area	The release is due to the UE not being within the serving area of its current PLMN (for NTN).

Protocol cause	Meaning
Transfer syntax error	The received message included a transfer syntax error.
Abstract syntax error (reject)	The received message included an abstract syntax error and the concerning criticality indicated "reject".
Abstract syntax error (ignore and notify)	The received message included an abstract syntax error and the concerning criticality indicated "ignore and notify".
Message not compatible with receiver state	The received message was not compatible with the receiver state.
Semantic error	The received message included a semantic error.
Abstract syntax error (falsely constructed message)	The received message contained IEs or IE groups in wrong order or with too many occurrences.
Unspecified	Sent when none of the above cause values applies but still the cause is Protocol related.

Miscellaneous cause	Meaning
Control processing overload	Control processing overload.
Not enough user plane processing resources	Not enough resources are available related to user plane processing.
Hardware failure	Action related to hardware failure.
O&M intervention	The action is due to O&M intervention.
Unknown PLMN or SNPN	The AMF does not identify any PLMN or SNPN provided by the NG-RAN node.
Unspecified failure	Sent when none of the above cause values applies and the cause is not related to any of the categories Radio Network Layer, Transport Network Layer, NAS or Protocol.

9.3.1.3 Criticality Diagnostics

The *Criticality Diagnostics* IE is sent by the NG-RAN node or the AMF when parts of a received message have not been comprehended or were missing, or if the message contained logical errors. When applicable, it contains information about which IEs were not comprehended or were missing.

For further details on how to use the *Criticality Diagnostics* IE, see clause 10.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Procedure Code	O		INTEGER (0..255)	Used if Criticality Diagnostics is part of Error Indication procedure, and not within the response message of the same procedure that caused the error.
Triggering Message	O		ENUMERATED (initiating message, successful outcome, unsuccessful outcome)	Used only if the Criticality Diagnostics is part of Error Indication procedure.
Procedure Criticality	O		ENUMERATED (reject, ignore, notify)	Used for reporting the Criticality of the Triggering message (Procedure).
Information Element Criticality Diagnostics		*0..<maxnoofErrors>*		
>IE Criticality	M		ENUMERATED (reject, ignore, notify)	Used for reporting the criticality of the triggering IE. The value 'ignore' is not applicable.
>IE ID	M		INTEGER (0..65535)	The IE ID of the not understood or missing IE.
>Type of Error	M		ENUMERATED (not understood, missing, …)	

Range bound	Explanation
maxnoofErrors	Maximum no. of IE errors allowed to be reported with a single message. Value is 256.

9.3.1.4 Bit Rate

This IE indicates the number of bits delivered by NG-RAN in UL or to NG-RAN in DL or by UE in sidelink within a period of time, divided by the duration of the period. It is used, for example, to indicate the maximum or guaranteed bit rate for a GBR QoS flow, or an aggregate maximum bit rate.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Bit Rate	M		INTEGER (0..4,000,000,000,000, …)	The unit is: bit/s

9.3.1.5 Global RAN Node ID

This IE is used to globally identify an NG-RAN node (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *NG-RAN node*	M				-	
>gNB						
>>Global gNB ID	M		9.3.1.6		-	
>ng-eNB						
>>Global ng-eNB ID	M		9.3.1.8		-	
>N3IWF						
>>Global N3IWF ID	M		9.3.1.57		-	
>TNGF						
>>Global TNGF ID	M		9.3.1.161		YES-	reject
>TWIF						
>>Global TWIF ID	M		9.3.1.163		YES-	reject
>W-AGF						
>>Global W-AGF ID	M		9.3.1.162		YES-	reject

9.3.1.6 Global gNB ID

This IE is used to globally identify a gNB (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE *gNB ID*	M			
>gNB ID				
>>gNB ID	M		BIT STRING (SIZE(22..32))	Equal to the leftmost bits of the *NR Cell Identity* IE contained in the *NR CGI* IE of each cell served by the gNB.

9.3.1.7 NR CGI

This IE is used to globally identify an NR cell (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
NR Cell Identity	M		BIT STRING (SIZE(36))	The leftmost bits of the *NR Cell Identity* IE correspond to the gNB ID (defined in subclause 9.3.1.6).

9.3.1.8 Global ng-eNB ID

This IE is used to globally identify an ng-eNB (see TS 38.300 [8]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE *ng-eNB ID*	M			
>Macro ng-eNB ID				
>>Macro ng-eNB ID	M		BIT STRING (SIZE(20))	Equal to the 20 leftmost bits of the *E-UTRA Cell Identity* IE contained in the *E-UTRA CGI* IE of each cell served by the ng-eNB.
>Short Macro ng-eNB ID				
>>Short Macro ng-eNB ID	M		BIT STRING (SIZE(18))	Equal to the 18 leftmost bits of the *E-UTRA Cell Identity* IE contained in the *E-UTRA CGI* IE of each cell served by the ng-eNB.
>Long Macro ng-eNB ID				
>>Long Macro ng-eNB ID	M		BIT STRING (SIZE(21))	Equal to the 21 leftmost bits of the *E-UTRA Cell Identity* IE contained in the *E-UTRA CGI* IE of each cell served by the ng-eNB.

9.3.1.9 E-UTRA CGI

This IE is used to globally identify an E-UTRA cell (see TS 36.300 [17]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
E-UTRA Cell Identity	M		BIT STRING (SIZE(28))	The leftmost bits of the *E-UTRA Cell Identity* IE correspond to the ng-eNB ID (defined in subclause 9.3.1.8) and to the eNB ID (defined in subclause 9.3.1.165).

9.3.1.10 GBR QoS Flow Information

This IE indicates QoS parameters for a GBR QoS flow for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Maximum Flow Bit Rate Downlink	M		Bit Rate 9.3.1.4	Maximum Bit Rate in DL. Details in TS 23.501 [9].	-	
Maximum Flow Bit Rate Uplink	M		Bit Rate 9.3.1.4	Maximum Bit Rate in UL. Details in TS 23.501 [9].	-	
Guaranteed Flow Bit Rate Downlink	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate (provided there is data to deliver) in DL. Details in TS 23.501 [9].	-	
Guaranteed Flow Bit Rate Uplink	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate (provided there is data to deliver) in UL. Details in TS 23.501 [9].	-	
Notification Control	O		ENUMERATED (notification requested, ...)	Details in TS 23.501 [9].	-	
Maximum Packet Loss Rate Downlink	O		Packet Loss Rate 9.3.1.79	Indicates the maximum rate for lost packets that can be tolerated in the downlink direction. Details in TS 23.501 [9].	-	
Maximum Packet Loss Rate Uplink	O		Packet Loss Rate 9.3.1.79	Indicates the maximum rate for lost packets that can be tolerated in the uplink direction. Details in TS 23.501 [9].	-	
Alternative QoS Parameters Set List	O		9.3.1.151	Indicates alternative sets of QoS parameters for the QoS flow.	YES	ignore

9.3.1.11 Void

9.3.1.12 QoS Flow Level QoS Parameters

This IE defines the QoS parameters to be applied to a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *QoS Characteristics*	M				-	
>Non-dynamic 5QI						
>>Non Dynamic 5QI Descriptor	M		9.3.1.28		-	
>Dynamic 5QI						
>>Dynamic 5QI Descriptor	M		9.3.1.18		-	
Allocation and Retention Priority	M		9.3.1.19		-	
GBR QoS Flow Information	O		9.3.1.10	This IE shall be present for GBR QoS flows and is ignored otherwise.	-	
Reflective QoS Attribute	O		ENUMERATED (subject to, …)	Details in TS 23.501 [9]. This IE may be present in case of Non-GBR QoS flows and is ignored otherwise.	-	
Additional QoS Flow Information	O		ENUMERATED (more likely, …)	This IE indicates that traffic for this QoS flow is likely to appear more often than traffic for other flows established for the PDU session. This IE may be present in case of Non-GBR QoS flows and is ignored otherwise.	-	
QoS Monitoring Request	O		ENUMERATED (UL, DL, Both, …, stop)	Indicates to measure UL, or DL, or both UL/DL delays for the associated QoS flow or stop the corresponding QoS monitoring.	YES	ignore
QoS Monitoring Reporting Frequency	O		INTEGER (1.. 1800, …)	Indicates the reporting frequency for RAN part delay for QoS monitoring. Units: second	YES	ignore

9.3.1.13 QoS Flow List with Cause

This IE contains a list of QoS flows with a cause value. It is used for example to indicate failed QoS flow(s) or QoS flow(s) to be released.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Item		*1..<maxnoofQoSFlows>*		
>QoS Flow Identifier	M		9.3.1.51	
>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.1.14 Trace Activation

This IE defines parameters related to a trace session activation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [11] (leftmost 6 octets, with PLMN information encoded as in 9.3.3.5), and Trace Recording Session Reference defined in TS 32.422 [11] (last 2 octets).	-	
Interfaces to Trace	M		BIT STRING (SIZE(8))	Each position in the bitmap represents an NG-RAN node interface: first bit = NG-C, second bit = Xn-C, third bit = Uu, fourth bit = F1-C, fifth bit = E1: other bits reserved for future use. Value '1' indicates 'should be traced'. Value '0' indicates 'should not be traced'.	-	
Trace Depth	M		ENUMERATED (minimum, medium, maximum, minimumWithoutVendorSpecificExtension, mediumWithoutVendorSpecificExtension, maximumWithoutVendorSpecificExtension, …)	Defined in TS 32.422 [11].	-	
Trace Collection Entity IP Address	M		Transport Layer Address 9.3.2.4	For File based Reporting. Defined in TS 32.422 [11]. This IE is ignored if the *Trace Collection Entity URI* IE is present.	-	
MDT Configuration	O		9.3.1.167		YES	ignore
Trace Collection Entity URI	O		URI 9.3.2.14	For Streaming based Reporting. Defined in TS 32.422 [11].	YES	ignore

9.3.1.15 Core Network Assistance Information for RRC INACTIVE

This IE provides assistance information for RRC_INACTIVE configuration.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UE Identity Index Value	M		9.3.3.23		-	
UE Specific DRX	O		Paging DRX 9.3.1.90		-	
Periodic Registration Update Timer	M		9.3.3.24		-	
MICO Mode Indication	O		9.3.1.23		-	
TAI List for RRC Inactive		*1*			-	
>TAI List for RRC Inactive Item		*1..<maxnoofTAIforInactive>*			-	
>>TAI	M		9.3.3.11		-	
Expected UE Behaviour	O		9.3.1.93		-	
E-UTRA Paging eDRX Information	O		9.3.1.154		YES	ignore
Extended UE Identity Index Value	O		9.3.3.52		YES	ignore
UE Radio Capability for Paging	O		9.3.1.68		YES	ignore
MICO All PLMN	O		9.3.1.194		YES	ignore
NR Paging eDRX Information	O		9.3.1.227		YES	ignore
Paging Cause Indication for Voice Service	O		ENUMERATED (supported, …)	This IE indicates whether the UE supports the feature of indication of paging cause for voice service.	YES	ignore
PEIPS Assistance Information	O		9.3.1.232		YES	ignore
Hashed UE Identity Index Value	O		9.3.3.62		YES	ignore

Range bound	Explanation
maxnoofTAIforInactive	Maximum no. of TAIs for RRC Inactive. Value is 16.

9.3.1.16 User Location Information

This IE is used to provide location information of the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *User Location Information*	M				-	
>E-UTRA user location information						
>>E-UTRA CGI	M		9.3.1.9		-	
>>TAI	M		9.3.3.11		-	
>>Age of Location	O		Time Stamp 9.3.1.75	Indicates the UTC time when the location information was generated.	-	
>>PSCell Information	O		NG-RAN CGI 9.3.1.73		YES	ignore
>NR user location information						
>>NR CGI	M		9.3.1.7		-	
>>TAI	M		9.3.3.11	This IE is ignored if the NR NTN TAI Information IE is present.	-	
>>Age of Location	O		Time Stamp 9.3.1.75	Indicates the UTC time when the location information was generated.	-	
>>PSCell Information	O		NG-RAN CGI 9.3.1.73		YES	ignore
>>NID	O		9.3.3.42		YES	reject
>>NR NTN TAI Information	O		9.3.3.53		YES	ignore
>N3IWF user location information						
>>IP Address	M		Transport Layer Address 9.3.2.4	UE's local IP address used to reach the N3IWF	-	
>>Port Number	O		OCTET STRING (SIZE(2))	UDP or TCP source port number if NAT is detected.	-	
>>TAI	O		9.3.3.11		YES	ignore
>TNGF user location information					YES	ignore
>>TNAP ID	M		OCTET STRING	TNAP Identifier used to identify the TNAP. Details in TS 29.571 [35].	-	
>>IP Address	M		Transport Layer Address 9.3.2.4	UE's local IP address used to reach the TNGF.	-	
>>Port Number	O		OCTET STRING (SIZE(2))	UDP or TCP source port number if NAT is detected.	-	
>>TAI	O		9.3.3.11		YES	ignore
>TWIF user location information					YES	ignore
>>TWAP ID	M		OCTET STRING	TWAP Identifier used to identify the TWAP. Details in TS 29.571 [35].	-	
>>IP Address	M		Transport Layer Address	Non-5G-Capable over WLAN	-	

			9.3.2.4	device's local IP address used to reach the TWIF.		
>>Port Number	O		OCTET STRING (SIZE(2))	UDP or TCP source port number if NAT is detected.	-	
>>TAI	O		9.3.3.11		YES	ignore
>W-AGF user location information				Indicates the location information via wireline access as specified in TS 23.316 [34].	YES	ignore
>>W-AGF user location information	M		9.3.1.164		-	

9.3.1.17 Slice Support List

This IE indicates the list of supported slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Support Item		*1..<maxnoofSliceItems>*		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.

9.3.1.18 Dynamic 5QI Descriptor

This IE indicates the QoS Characteristics for a Non-standardised or not pre-configured 5QI for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Priority Level	M		9.3.1.84	Priority Level is specified in TS 23.501 [9].	-	
Packet Delay Budget	M		9.3.1.80	Packet Delay Budget is specified in TS 23.501 [9]. This IE is ignored if the *Extended Packet Delay Budget* IE is present.	-	
Packet Error Rate	M		9.3.1.81	Packet Error Rate is specified in TS 23.501 [9].	-	
5QI	O		INTEGER (0..255, …)	Indicates the dynamically assigned 5QI as specified in TS 23.501 [9].	-	
Delay Critical	C-ifGBRflow		ENUMERATED (delay critical, non-delay critical, …)	Indicates whether the GBR QoS flow is delay critical as specified in TS 23.501 [9].	-	
Averaging Window	C-ifGBRflow		9.3.1.82	Averaging Window is specified in TS 23.501 [9].	-	
Maximum Data Burst Volume	O		9.3.1.83	Maximum Data Burst Volume is specified in TS 23.501 [9]. This IE shall be included if the *Delay Critical* IE is set to “delay critical” and is ignored otherwise.	-	
Extended Packet Delay Budget	O		9.3.1.135	Packet Delay Budget is specified in TS 23.501 [9].	YES	ignore
CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore

Condition	Explanation
ifGBRflow	This IE shall be present if the *GBR QoS Flow Information* IE is present in the *QoS Flow Level QoS Parameters* IE.

9.3.1.19 Allocation and Retention Priority

This IE specifies the relative importance of a QoS flow compared to other QoS flows for allocation and retention of NG-RAN resources.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Priority Level	M		INTEGER (1..15)	**Desc**.: This IE defines the relative importance of a resource request (see TS 23.501 [9]). **Usage**: Values are ordered in decreasing order of priority, i.e., with 1 as the highest priority and 15 as the lowest priority.
Pre-emption Capability	M		ENUMERATED (shall not trigger pre-emption, may trigger pre-emption, …)	**Desc.:** This IE indicates the pre-emption capability of the request on other QoS flows (see TS 23.501 [9]). **Usage**: The QoS flow shall not pre-empt other QoS flows or, the QoS flow may pre-empt other QoS flows. Note: The Pre-emption Capability indicator applies to the allocation of resources for a QoS flow and as such it provides the trigger to the pre-emption procedures/processes of the NG-RAN node.
Pre-emption Vulnerability	M		ENUMERATED (not pre-emptable, pre-emptable, …)	**Desc.**: This IE indicates the vulnerability of the QoS flow to pre-emption of other QoS flows (see TS 23.501 [9]). **Usage**: The QoS flow shall not be pre-empted by other QoS flows or the QoS flow may be pre-empted by other QoS flows. Note: The Pre-emption Vulnerability indicator applies for the entire duration of the QoS flow, unless modified and as such indicates whether the QoS flow is a target of the pre-emption procedures/processes of the NG-RAN node.

9.3.1.20 Source to Target Transparent Container

This IE is used to transparently pass radio related information from the handover source to the handover target through the core network; it is produced by the source RAN node and is transmitted to the target RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Source to Target Transparent Container	M		OCTET STRING	This IE includes a transparent container from the source RAN node to the target RAN node. The octets of the OCTET STRING are encoded according to the specifications of the target system. Note: In the current version of the specification, this IE may carry either the *Source NG-RAN Node to Target NG-RAN Node Transparent Container* IE or the *Source eNB to Target eNB Transparent Container* IE as defined in TS 36.413 [16] or the *Source RNC to Target RNC Transparent Container* IE as defined in TS 25.413 [28].

9.3.1.21 Target to Source Transparent Container

This IE is used to transparently pass radio related information from the handover target to the handover source through the core network; it is produced by the target RAN node and is transmitted to the source RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target to Source Transparent Container	M		OCTET STRING	This IE includes a transparent container from the target RAN node to the source RAN node. The octets of the OCTET STRING are encoded according to the specifications of the target system. Note: In the current version of the specification, this IE may carry either the *Target NG-RAN Node to Source NG-RAN Node Transparent Container* IE or the *Target eNB to Source eNB Transparent Container* IE as defined in TS 36.413 [16], or the *Target RNC to Source RNC Transparent Container* IE as defined in TS 25.413 [28].

9.3.1.22 Handover Type

This IE indicates which kind of handover was triggered in the source side.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Handover Type	M		ENUMERATED (Intra5GS, 5GStoEPS, EPSto5GS, …, 5GStoUTRA)	Intra5GS: NG-RAN node to NG-RAN node 5GStoEPS: NG-RAN node to eNB EPSto5GS: eNB to NG-RAN node 5GStoUTRA: NG-RAN node to UTRA

9.3.1.23 MICO Mode Indication

This IE indicates that the UE is configured with MICO mode by the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MICO Mode Indication	M		ENUMERATED (true, …)	

9.3.1.24 S-NSSAI

This IE indicates the S-NSSAI as defined in TS 23.003 [23].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SST	M		OCTET STRING (SIZE(1))	
SD	O		OCTET STRING (SIZE(3))	

9.3.1.25 Target ID

This IE identifies the target for the handover.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *Target ID*	M				-	
>NG-RAN						
>>Global RAN Node ID	M		9.3.1.5		-	
>>Selected TAI	M		TAI 9.3.3.11		-	
>E-UTRAN						
>>Global eNB ID	M		Global ng-eNB ID 9.3.1.8		-	
>>Selected EPS TAI	M		EPS TAI 9.3.3.17		-	
>Target RNC-ID					YES	reject
>>LAI	M		9.3.3.30		-	
>>RNC-ID	M		9.3.1.123	This IE is ignored if the Extended RNC-ID IE is included in the Target ID IE.	-	
>>Extended RNC-ID	O		9.3.1.124	The Extended RNC-ID IE is used if the RNC identity has a value larger than 4095.	-	
>Target Home eNB ID					YES	reject
>>PLMN Identity	M		9.3.3.5		-	
>>Home eNB ID	M		BIT STRING (SIZE(28))	Equal to the E-UTRA Cell Identity IE contained in the E-UTRA CGI IE of the cell served by the eNB.	-	
>>Selected EPS TAI	M		EPS TAI 9.3.3.17		-	

9.3.1.26 Emergency Fallback Indicator

The IE indicates emergency service fallback.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Emergency Fallback Request Indicator	M		ENUMERATED (emergency fallback requested, …)	
Emergency Service Target CN	O		ENUMERATED (5GC, EPC, …)	

9.3.1.27 Security Indication

This IE contains the user plane integrity protection indication and confidentiality protection indication which indicates the requirements on UP integrity protection and ciphering for corresponding PDU sessions, respectively. Additionally, this IE contains the maximum integrity protected data rate per UE for integrity protection for DRBs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Integrity Protection Indication	M		ENUMERATED (required, preferred, not needed, …)	Indicates whether UP integrity protection shall apply, should apply or shall not apply for the concerned PDU session.	-	
Confidentiality Protection Indication	M		ENUMERATED (required, preferred, not needed, …)	Indicates whether UP ciphering shall apply, should apply or shall not apply for the concerned PDU session.	-	
Maximum Integrity Protected Data Rate Uplink	C-ifIntegrityProtectionRequiredorPreferred		Maximum Integrity Protected Data Rate 9.3.1.103	Indicates the maximum aggregate rate for integrity protected DRBs supported by the UE in UL. If the *Maximum Integrity Protected Data Rate Downlink* IE is absent, this IE applies to both UL and DL.	-	
Maximum Integrity Protected Data Rate Downlink	O		Maximum Integrity Protected Data Rate 9.3.1.103	Indicates the maximum aggregate rate for integrity protected DRBs supported by the UE in the DL.	YES	ignore

Condition	Explanation
ifIntegrityProtectionRequiredorPreferred	This IE shall be present if the *Integrity Protection Indication* IE within the *Security Indication* IE is set to “required” or “preferred”.

9.3.1.28 Non Dynamic 5QI Descriptor

This IE indicates the QoS Characteristics for a standardized or pre-configured 5QI for downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
5QI	M		INTEGER (0..255, …)	Indicates the standardized or pre-configured 5QI as specified in TS 23.501 [9].	-	
Priority Level	O		9.3.1.84	Priority Level is specified in TS 23.501 [9]. When included, it overrides standardized or pre-configured value.	-	
Averaging Window	O		9.3.1.82	Averaging Window is specified in TS 23.501 [9]. When included, it overrides standardized or pre-configured value.	-	
Maximum Data Burst Volume	O		9.3.1.83	Maximum Data Burst Volume is specified in TS 23.501 [9]. When included, it overrides standardized or pre-configured value.	-	
CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore

9.3.1.29 Source NG-RAN Node to Target NG-RAN Node Transparent Container

This IE is produced by the source NG-RAN node and is transmitted to the target NG-RAN node. For inter-system handovers to 5G, the IE is transmitted from the external handover source to the target NG-RAN node.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RRC Container	M		OCTET STRING	Includes the *HandoverPreparationInformation* message as defined in TS 38.331 [18] if the target is a gNB. Includes the *HandoverPreparationInformation* message as defined in TS 36.331 [21] if the target is an ng-eNB.	-	
PDU Session Resource Information List		*0..1*		For intra-system handovers in NG-RAN.	-	
>PDU Session Resource Information Item		*1..<maxnoofPDUSessions>*			-	
>>PDU Session ID	M		9.3.1.50		-	
>>QoS Flow Information List		*1*			-	
>>>QoS Flow Information Item		*1..<maxnoofQoSFlows>*			-	
>>>>QoS Flow Identifier	M		9.3.1.51		-	
>>>>DL Forwarding	O		9.3.1.33		-	
>>>>UL Forwarding	O		9.3.1.118		YES	ignore
>>>>Source Transport Layer Address	O		Transport Layer Address 9.3.2.4	Identifies the TNL address used by the sending node for direct data forwarding towards the target NG-RAN node	YES	ignore
>>>>Source Node Transport Layer Address	O		Transport Layer Address 9.3.2.4	Identifies the TNL address used by the source SN node for direct data forwarding towards the target NG-RAN node	YES	ignore
>>DRBs to QoS Flows Mapping List	O		9.3.1.34		-	
E-RAB Information List		*0..1*		For inter-system handovers to 5G.	-	
>E-RAB Information Item		*1..<maxnoofE-RABs>*			-	
>>E-RAB ID	M		9.3.2.3		-	
>>DL Forwarding	O		9.3.1.33		-	
>>Source Transport Layer Address	O		Transport Layer Address 9.3.2.4	Identifies the TNL address used by the sending node for direct data forwarding towards the target NG-RAN node	YES	ignore
>>Source Node Transport Layer	O		Transport Layer	Identifies the TNL address used by	YES	ignore

Address			Address 9.3.2.4	the source SN node for direct data forwarding towards the target NG-RAN node		
Target Cell ID	M		NG-RAN CGI 9.3.1.73		-	
Index to RAT/Frequency Selection Priority	O		9.3.1.61		-	
UE History Information	M		9.3.1.95		-	
SgNB UE X2AP ID	O		9.3.1.127	Allocated at the Source en-gNB	-	
UE History Information from UE	O		9.3.1.166		YES	ignore
Source Node ID	O		9.3.1.195	Source SN ID	YES	ignore
UE Context Reference at Source	O		RAN UE NGAP ID 9.3.3.2		YES	ignore
MBS Active Session Information Source to Target List		*0..1*			YES	ignore
>MBS Active Session Information Source to Target Item		*1..<maxnoofMBSSessionsofUE>*			-	
>>MBS Session ID	M		9.3.1.206		-	
>>MBS Area Session ID	O		9.3.1.207	If included, this IE indicates the MBS Area Session ID of the UE at the NG-RAN node from which the UE context is transferred	-	
>>MBS Service Area	O		9.3.1.208	Included if available in source NG-RAN node.	-	
>>MBS QoS Flows To Be Setup List	M		9.3.1.236		-	
>>MBS Mapping and Data Forwarding Request List		*0..1*			-	
>>>MBS Mapping and Data Forwarding Request Item		*1..<maxnoofMRBs>*			-	
>>>>MRB ID	M		9.3.1.218	Contains the MRB ID value allocated at the source NG-RAN node.	-	
>>>>MBS QoS Flow List		*1..<maxnoofMBSQoSflows>*			-	
>>>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51		-	
>>>>MRB Progress Information	O		9.3.1.219	The SN information of the last packet which has already been delivered for the MRB.	-	
QMC Configuration Information	O		9.3.1.223	Used for passing the QoE	YES	ignore

				measurement information from the source NG-RAN node to the target NG-RAN node.		
NGAP IE Support Information Request List		*0..1*			YES	ignore
>NGAP IE Support Information Request Item		*1..<maxnoofIESupportInfo>*			-	
>>NGAP Protocol IE-Id	M		9.3.1.239		-	

Range bound	Explanation
maxnoofPDUSessions	Maximum no. of PDU sessions allowed towards one UE. Value is 256.
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.
maxnoofE-RABs	Maximum no. of E-RABs allowed towards one UE. Value is 256.
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.
maxnoofMBSSessionsofUE	Maximum no. of MBS sessions allowed towards one UE. Value is 256.
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.
maxnoofMRBs	Maximum no. of MRBs. Value is 32.
maxnoofIESupportInfo	Maximum no. of IE Support Information. Value is 32.

9.3.1.30 Target NG-RAN Node to Source NG-RAN Node Transparent Container

This IE is produced by the target NG-RAN node and is transmitted to the source NG-RAN node. For inter-system handovers to 5G, the IE is transmitted from the target NG-RAN node to the external relocation source.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RRC Container	M		OCTET STRING	Includes the *HandoverCommand* message as defined in TS 38.331 [18] if the target is a gNB. Includes the *HandoverCommand* message as defined in TS 36.331 [21] if the target is an ng-eNB.	-	
DAPS Response Information List		*0..1*			YES	ignore
>DAPS Response Information Item		*1..<maxnoofDRBs>*			-	
>>DRB ID	M		9.3.1.53		-	
>>DAPS Response Information	M		9.3.1.189	Indicates the response to a requested DAPS Handover	-	
Direct Forwarding Path Availability	O		9.3.1.64	Indicates whether a direct forwarding path between the source SN and the target NG-RAN node is available for inter-system handover	YES	ignore
MBS Active Session Information Target to Source List		*0..1*			YES	ignore
>MBS Active Session Information Target to Source Item		*1..<maxnoofMBSSessionsofUE>*			-	
>>MBS Session ID	M		9.3.1.206		-	
>>Data Forwarding Response MRB List		*0..1*			-	
>>>Data Forwarding Response MRB Item		*1..<maxnoofMRBs>*			-	
>>>>MRB ID	M		9.3.1.218	Contains the MRB ID value allocated at the source NG-RAN node.	-	
>>>>DL Forwarding UP TNL Information	M		UP Transport Layer Information 9.3.2.2		-	
>>>>MRB Progress Information	O		9.3.1.219	This IE includes the information of the oldest packet available at the target NG-RAN node for the MRB.	-	
NGAP IE Support Information Response List	O		9.3.1.242		YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.
maxnoofMBSSessionsofUE	Maximum no. of MBS sessions allowed towards one UE. Value is 256.
maxnoofMRBs	Maximum no. of MRBs. Value is 32.

9.3.1.31 Allowed NSSAI

This IE contains the allowed NSSAI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Allowed S-NSSAI List		*1*		
>Allowed S-NSSAI Item		*1..<maxnoofAllowedS-NSSAIs>*		
>>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofAllowedS-NSSAIs	Maximum no. of allowed S-NSSAI. Value is 8.

9.3.1.32 Relative AMF Capacity

This IE indicates the relative processing capacity of an AMF with respect to the other AMFs in the AMF Set in order to load-balance AMFs within an AMF Set defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Relative AMF Capacity	M		INTEGER (0..255)	

9.3.1.33 DL Forwarding

This IE indicates that the QoS flow or E-RAB is proposed for forwarding of downlink packets.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL Forwarding	M		ENUMERATED (DL forwarding proposed, …)	

9.3.1.34 DRBs to QoS Flows Mapping List

This IE contains a list of DRBs containing information about the mapped QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DRBs to QoS Flows Mapping Item		*1..<maxnoofDRBs>*			-	
>DRB ID	M		9.3.1.53		-	
>Associated QoS Flow List	M		9.3.1.99	Contains information of the QoS flows mapped to the DRB	-	
>DAPS Request Information	O		9.3.1.188		YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.35 Message Identifier

This IE identifies the warning message. It is set by the AMF and transferred to the UE by the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Message Identifier	M		BIT STRING (SIZE(16))	This IE is set by the 5GC, transferred to the UE by the NG-RAN node.

9.3.1.36 Serial Number

This IE identifies a particular message from the source and type indicated by the Message Identifier and is altered every time the message with a given Message Identifier is changed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Serial Number	M		BIT STRING (SIZE(16))	

9.3.1.37 Warning Area List

This IE indicates the areas where the warning message needs to be broadcast or cancelled.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Warning Area*	M			
>E-UTRA Cell IDs				
>>EUTRA CGI List for Warning		*1..<maxnoofCellIDforWarning>*		
>>>E-UTRA CGI	M		9.3.1.9	
>NR Cell IDs				
>>NR CGI List for Warning		*1..<maxnoofCellIDforWarning>*		
>>>NR CGI	M		9.3.1.7	
>TAIs for Warning				
>>TAI List for Warning		*1..<maxnoofTAIforWarning>*		
>>>TAI	M		9.3.3.11	
>Emergency Area IDs				
>>Emergency Area ID List		*1..<maxnoofEmergencyAreaID>*		
>>>Emergency Area ID	M		9.3.1.48	

Range bound	Explanation
maxnoofCellIDforWarning	Maximum no. of Cell ID subject for warning message broadcast. Value is 65535.
maxnoofTAIforWarning	Maximum no. of TAI subject for warning message broadcast. Value is 65535.
maxnoofEmergencyAreaID	Maximum no. of Emergency Area ID subject for warning message broadcast. Value is 65535.

9.3.1.38 Number of Broadcasts Requested

This IE indicates the number of times a message is to be broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Number of Broadcasts Requested	M		INTEGER (0..65535)	

9.3.1.39 Warning Type

This IE indicates types of the disaster. This IE also indicates that a Primary Notification is included. This IE can be used by the UE to differentiate the type of alert according to the type of disaster.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Warning Type	M		OCTET STRING (SIZE(2))	

9.3.1.40 Void

9.3.1.41 Data Coding Scheme

This IE identifies the alphabet or coding employed for the message characters and message handling at the UE (it is passed transparently from the 5GC to the UE).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Coding Scheme	M		BIT STRING (SIZE(8))	

9.3.1.42 Warning Message Contents

This IE contains user information, e.g., the message with warning contents, and will be broadcast over the radio interface.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Warning Message Contents	M		OCTET STRING (SIZE(1..9600))	

9.3.1.43 Broadcast Completed Area List

This IE indicates the areas where either resources are available to perform the broadcast or where broadcast is performed successfully.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Broadcast Completed Area*	M			
>Cell ID Broadcast E-UTRA				
>>Completed Cell List		1..<maxnoofCellIDforWarning>		
>>>E-UTRA CGI	M		9.3.1.9	
>TAI Broadcast E-UTRA				
>>TAI Broadcast		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Completed Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>Emergency Area ID Broadcast E-UTRA				
>>Emergency Area ID Broadcast		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Completed Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>Cell ID Broadcast NR				
>>Completed Cell List		1..<maxnoofCellIDforWarning>		
>>>NR-CGI	M		9.3.1.7	
>TAI Broadcast NR				
>>TAI Broadcast		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Completed Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>NR-CGI	M		9.3.1.7	
>Emergency Area ID Broadcast NR				
>>Emergency Area ID Broadcast		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Completed Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>NR-CGI	M		9.3.1.7	

Range bound	Explanation
maxnoofCellIDforWarning	Maximum no. of Cell ID subject for warning message broadcast. Value is 65535.
maxnoofTAIforWarning	Maximum no. of TAI subject for warning message broadcast. Value is 65535.
maxnoofEmergencyAreaID	Maximum no. of Emergency Area ID subject for warning message broadcast. Value is 65535.
maxnoofCellinTAI	Maximum no. of Cell ID within a TAI. Value is 65535.
maxnoofCellinEAI	Maximum no. of Cell ID within an Emergency Area. Value is 65535.

9.3.1.44 Broadcast Cancelled Area List

This IE indicates the areas where broadcast was stopped successfully.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Broadcast Cancelled Area*	M			
>Cell ID Cancelled E-UTRA				
>>Cancelled Cell List		1..<maxnoofCellIDforWarning>		
>>>E-UTRA CGI	M		9.3.1.9	
>>>Number of Broadcasts	M		9.3.1.45	
>TAI Cancelled E-UTRA				
>>TAI Cancelled		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Cancelled Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>>>>Number of Broadcasts	M		9.3.1.45	
>Emergency Area ID Cancelled E-UTRA				
>>Emergency Area ID Cancelled		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Cancelled Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>E-UTRA CGI	M		9.3.1.9	
>>>>Number of Broadcasts	M		9.3.1.45	
>Cell ID Cancelled NR				
>>Cancelled Cell List		1..<maxnoofCellIDforWarning>		
>>>NR-CGI	M		9.3.1.7	
>>>Number of Broadcasts	M		9.3.1.45	
>TAI Cancelled NR				
>>TAI Cancelled		1..<maxnoofTAIforWarning>		
>>>TAI	M		9.3.3.11	
>>>Cancelled Cell in TAI List		1..<maxnoofCellinTAI>		
>>>>NR-CGI	M		9.3.1.7	
>>>>Number of Broadcasts	M		9.3.1.45	
>Emergency Area ID Cancelled NR				
>>Emergency Area ID Cancelled		1..<maxnoofEmergencyAreaID>		
>>>Emergency Area ID	M		9.3.1.48	
>>>Cancelled Cell in Emergency Area ID List		1..<maxnoofCellinEAI>		
>>>>NR-CGI	M		9.3.1.7	
>>>>Number of Broadcasts	M		9.3.1.45	

Range bound	Explanation
maxnoofCellIDforWarning	Maximum no. of Cell ID subject for warning message broadcast. Value is 65535.
maxnoofTAIforWarning	Maximum no. of TAI subject for warning message broadcast. Value is 65535.
maxnoofEmergencyAreaID	Maximum no. of Emergency Area ID subject for warning message broadcast. Value is 65535.
maxnoofCellinTAI	Maximum no. of Cell ID within a TAI. Value is 65535.
maxnoofCellinEAI	Maximum no. of Cell ID within an Emergency Area. Value is 65535.

9.3.1.45 Number of Broadcasts

This IE indicates the number of times that a particular message has been broadcast in a given warning area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Number of Broadcasts	M		INTEGER (0..65535)	This IE is set to '0' if valid results are not known or not available. It is set to 65535 if the counter results have overflowed.

9.3.1.46 Concurrent Warning Message Indicator

This IE indicates to the NG-RAN node that the received warning message is a new message to be scheduled for concurrent broadcast with any other ongoing broadcast of warning messages.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Concurrent Warning Message Indicator	M		ENUMERATED (true, …)	This IE is used to identify a PWS type warning system which allows the broadcast of multiple concurrent warning messages over the radio.

9.3.1.47 Cancel-All Warning Messages Indicator

This IE indicates to the NG-RAN node to stop all already ongoing broadcast of warning messages in the NG-RAN node or in an area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cancel-All Warning Messages Indicator	M		ENUMERATED (true, …)	

9.3.1.48 Emergency Area ID

This IE is used to indicate the area which has the emergency impact.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Emergency Area ID	M		OCTET STRING (SIZE(3))	Emergency Area ID may consist of several cells. Emergency Area ID is defined by the operator.

9.3.1.49 Repetition Period

This IE indicates the periodicity of the warning message to be broadcast.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Repetition Period	M		INTEGER (0..2^{17}-1)	The unit of value 1 to 2^{17}-1 is [second].

9.3.1.50 PDU Session ID

This IE identifies a PDU Session for a UE. The definition and use of the PDU Session ID is specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session ID	M		INTEGER (0..255)	

9.3.1.51 QoS Flow Identifier

This IE identifies a QoS flow within a PDU Session, or a MBS QoS flow within a MBS session. The definition and use of the QoS Flow Identifier is specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Identifier	M		INTEGER (0..63, …)	

9.3.1.52 PDU Session Type

This IE indicates the PDU Session Type as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session Type	M		ENUMERATED (Ipv4, Ipv6, Ipv4v6, ethernet, unstructured, ...)	

9.3.1.53 DRB ID

This IE contains the DRB ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DRB ID	M		INTEGER (1..32, …)	

9.3.1.54 Masked IMEISV

This IE contains the IMEISV value with a mask, to identify a terminal model without identifying an individual Mobile Equipment.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Masked IMEISV	M		BIT STRING (SIZE(64))	Coded as the International Mobile station Equipment Identity and Software Version Number (IMEISV) defined in TS 23.003 [23] with the last 4 digits of the SNR masked by setting the corresponding bits to 1. The first to fourth bits correspond to the first digit of the IMEISV, the fifth to eighth bits correspond to the second digit of the IMEISV, and so on.

9.3.1.55 New Security Context Indicator

This IE indicates that the AMF has activated a new 5G NAS security context as described in TS 33.501 [13].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
New Security Context Indicator	M		ENUMERATED (true, …)	The NSCI as defined in TS 33.501 [13].

9.3.1.56 Time to Wait

This IE defines the minimum allowed waiting time.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time to Wait	M		ENUMERATED (1s, 2s, 5s, 10s, 20s, 60s, …)	

9.3.1.57 Global N3IWF ID

This IE is used to globally identify an N3IWF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE *N3IWF ID*	M			
>N3IWF ID				
>>N3IWF ID	M		BIT STRING (SIZE(16))	

9.3.1.58 UE Aggregate Maximum Bit Rate

This IE is applicable for all Non-GBR QoS flows per UE which is defined for the downlink and the uplink direction and a subscription parameter provided by the AMF to the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Aggregate Maximum Bit Rate		*1*		Applicable for Non-GBR QoS flows.
>UE Aggregate Maximum Bit Rate Downlink	M		Bit Rate 9.3.1.4	This IE indicates the UE Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the downlink direction.
>UE Aggregate Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.4	This IE indicates the UE Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the uplink direction.

9.3.1.59 Security Result

This IE indicates whether the security policy indicated as "preferred" in the *Security Indication* IE is performed or not.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Integrity Protection Result	M		ENUMERATED (performed, not performed, …)	Indicates whether UP integrity protection is performed or not for the concerned PDU session.
Confidentiality Protection Result	M		ENUMERATED (performed, not performed, …)	Indicates whether UP ciphering is performed or not for the concerned PDU session.

9.3.1.60 User Plane Security Information

This IE indicates user plane security information related to security policy.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Security Result	M		9.3.1.59	
Security Indication	M		9.3.1.27	

9.3.1.61 Index to RAT/Frequency Selection Priority

This IE is used to define local configuration for RRM strategies such as camp priorities in Idle mode and control of inter-RAT/inter-frequency handover in Active mode (see TS 23.501 [9]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Index to RAT/Frequency Selection Priority	M		INTEGER (1..256, …)	

9.3.1.62 Data Forwarding Accepted

This IE indicates that the NG-RAN node accepts the proposed DL data forwarding for the QoS flow which is subject to data forwarding.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Accepted	M		ENUMERATED (data forwarding accepted, …)	

9.3.1.63 Data Forwarding Not Possible

This IE indicates that the 5GC decided that the corresponding PDU session will not be subject to data forwarding.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Not Possible	M		ENUMERATED (data forwarding not possible, …)	

9.3.1.64 Direct Forwarding Path Availability

This IE indicates whether a direct forwarding path is available.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Direct Forwarding Path Availability	M		ENUMERATED (direct path available, …)	

9.3.1.65 Location Reporting Request Type

This IE indicates the type of location request to be handled by the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Event Type	M		ENUMERATED (direct, change of serving cell, UE presence in the area of interest, stop change of serving cell, stop UE presence in the area of interest, cancel location reporting for the UE, …)		-	
Report Area	M		ENUMERATED (cell, …)		-	
Area of Interest List		*0..1*			-	
>Area of Interest Item		*1..<maxnoofAoI>*			-	
>>Area of Interest	M		9.3.1.66		-	
>>Location Reporting Reference ID	M		9.3.1.76		-	
Location Reporting Reference ID to be Cancelled	C-ifEventTypeisStopUEPresinAoI		Location Reporting Reference ID 9.3.1.76		-	
Additional Location Information	O		ENUMERATED (Include PSCell, ...)		YES	ignore

Range bound	Explanation
maxnoofAoI	Maximum no. of areas of interest. Value is 64.

Condition	Explanation
ifEventTypeisStopUEPresinAoI	This IE shall be present if the *Event Type* IE is set to "stop UE presence in the area of interest".

9.3.1.66 Area of Interest

This IE indicates the area of interest.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Area of Interest TAI List		*0..1*		
>Area of Interest TAI Item		*1..<maxnoofTAIinAoI>*		
>>TAI	M		9.3.3.11	
Area of Interest Cell List		*0..1*		
>Area of Interest Cell Item		*1..<maxnoofCellinAoI>*		
>>NG-RAN CGI	M		9.3.1.73	
Area of Interest RAN Node List		*0..1*		
>Area of Interest RAN Node Item		*1..<maxnoofRANNodeinAoI>*		
>>Global RAN Node ID	M		9.3.1.5	

Range bound	Explanation
maxnoofTAlinAoI	Maximum no. of tracking areas in an area of interest. Value is 16.
maxnoofCellinAoI	Maximum no. of cells in an area of interest. Value is 256.
maxnoofRANNodeinAoI	Maximum no. of NG-RAN nodes in an area of interest. Value is 64.

9.3.1.67 UE Presence in Area of Interest List

This IE indicates the UE presence in the area of interest.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Presence in Area of Interest Item		*1..<maxnoofAoI>*		
>Location Reporting Reference ID	M		9.3.1.76	
>UE Presence	M		ENUMERATED (in, out, unknown, …)	

Range bound	Explanation
maxnoofAoI	Maximum no. of areas of interest. Value is 64.

9.3.1.68 UE Radio Capability for Paging

This IE contains paging specific UE Radio Capability information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability for Paging of NR	O		OCTET STRING	Includes the *UERadioPagingInformation* message as defined in TS 38.331 [18].
UE Radio Capability for Paging of E-UTRA	O		OCTET STRING	Includes the *UERadioPagingInformation* message as defined in TS 36.331 [21].
UE Radio Capability for Paging of NB-IoT	O		OCTET STRING	Includes the *UERadioPagingInformation-NB* message as defined in TS 36.331 [21].

9.3.1.69 Assistance Data for Paging

This IE provides assistance information for paging optimisation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Assistance Data for Recommended Cells	O		9.3.1.70		-	
Paging Attempt Information	O		9.3.1.72		-	
NPN Paging Assistance Information	O		9.3.1.183		YES	ignore
Paging Assistance Data for CE Capable UE	O		9.3.1.141		YES	ignore

9.3.1.70 Assistance Data for Recommended Cells

This IE provides assistance information for paging in recommended cells.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended Cells for Paging	M		9.3.1.71	

9.3.1.71 Recommended Cells for Paging

This IE contains the recommended cells for paging.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended Cell List		*1*		
>Recommended Cell Item		*1..<maxnoofRecommendedCells>*		Includes visited and non-visited cells, where visited cells are listed in the order the UE visited them with the most recent cell being the first in the list. Non-visited cells are included immediately after the visited cell they are associated with.
>>NG-RAN CGI	M		9.3.1.73	
>>Time Stayed in Cell	O		INTEGER (0..4095)	This is included for visited cells and indicates the time a UE stayed in a cell in seconds. If the UE stays in a cell more than 4095 seconds, this IE is set to 4095.

Range bound	Explanation
maxnoofRecommendedCells	Maximum no. of recommended Cells. Value is 16.

9.3.1.72 Paging Attempt Information

This IE includes information related to the paging count over NG.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Attempt Count	M		INTEGER (1..16, ...)	Paging attempt count (see TS 38.300 [8]).
Intended Number of Paging Attempts	M		INTEGER (1..16, …)	Intended number of paging attempts (see TS 38.300 [8]).
Next Paging Area Scope	O		ENUMERATED (same, changed, …)	Indicates whether the paging area scope will change or not at next paging attempt. Usage specified in TS 38.300 [8].

9.3.1.73 NG-RAN CGI

This IE is used to globally identify a cell in NG-RAN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NG-RAN CGI*	M			
>NR				
>>NR CGI	M		9.3.1.7	
>E-UTRA				
>>E-UTRA CGI	M		9.3.1.9	

9.3.1.74 UE Radio Capability

This IE contains UE Radio Capability information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability	M		OCTET STRING	Includes either the *UERadioAccessCapabilityInformation* message as defined in TS 38.331 [18], or the *UERadioAccessCapabilityInformation-NB* message as defined in TS 36.331 [21].

9.3.1.74a UE Radio Capability – E-UTRA Format

This IE contains UE Radio Capability information to support Mode of operation A as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability – E-UTRA Format	M		OCTET STRING	Includes the *UERadioAccessCapabilityInformation* message as defined in TS 36.331 [21].

9.3.1.75 Time Stamp

This IE contains UTC time information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time Stamp	M		OCTET STRING (SIZE(4))	Encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [25].

9.3.1.76 Location Reporting Reference ID

This IE contains the Location Reporting Reference ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Location Reporting Reference ID	M		INTEGER (1..64, …)	

9.3.1.77 Data Forwarding Response DRB List

This IE indicates data forwarding related information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Response DRB Item		*1..<maxnoofDRBs>*		
>DRB ID	M		9.3.1.53	
>DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	
>UL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.78 Paging Priority

This element indicates the paging priority for paging a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Priority	M		ENUMERATED (PrioLevel1, PrioLevel2, PrioLevel3, PrioLevel4, PrioLevel5, PrioLevel6, PrioLevel7, PrioLevel8, …)	Lower value codepoint indicates higher priority.

9.3.1.79 Packet Loss Rate

This IE indicates the Packet Loss Rate for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Packet Loss Rate	M		INTEGER (0..1000, …)	Ratio of lost packets per number of packets sent, expressed in tenth of percent.

9.3.1.80 Packet Delay Budget

This IE indicates the Packet Delay Budget for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Packet Delay Budget	M		INTEGER (0..1023, …)	Upper bound value for the delay that a packet may experience expressed in unit of 0.5ms.

9.3.1.81 Packet Error Rate

This IE indicates the Packet Error Rate for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Scalar	M		INTEGER (0..9, …)	The packet error rate is expressed as *Scalar* x 10-k where k is the *Exponent*.
Exponent	M		INTEGER (0..9, …)	

9.3.1.82 Averaging Window

This IE indicates the Averaging Window for a QoS flow, and applies to GBR QoS flows only.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Averaging Window	M		INTEGER (0..4095, …)	Unit: ms. The default value of the IE is 2000ms.

9.3.1.83 Maximum Data Burst Volume

This IE indicates the Maximum Data Burst Volume for a QoS flow, and applies to delay critical GBR QoS flows only.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Maximum Data Burst Volume	M		INTEGER (0..4095, …, 4096.. 2000000)	Unit: byte.

9.3.1.84 Priority Level

This IE indicates the Priority Level for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Priority Level	M		INTEGER (1..127, …)	Values ordered in decreasing order of priority, i.e. with 1 as the highest priority and 127 as the lowest priority.

9.3.1.85 Mobility Restriction List

This IE defines roaming or access restrictions for subsequent mobility action for which the NG-RAN provides information about the target of the mobility action towards the UE, e.g., handover, or for SCG selection during dual connectivity operation or for assigning proper RNAs. NG-RAN behaviour upon receiving this IE is specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Serving PLMN	M		PLMN Identity 9.3.3.5		-	
Equivalent PLMNs		*0..<maxnoofEPLMNs>*		Allowed PLMNs in addition to Serving PLMN. This list corresponds to the list of "equivalent PLMNs" as defined in TS 24.501 [26]. This list is part of the roaming restriction information. Roaming restrictions apply to PLMNs other than the Serving PLMN and Equivalent PLMNs.	-	
>PLMN Identity	M		9.3.3.5		-	
RAT Restrictions		*0..<maxnoofEPLMNsPlusOne>*		This IE contains RAT restriction related information as specified in TS 23.501 [9].	-	
>PLMN Identity	M		9.3.3.5		-	
>RAT Restriction Information	M		BIT STRING { e-UTRA (0), nR (1), nR-unlicensed (2), nR-LEO (3), nR-MEO (4), nR-GEO (5), nR-OTHERSAT (6)} (SIZE(8, …))	Each position in the bitmap represents a RAT. If a bit is set to "1", the respective RAT is restricted for the UE. If a bit is set to "0", the respective RAT is not restricted for the UE. Bit 7 reserved for future use.	-	
>Extended RAT Restriction Information	O		9.3.1.126	If this IE is included, the *RAT Restriction Information* IE is ignored.	YES	ignore
Forbidden Area Information		*0..<maxnoofEPLMNsPlusOne>*		This IE contains Forbidden Area information as specified in TS 23.501 [9].	-	
>PLMN Identity	M		9.3.3.5		-	
>Forbidden TACs		*1..<maxnoofForbTACs>*			-	
>>TAC	M		9.3.3.10	The TAC of the forbidden TAI.	-	
Service Area Information		*0..<maxnoofEPLMNsPlusOne>*		This IE contains Service Area Restriction information as specified in TS 23.501 [9].	-	
>PLMN Identity	M		9.3.3.5		-	

>Allowed TACs		*0..<maxnoofAllowedAreas>*			-	
>>TAC	M		9.3.3.10	The TAC of the allowed TAI.	-	
>Not Allowed TACs		*0..<maxnoofAllowedAreas>*			-	
>>TAC	M		9.3.3.10	The TAC of the not-allowed TAI.	-	
Last E-UTRAN PLMN Identity	O		PLMN Identity 9.3.3.5	Indicates the E-UTRAN PLMN ID from where the UE formerly handed over to 5GS and which is preferred in case of subsequent mobility to EPS.	YES	ignore
Core Network Type Restriction for Serving PLMN	O		ENUMERATED (EPCForbidden, …)	Indicates whether the UE is restricted to connect to EPC for the Serving PLMN as specified in TS 23.501 [9].	YES	ignore
Core Network Type Restriction for Equivalent PLMNs		*0..<maxnoofEPLMNs>*			YES	ignore
>PLMN Identity	M		9.3.3.5	Includes any of the Equivalent PLMNs listed in the *Mobility Restriction List* IE for which CN Type restriction applies as specified in TS 23.501 [9].	-	
>Core Network Type Restriction	M		ENUMERATED (EPCForbidden, 5GCForbidden, …)	Indicates whether the UE is restricted to connect to EPC or to 5GC for this PLMN.	-	
NPN Mobility Information	O		9.3.1.184		YES	reject

Range bound	Explanation
maxnoofEPLMNs	Maximum no. of equivalent PLMNs. Value is 15.
maxnoofEPLMNsPlusOne	Maximum no. of allowed PLMNs. Value is 16.
maxnoofForbTACs	Maximum no. of forbidden Tracking Area Codes. Value is 4096.
maxnoofAllowedAreas	Maximum no. of allowed or not allowed Tracking Areas. Value is 16.

9.3.1.86 UE Security Capabilities

This IE defines the supported algorithms for encryption and integrity protection in the UE. The Security Capabilities received from NAS signaling shall not be modified or truncated when forwarded to NG-RAN nodes and the NG-RAN nodes store and send the complete bitmaps without modification or truncation as specified in TS 38.300 [8].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Encryption Algorithms	M		BIT STRING (SIZE(16, …))	Each position in the bitmap represents an encryption algorithm: "all bits equal to 0" – UE supports no other algorithm than NEA0, "first bit" – 128-NEA1, "second bit" – 128-NEA2, "third bit" – 128-NEA3, “fourth to seventh bit” are mapped from bit 4 to bit 1 of octet 3 in the *UE Security Capability* IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.501 [13].
NR Integrity Protection Algorithms	M		BIT STRING (SIZE(16, …))	Each position in the bitmap represents an integrity protection algorithm: "all bits equal to 0" – UE supports no other algorithm than NIA0, "first bit" – 128-NIA1, "second bit" – 128-NIA2, "third bit" – 128-NIA3, “fourth to seventh bit” are mapped from bit 4 to bit 1 of octet 4 in the *UE Security Capability* IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.501 [13].
E-UTRA Encryption Algorithms	M		BIT STRING (SIZE(16, …))	Each position in the bitmap represents an encryption algorithm: "all bits equal to 0" – UE supports no other algorithm than EEA0, "first bit" – 128-EEA1, "second bit" – 128-EEA2, "third bit" – 128-EEA3, “fourth to seventh bit” are mapped from bit 4 to bit 1 of octet 5 in the *UE Security Capability* IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.401 [27].
E-UTRA Integrity Protection Algorithms	M		BIT STRING (SIZE(16, …))	Each position in the bitmap represents an encryption algorithm:

				"all bits equal to 0" – UE supports no other algorithm than EIA0, "first bit" – 128-EIA1, "second bit" – 128-EIA2, "third bit" – 128-EIA3, “fourth to seventh bit” are mapped from bit 4 to bit 1 of octet 6 in the *UE Security Capability* IE defined in TS 24.501 [26], other bits reserved for future use. Value '1' indicates support and value '0' indicates no support of the algorithm. Algorithms are defined in TS 33.401 [27].

9.3.1.87 Security Key

This IE is used to apply security in the NG-RAN for different scenarios as defined in TS 33.501 [13].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Security Key	M		BIT STRING (SIZE(256))	Key material for NG-RAN node or Next Hop Key as defined in TS 33.501 [13]

9.3.1.88 Security Context

This IE provides security related parameters to the NG-RAN node which are used to derive security keys for user plane traffic and RRC signalling messages and for security parameter generation for subsequent mobility, see TS 33.501 [13].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Next Hop Chaining Count	M		INTEGER (0..7)	Next Hop Chaining Counter (NCC) defined in TS 33.501 [13].
Next-Hop NH	M		Security Key 9.3.1.87	The NH together with the NCC is used to derive the security configuration as defined in TS 33.501 [13].

9.3.1.89 IMS Voice Support Indicator

This IE is set by the NG-RAN node to indicate whether the UE radio capabilities are compatible with the network configuration for IMS voice.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IMS Voice Support Indicator	M		ENUMERATED (Supported, Not Supported, …)	

9.3.1.90 Paging DRX

This IE indicates the Paging DRX as defined in TS 38.304 [12] and TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging DRX	M		ENUMERATED (32, 64, 128, 256, …)	Unit: [number of radioframes]

9.3.1.91 RRC Inactive Transition Report Request

This IE is used to request the NG-RAN node to report or stop reporting to the 5GC when the UE enters or leaves RRC_INACTIVE state.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC Inactive Transition Report Request	M		ENUMERATED (Subsequent state transition report, Single RRC connected state report, Cancel report, …)	

9.3.1.92 RRC State

This IE indicates the RRC state of the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC State	M		ENUMERATED (Inactive, Connected, …)	Indicates the current RRC state of the UE.

9.3.1.93 Expected UE Behaviour

This IE indicates the behaviour of a UE with predictable activity and/or mobility behaviour, to assist the NG-RAN node in e.g. determining the optimum RRC connection time or helping with the RRC_INACTIVE state transition and RNA configuration (e.g. size and shape of the RNA).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Expected UE Activity Behaviour	O		9.3.1.94	
Expected HO Interval	O		ENUMERATED (sec15, sec30, sec60, sec90, sec120, sec180, long-time, …)	Indicates the expected time interval between inter NG-RAN node handovers. If "long-time" is included, the interval between inter NG-RAN node handovers is expected to be longer than 180 seconds.
Expected UE Mobility	O		ENUMERATED (stationary, mobile, ...)	Indicates whether the UE is expected to be stationary or mobile.
Expected UE Moving Trajectory		*0..1*		Indicates the UE's expected geographical movement.
>Expected UE Moving Trajectory Item		*1..<maxnoofCellsUEMovingTrajectory>*		Includes list of visited and non-visited cells, where visited cells are listed in the order the UE visited them with the most recent cell being the first in the list. Non-visited cells are included immediately after the visited cell they are associated with.
>>NG-RAN CGI	M		9.3.1.73	
>>Time Stayed in Cell	O		INTEGER (0..4095)	Included for visited cells and indicates the time a UE stayed in a cell in seconds. If the UE stays in a cell more than 4095 seconds, this IE is set to 4095.

Range bound	Explanation
maxnoofCellsUEMovingTrajectory	Maximum no. of cells of UE moving trajectory. Value is 16.

9.3.1.94 Expected UE Activity Behaviour

This IE indicates information about the expected "UE activity behaviour" of the UE or the PDU session as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Expected Activity Period	O		INTEGER (1..30|40| 50|60|80| 100|120| 150|180| 181, ...)	If set to "181" the expected activity time is longer than 180 seconds. The remaining values indicate the expected activity time in [seconds].
Expected Idle Period	O		INTEGER (1..30|40| 50|60|80| 100|120| 150|180| 181, ...)	If set to "181" the expected idle time is longer than 180 seconds. The remaining values indicate the expected idle time in [seconds].
Source of UE Activity Behaviour Information	O		ENUMERATED (subscription information, statistics, ...)	If "subscription information" is indicated, the information contained in the *Expected Activity Period* IE and the *Expected Idle Period* IE, if present, is derived from subscription information. If "statistics" is indicated, the information contained in the *Expected Activity Period* IE and the *Expected Idle Period* IE, if present, is derived from statistical information.

9.3.1.95 UE History Information

This IE contains information about cells that a UE has been served by in active state prior to the target cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Last Visited Cell Item		*1..<maxnoofCellsinUEHistoryInfo>*		Most recent information is added to the top of this list.
>Last Visited Cell Information	M		9.3.1.96	

Range bound	Explanation
maxnoofCellsinUEHistoryInfo	Maximum no. of cells in the UE history information. Value is 16.

9.3.1.96 Last Visited Cell Information

This IE may contain cell specific information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Last Visited Cell Information*	M			
>NG-RAN Cell				
>>Last Visited NG-RAN Cell Information	M		9.3.1.97	
>E-UTRAN Cell				
>>Last Visited E-UTRAN Cell Information	M		OCTET STRING	Defined in TS 36.413 [16].
>UTRAN Cell				
>>Last Visited UTRAN Cell Information	M		OCTET STRING	Defined in TS 25.413 [28].
>GERAN Cell				
>>Last Visited GERAN Cell Information	M		OCTET STRING	Defined in TS 36.413 [16].

9.3.1.97 Last Visited NG-RAN Cell Information

This IE contains information about a cell. In case of NR cell, this IE contains information about a set of NR cells with the same NR ARFCN for reference point A, and the *Global Cell ID* IE identifies one of the NR cells in the set. The information is to be used for RRM purposes.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Global Cell ID	M		NG-RAN CGI 9.3.1.73		-	
Cell Type	M		9.3.1.98		-	
Time UE Stayed in Cell	M		INTEGER (0..4095)	The duration of time the UE stayed in the cell, or set of NR cells with the same NR ARFCN for reference point A, in seconds. If the duration is more than 4095s, this IE is set to 4095.	-	
Time UE Stayed in Cell Enhanced Granularity	O		INTEGER (0..40950)	The duration of time the UE stayed in the cell, or set of NR cells with the same NR ARFCN for reference point A, in 1/10 seconds. If the duration is more than 4095s, this IE is set to 40950.	-	
HO Cause Value	O		Cause 9.3.1.2	The cause for the handover.	-	
Last Visited PSCell List		*0..<maxnoofPSCellsPerPrimaryCellinUEHistoryInfo>*		List of cells configured as PSCells. Most recent PSCell related information is added to the top of the list.	YES	ignore
>Last Visited PSCell Information	M		9.3.1.235	The PSCell related information.	-	

Range bound	Explanation
maxnoofPSCellsPerPrimaryCellinUEHistoryInfo	Maximum number of last visited PSCell information records that can be reported in the IE. Value is 8.

9.3.1.98 Cell Type

This IE provides the cell coverage area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Size	M		ENUMERATED (verysmall, small, medium, large, …)	

9.3.1.99 Associated QoS Flow List

This IE indicates the list of QoS flows associated with e.g. a DRB or UP TNL endpoint.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Associated QoS Flow Item		*1..<maxnoofQoSFlows>*			-	
>QoS Flow Identifier	M		9.3.1.51		-	
>QoS Flow Mapping Indication	O		ENUMERATED (ul, dl, …)		-	
>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.1.100 Information on Recommended Cells and RAN Nodes for Paging

This IE provides information on recommended cells and NG-RAN nodes for paging.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended Cells for Paging	M		9.3.1.71	
Recommended RAN Nodes for Paging	M		9.3.1.101	

9.3.1.101 Recommended RAN Nodes for Paging

This IE contains recommended NG-RAN nodes for paging.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Recommended RAN Node List		*1*		
>Recommended RAN Node Item		*1..<maxnoofRecommendedRANNodes>*		Includes visited and non-visited NG-RAN nodes, where visited NG-RAN nodes are listed in the order the UE visited them with the most recent NG-RAN node being the first in the list. Non-visited NG-RAN nodes are included after the visited NG-RAN node they are associated with.
>>CHOICE *AMF Paging Target*				The AMF paging target is either an NG-RAN node identity or a TAI as specified in TS 38.300 [8].
>>>RAN Node				
>>>>Global RAN Node ID	M		9.3.1.5	
>>>TAI				
>>>>TAI	M		9.3.3.11	

Range bound	Explanation
maxnoofRedommendedRANNodes	Maximum no. of recommended NG-RAN nodes. Value is 16.

9.3.1.102 PDU Session Aggregate Maximum Bit Rate

This IE is applicable for all Non-GBR QoS flows per PDU session which is defined for the downlink and the uplink direction and is provided by the SMF to the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session Aggregate Maximum Bit Rate		*1*		Applicable for Non-GBR QoS flows.
>PDU Session Aggregate Maximum Bit Rate Downlink	M		Bit Rate 9.3.1.4	Indicates the PDU session Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the downlink direction.
>PDU Session Aggregate Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.4	Indicates the PDU session Aggregate Maximum Bit Rate as specified in TS 23.501 [9] in the uplink direction.

9.3.1.103 Maximum Integrity Protected Data Rate

This IE indicates the maximum aggregate data rate for integrity protected DRBs for a UE as defined in TS 38.300 [8].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Maximum Integrity Protected Data Rate	M		ENUMERATED (64kbps, max UE rate, …)	Defines the upper bound of the aggregate data rate of user plane integrity protected data for either UL or DL.

9.3.1.104 Overload Response

This IE indicates the required behaviour of the NG-RAN node in an overload situation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Overload Response*	M			
>Overload Action				
>>Overload Action	M		9.3.1.105	

9.3.1.105 Overload Action

This IE indicates which signalling traffic is subject to rejection by the NG-RAN node in an AMF overload situation as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Overload Action	M		ENUMERATED (Reject RRC connection establishments for non-emergency MO DT, Reject RRC connection establishments for Signalling, Permit Emergency Sessions and mobile terminated services only, Permit High Priority Sessions and mobile terminated services only, …)	

9.3.1.106 Traffic Load Reduction Indication

This IE indicates the percentage of the type of traffic relative to the instantaneous incoming rate at the NG-RAN node, as indicated in the *Overload Action* IE, to be rejected.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Traffic Load Reduction Indication	M		INTEGER (1..99)	

9.3.1.107 Slice Overload List

This IE indicates the list of overloaded slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Overload Item		*1..<maxnoofSliceItems>*		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofSliceItems	Maximum no. of signalled slice support items. Value is 1024.

9.3.1.108 RAN Status Transfer Transparent Container

This IE is produced by the source NG-RAN node and is transmitted to the target NG-RAN node. It is used for intra 5GC NG handover.

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DRBs Subject to Status Transfer List		*1*			-	
>DRBs Subject to Status Transfer Item		*1..<maxnoof DRBs>*			-	
>>DRB ID	M		9.3.1.53		-	
>>CHOICE *UL DRB Status*	M				-	
>>>12 bits						
>>>>UL COUNT Value	M		COUNT Value for PDCP SN Length 12 9.3.1.109	PDCP-SN and HFN of the first missing UL PDCP SDU in case of 12 bit long PDCP-SN.	-	
>>>>Receive Status of UL PDCP SDUs	O		BIT STRING (SIZE(1..2048))	The IE is used in case of 12 bit long PDCP-SN. The first bit indicates the status of the SDU after the First Missing UL PDCP SDU. The *N*th bit indicates the status of the UL PDCP SDU in position (*N* + First Missing SDU Number) modulo (1 + the maximum value of the PDCP-SN). 0: PDCP SDU has not been received. 1: PDCP SDU has been received correctly.	-	
>>>18 bits						
>>>>UL COUNT Value	M		COUNT Value for PDCP SN Length 18 9.3.1.110	PDCP-SN and HFN of the first missing UL PDCP SDU in case of 18 bit long PDCP-SN.	-	
>>>>Receive Status of UL PDCP SDUs	O		BIT STRING (SIZE(1..131072))	The IE is used in case of 18 bit long PDCP-SN. The first bit indicates the status of the SDU after the First Missing UL PDCP SDU. The *N*th bit indicates the status of the UL PDCP SDU in position (*N* + First Missing SDU Number) modulo (1 + the maximum value of the PDCP-	-	

				SN). 0: PDCP SDU has not been received. 1: PDCP SDU has been received correctly.		
>>CHOICE *DL DRB Status*	M				-	
>>>12 bits						
>>>>DL COUNT Value	M		COUNT Value for PDCP SN Length 12 9.3.1.109	PDCP-SN and HFN that the target NG-RAN node should assign for the next DL PDCP SDU not having an SN yet in case of 12 bit long PDCP-SN.	-	
>>>18 bits						
>>>>DL COUNT Value	M		COUNT Value for PDCP SN Length 18 9.3.1.110	PDCP-SN and HFN that the target NG-RAN node should assign for the next DL PDCP SDU not having an SN yet in case of 18 bit long PDCP-SN.	-	
>>Old Associated QoS Flow List - UL End Marker Expected	O		Associated QoS Flow List 9.3.1.99	Indicates that the source NG-RAN node has initiated QoS flow re-mapping and has not yet received SDAP end markers, as described in TS 38.300 [8].	YES	ignore

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.109 COUNT Value for PDCP SN Length 12

This IE contains a PDCP sequence number and a hyper frame number in case of 12 bit long PDCP-SN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDCP SN Length 12	M		INTEGER (0..4095)	
HFN for PDCP SN Length 12	M		INTEGER (0..1048575)	

9.3.1.110 COUNT Value for PDCP SN Length 18

This IE contains a PDCP sequence number and a hyper frame number in case of 18 bit long PDCP-SN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDCP SN Length 18	M		INTEGER (0..262143)	
HFN for PDCP SN Length 18	M		INTEGER (0..16383)	

9.3.1.111 RRC Establishment Cause

This IE indicates the reason for RRC Connection Establishment as received from the UE in the *EstablishmentCause* IE defined in TS 38.331 [18] or the *EstablishmentCause-5GC* IE defined in TS 36.331 [21], or the reason for RRC Connection Resume as received from the UE in the *ResumeCause* IE defined in TS 38.331 [18] or the *ResumeCause-r15* IE defined in TS 36.331 [21], or the reason for RRC Connection Establishment as received from the UE in the *EstablishmentCause-NB* IE defined in TS 36.331 [21].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RRC Establishment Cause	M		ENUMERATED (emergency, highPriorityAccess, mt-Access, mo-Signalling, mo-Data, mo-VoiceCall, mo-VideoCall, mo-SMS, mps-PriorityAccess, mcs-PriorityAccess, …, notAvailable, mo-ExceptionData)	The *notAvailable* value is used in case the UE is re-establishing an RRC connection but there is fallback to RRC connection establishment as described in TS 38.331 [18], or the UE is resuming an RRC connection and the cause value received from the UE does not map to any other value of the *RRC Establishment Cause* IE.

9.3.1.112 Warning Area Coordinates

This IE contains the affected alert area coordinates of a warning message, and will be broadcast over the radio interface.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Warning Area Coordinates	M		OCTET STRING (SIZE(1..1024))	

9.3.1.113 Network Instance

This IE provides the network instance to be used by the NG-RAN node when selecting a particular transport network resource as described in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Network Instance	M		INTEGER (1..256, …)	

9.3.1.114 Secondary RAT Usage Information

This IE provides information on the secondary resources used with MR-DC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PDU Session Usage Report		*0..1*		
>RAT Type	M		ENUMERATED (nR, e-UTRA, …, nR-unlicensed, eUTRA-unlicensed)	
>PDU Session Timed Report List	M		Volume Timed Report List 9.3.1.115	
QoS Flows Usage Report List		*0..1*		
>QoS Flow Usage Report Item		*1..<maxnoofQoSFlows>*		
>>QoS Flow Indicator	M		9.3.1.51	
>>RAT Type	M		ENUMERATED (nR, e-UTRA, …, nR-unlicensed, eUTRA-unlicensed)	
>>QoS Flows Timed Report List	M		Volume Timed Report List 9.3.1.115	

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.1.115 Volume Timed Report List

This IE provides information on the data usage.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Volume Timed Report Item		*1..<maxnoofTimePeriods>*		
>Start Timestamp	M		OCTET STRING (SIZE(4))	UTC time encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [25]. It indicates the start time of the collecting period of the included Usage Count UL IE and Usage Count DL IE.
>End Timestamp	M		OCTET STRING (SIZE(4))	UTC time encoded in the same format as the first four octets of the 64-bit timestamp format as defined in section 6 of IETF RFC 5905 [25]. It indicates the end time of the collecting period of the included Usage Count UL IE and Usage Count DL IE.
>Usage Count UL	M		INTEGER (0..2^{64}-1)	The unit is: octets.
>Usage Count DL	M		INTEGER (0..2^{64}-1)	The unit is: octets.

Range bound	Explanation
maxnoofTimePeriods	Maximum no. of time reporting periods. Value is 2.

9.3.1.116 Redirection for Voice EPS Fallback

This IE is used to indicate that the AMF and the UE support the redirection for voice for EPS Fallback.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Redirection for Voice EPS Fallback	M		ENUMERATED (possible, not-possible, …)	

9.3.1.117 UE Retention Information

This IE allows the NG-RAN node and the AMF to indicate whether prior UE related contexts and related UE-associated logical NG-connections and RRC connections are intended to be retained.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Retention Information	M		ENUMERATED (ues-retained, ...)	

9.3.1.118 UL Forwarding

This IE indicates that the QoS flow is proposed for forwarding of uplink packets.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UL Forwarding	M		ENUMERATED (UL forwarding proposed, …)	

9.3.1.119 CN Assisted RAN Parameters Tuning

This IE provides information for assisting in parameters tuning of the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Expected UE Behaviour	O		9.3.1.93	This IE may be present in case the *Core Network Assistance Information for RRC INACTIVE* IE is not included and is ignored otherwise.

9.3.1.120 Common Network Instance

This IE provides the common network instance to be used by the NG-RAN node when selecting a particular transport network resource as described in TS 23.501 [9] in a format common with 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Common Network Instance	M		OCTET STRING	The octets of OCTET STRING are encoded as the Network Instance field of the *Network Instance* IE specified in TS 29.244 [43]

9.3.1.121 Data Forwarding Response E-RAB List

This IE is used at inter-system HO to provide DL data forwarding address information, if direct data forwarding is applied.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Data Forwarding Response E-RAB List		*1..<maxnoofE RABs>*		The list may include the same DL Forwarding UP TNL Information for multiple E-RABs.
>E-RAB ID	M		9.3.2.3	
>DL Forwarding UP TNL Information	M		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofE-RABs	Maximum no. of E-RABs. Value is 256.

9.3.1.122 gNB Set ID

The gNB Set ID IE is used to identify a group of gNBs which transmit the same RIM-RS.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
gNB Set ID	M		BIT STRING (SIZE(22))	

9.3.1.123 RNC-ID

The RNC-ID is used to identify an RNC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RNC-ID	M		INTEGER (0..4095)	

9.3.1.124 Extended RNC-ID

This IE is used to identify an RNC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended RNC-ID	M		INTEGER (4096..65535)	This IE is used if the RNC identity has a value larger than 4095.

9.3.1.125 RAT Information

This IE provides RAT related information associated with a TAC, used as described in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAT Information	M		ENUMERATED (unlicensed, nb-IoT, ..., nR-LEO, nR-MEO, nR-GEO, nR-OTHERSAT)	

9.3.1.126 Extended RAT Restriction Information

This IE provides RAT restrictions as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Primary RAT Restriction	M		BIT STRING { e-UTRA (0), nR (1), nR-unlicensed (2), nR-LEO (3), nR-MEO (4), nR-GEO (5), nR-OTHERSAT (6)} (SIZE(8, …))	Each position in the bitmap represents a Primary RAT. If a bit is set to "1", the respective RAT is restricted for the UE. If a bit is set to "0", the respective RAT is not restricted for the UE. Bit 7 reserved for future use. The Primary RAT is the RAT used in the access cell, or target cell.
Secondary RAT Restriction	M		BIT STRING { e-UTRA (0), nR (1), e-UTRA-unlicensed (2), nR-unlicensed (3)} (SIZE(8, …))	Each position in the bitmap represents a Secondary RAT. If a bit is set to "1", the respective RAT is restricted for the UE. If a bit is set to "0", the respective RAT is not restricted for the UE. Bits 4-7 reserved for future use. A Secondary RAT is a RAT, distinct from the UE's primary RAT, used in any cell serving the UE excluding the PCell.

9.3.1.127 SgNB UE X2AP ID

This IE uniquely identifies an UE over the X2 interface within an en-gNB. The usage of this IE is defined in TS 37.340 [32].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SgNB UE X2AP ID	M		INTEGER (0.. 2^{32} -1)	

9.3.1.128 SRVCC Operation Possible

This IE indicates that both UE and AMF are SRVCC-capable. NG-RAN behaviour on receipt of this IE is specified in TS 23.216 [31].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
SRVCC Operation Possible	M		ENUMERATED (Possible, not Possible, …)	The value "Possible" indicates that UE and AMF are SRVCC capable.

9.3.1.129 IAB Authorized

This IE provides information about the authorization status of the IAB node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IAB Authorized	M		ENUMERATED (authorized, not authorized, ...)	Indicates the IAB node authorization status.

9.3.1.130 TSC Traffic Characteristics

This IE provides the traffic characteristics of TSC QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TSC Assistance Information Downlink	O		TSC Assistance Information 9.3.1.131	
TSC Assistance Information Uplink	O		TSC Assistance Information 9.3.1.131	

9.3.1.131 TSC Assistance Information

This IE provides the TSC assistance information for a TSC QoS flow in the uplink or downlink (see TS 23.501 [9]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Periodicity	M		9.3.1.132		-	
Burst Arrival Time	O		9.3.1.133		-	
Survival Time	O		9.3.1.221		YES	ignore

9.3.1.132 Periodicity

This IE indicates the Periodicity of the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodicity	M		INTEGER (0..640000, …)	Periodicity expressed in units of 1 us.

9.3.1.133 Burst Arrival Time

This IE indicates the Burst Arrival Time of the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Burst Arrival Time	M		OCTET STRING	Encoded in the same format as the *ReferenceTime* IE as defined in TS 38.331 [18]. The value is provided with 1 us accuracy.

9.3.1.134 Redundant QoS Flow Indicator

This IE provides the redundant QoS flow indicator for a QoS flow as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Redundant QoS Flow Indicator	M		ENUMERATED (true, false)	This IE indicates whether this QoS flow is requested for the redundant transmission. Value “true” indicates that redundant transmission is requested for this QoS flow. Value “false” indicates that redundant transmission is requested to be stopped if started.

9.3.1.135 Extended Packet Delay Budget

This IE indicates the Packet Delay Budget for a QoS flow.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended Packet Delay Budget	M		INTEGER (0..65535, …, 65536..109999)	Upper bound value for the delay that a packet may experience expressed in unit of 0.01ms.

9.3.1.136 Redundant PDU Session Information

This IE defines Redundancy information to be applied to a PDU session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
RSN	M		ENUMERATED (v1, v2, …)		-	
PDU Session Pair ID	O		INTEGER (0.. 255, …)	as defined in TS 23.501 [9]. This IE may be present in the request message and is ignored otherwise.	YES	ignore

9.3.1.137 NB-IoT Default Paging DRX

This IE indicates the NB-IoT Default Paging DRX as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT Default Paging DRX	M		ENUMERATED (128, 256, 512, 1024, …)	Unit: [number of radioframes]

9.3.1.138 NB-IoT Paging eDRX Information

This IE indicates the NB-IoT Paging eDRX parameters as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT Paging eDRX Cycle	M		ENUMERATED (hf2, hf4, hf6, hf8, hf10, hf12, hf14, hf16, hf32, hf64, hf128, hf256, hf512, hf1024, …)	T_{eDRX} defined in TS 36.304 [29]. Unit: [number of hyperframes].
NB-IoT Paging Time Window	O		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, …)	Unit: [2.56 seconds]

9.3.1.139 NB-IoT Paging DRX

This IE indicates the NB-IoT UE specific Paging DRX as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT Paging DRX	M		ENUMERATED (32, 64, 128, 256, 512, 1024, …)	Unit: [number of radioframes]

9.3.1.140 Enhanced Coverage Restriction

This IE provides information on the restriction information of using Coverage Enhancement.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Enhanced Coverage Restriction	O		ENUMERATED (restricted, ...)	Indicates whether the UE is restricted to use coverage enhancement. Value “restricted” indicates that the UE is not allowed to use coverage enhancement.

9.3.1.141 Paging Assistance Data for CE Capable UE

This IE provides Assistance Data for paging CE capable UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Global Cell ID	M		E-UTRA CGI 9.3.1.9	
Coverage Enhancement Level	M		OCTET STRING	Includes either the *UEPagingCoverageInformation* message as defined in TS 36.331 [21], or the *UEPagingCoverageInformation-NB* message as defined in TS 36.331 [21].

9.3.1.142 UE Radio Capability ID

This IE contains the UE Radio Capability ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Radio Capability ID	M		OCTET STRING	Defined in TS 23.003 [23].

9.3.1.143 WUS Assistance Information

This IE provides WUS Assistance Information to be used by the NG-RAN node for determining the WUS group for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Probability Information	M		ENUMERATED (p00, p05, p10, p15, p20, p25, p30, p35, p40, p45, p50, p55, p60, p65, p70, p75, p80, p85, p90, p95, p100, …)	Unit: percentage

9.3.1.144 UE Differentiation Information

This IE is generated by the AMF based on the UE subscription information, it provides the Expected UE Behavior Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodic Communication Indicator	O		ENUMERATED (periodically, on demand, …)	This IE indicates whether the UE communicates periodically or not, e.g. only on demand.
Periodic Time	O		INTEGER (1..3600, …)	This IE indicates the interval time of periodic communication, the unit is: second
Scheduled Communication Time		*0..1*		This IE indicates the time zone and day of the week when the UE is available for communication.
>Day of Week	O		BIT STRING (SIZE(7))	Each position in the bitmap represents a day of the week: first bit = Mon, second bit =Tue, third bit =Wed, and so on. Value '1' indicates 'scheduled. Value '0' indicates 'not scheduled'. If Day-Of-Week is not provided, this is interpreted as every day of the week.
>Time of Day Start	O		INTEGER (0..86399, …)	This IE indicates the time to start of the day, each value represents the corresponding second since 00:00 of the day. If Time-Of-Day-Start is not provided, starting time is start of the day(s) indicated by Day-Of-Week-Mask.
>Time of Day End	O		INTEGER (0..86399, …)	This IE indicates the time to start of the day, each value represents the corresponding second since 00:00 of the day. The value of this IE should be bigger than the value of Time of Day Start IE. If Time-Of-Day-End is not provided, ending time is end of the day(s) indicated by Day-Of-Week-Mask.
Stationary Indication	O		ENUMERATED (stationary, mobile, …)	
Traffic Profile	O		ENUMERATED (single packet, dual packets, multiple packets, …)	"single packet" indicates single packet transmission (UL or DL), "dual packets" indicates dual packet transmission (UL with subsequent DL, or DL with subsequent UL), "multiple packets" indicates multiple packets transmission.
Battery Indication	O		ENUMERATED (battery powered, battery powered not rechargeable or replaceable, not battery powered, …)	"battery powered" indicates that the UE is battery powered and the battery is rechargeable/replaceable, "battery powered not rechargeable or replaceable" indicates that the UE is battery powered but the battery is not rechargeable/replaceable, "not battery powered" indicates that the UE is not battery powered.

9.3.1.145 NB-IoT UE Priority

This IE provides the NB-IoT UE Priority and to be used by the NG-RAN to prioritise between UEs accessing via NB-IoT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NB-IoT UE Priority	M		INTEGER (0..255, …)	Lower value indicates higher priority.

9.3.1.146 NR V2X Services Authorized

This IE provides information on the authorization status of the UE to use the NR sidelink for V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Vehicle UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Vehicle UE
Pedestrian UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Pedestrian UE

9.3.1.147 LTE V2X Services Authorized

This IE provides information on the authorization status of the UE to use the LTE sidelink for V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Vehicle UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Vehicle UE
Pedestrian UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized as Pedestrian UE

9.3.1.148 NR UE Sidelink Aggregate Maximum Bit Rate

This IE provides information on the Aggregate Maximum Bitrate of the UE's sidelink communication.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR UE Sidelink Aggregate Maximum Bit Rate	M		Bit Rate 9.3.1.4	Value 0 is not valid, and considered as a logical error by the receiving NG-RAN node.

9.3.1.149 LTE UE Sidelink Aggregate Maximum Bit Rate

This IE provides information on the Aggregate Maximum Bitrate of the UE's sidelink communication for LTE V2X services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LTE UE Sidelink Aggregate Maximum Bit Rate	M		Bit Rate 9.3.1.4	Value 0 is not valid, and considered as a logical error by the receiving NG-RAN node.

9.3.1.150 PC5 QoS Parameters

This IE provides information on the PC5 QoS parameters of the UE's sidelink communication for NR PC5.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PC5 QoS Flow List		*1*		
>PC5 QoS Flow Item		*1..<maxnoofPC5QoSFlows>*		
>>PQI	M		INTEGER (0..255, …)	PQI is a special 5QI as specified in TS 23.501 [9].
>>PC5 Flow Bit Rates		*0..1*		Only applies for GBR QoS Flows.
>>>Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>>Maximum Flow Bit Rate	M		Bit Rate 9.3.1.4	Maximum Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>Range	O		ENUMERATED (m50, m80, m180, m200, m350, m400, m500, m700, m1000, …)	Only applies for groupcast.
PC5 Link Aggregate Bit Rates	O		Bit Rate 9.3.1.4	Only applies for non-GBR QoS Flows.

Range bound	Explanation
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flows allowed towards one UE. Value is 2048.

9.3.1.151 Alternative QoS Parameters Set List

This IE contains alternative sets of QoS parameters which the NG-RAN node can indicate to be fulfilled when notification control is enabled and it cannot fulfil the requested list of QoS parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Alternative QoS Parameters Set Item		*1..<maxnoofQoSparaSets>*		
>Alternative QoS Parameters Set Index	M		9.3.1.152	
>Guaranteed Flow Bit Rate Downlink	O		Bit Rate 9.3.1.4	
>Guaranteed Flow Bit Rate Uplink	O		Bit Rate 9.3.1.4	
>Packet Delay Budget	O		9.3.1.80	
>Packet Error Rate	O		9.3.1.81	

Range bound	Explanation
maxnoofQoSparaSets	Maximum no. of alternative sets of QoS Parameters allowed for the QoS profile. Value is 8.

9.3.1.152 Alternative QoS Parameters Set Index

This IE indicates the QoS parameters set which can currently be fulfilled.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Alternative QoS Parameters Set Index	M		INTEGER (1..8, ...)	Indicates the index of the item within the the *Alternative QoS Parameters Set List* IE corresponding to the currently fulfilled alternative QoS parameters set.

9.3.1.153 Alternative QoS Parameters Set Notify Index

This IE indicates the QoS parameters set which can currently be fulfilled.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Alternative QoS Parameters Set Notify Index	M		INTEGER (0..8, ...)	Indicates the index of the item within the the *Alternative QoS Parameters Set List* IE corresponding to the currently fulfilled alternative QoS parameters set. Value 0 indicates that NG-RAN cannot even fulfil the lowest alternative parameters set.

9.3.1.154 E-UTRA Paging eDRX Information

This IE indicates the E-UTRA Paging eDRX parameters as defined in TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-UTRA Paging eDRX Cycle	M		ENUMERATED (hfhalf, hf1, hf2, hf4, hf6, hf8, hf10, hf12, hf14, hf16, hf32, hf64, hf128, hf256, …)	TeDRX defined in TS 36.304 [29]. Unit: [number of hyperframes].
E-UTRA Paging Time Window	O		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, …)	Unit: [1.28 second].

9.3.1.155 CE-mode-B Restricted

This IE provides information on the restriction information of using Coverage Enhancement Mode B.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CE-mode-B Restricted	M		ENUMERATED (restricted, not-restricted, ...)	Indicates whether the UE is restricted to use coverage enhancement. Value “restricted” indicates that the UE is not allowed to use coverage enhancement mode B. Value “not-restricted” indicates that the UE is allowed to use coverage enhancement mode B.

9.3.1.156 CE-mode-B Support Indicator

This IE indicates whether CE-mode-B as specified in TS 36.306[42] is supported for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CE-mode-B Support Indicator	M		ENUMERATED (supported, …)	

9.3.1.157 LTE-M Indication

This IE is provided by the NG-RAN node to inform that the UE indicates category M1 or M2 in its UE Radio Capability.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LTE-M Indication	M		ENUMERATED (LTE-M, ...)	

9.3.1.158 Suspend Request Indication

This IE indicates that the NG-RAN node requests immediate transition to RRC idle with suspend, as specified in TS 23.502 [10].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Suspend Request Indication	M		ENUMERATED (suspend-requested, …)	

9.3.1.159 Suspend Response Indication

This IE is used by the AMF to inform the NG-RAN node to suspend the UE and send it to RRC_IDLE, as specified in TS 23.502 [10].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Suspend Response Indication	M		ENUMERATED (suspend-indicated, …)	

9.3.1.160 UE User Plane CIoT Support Indicator

This IE indicates whether User Plane CIoT 5GS Optimisation as specified in TS 23.501 [9] is supported for the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE User Plane CIoT Support Indicator	M		ENUMERATED (supported, …)	

9.3.1.161 Global TNGF ID

This IE is used to globally identify a TNGF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE *TNGF ID*	M			
>TNGF ID				
>>TNGF ID	M		BIT STRING (SIZE(32, …))	

9.3.1.162 Global W-AGF ID

This IE is used to globally identify a W-AGF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE *W-AGF ID*	M			
>W-AGF ID				
>>W-AGF ID	M		BIT STRING (SIZE(16, …))	

9.3.1.163 Global TWIF ID

This IE is used to globally identify a TWIF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE *TWIF ID*	M			
>TWIF ID				
>>TWIF ID	M		BIT STRING (SIZE(32, …))	

9.3.1.164 W-AGF User Location Information

This IE indicates the location information via wireline access as specified in TS 23.316 [34].

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *W-AGF User Location Information*	M				-	
>Global Line ID					-	
>>Global Line ID	M		OCTET STRING	Encoded as defined in TS 23.003 [23].	-	
>>Line Type	O		ENUMERATED (DSL, PON, …)		-	
>>TAI	O		9.3.3.11		YES	ignore
>HFC Node ID					-	
>>HFC Node ID	M		OCTET STRING	Indicates the identifier of the HFC node as specified in [37]. Encoded as defined in TS 23.003 [23].	-	
>Global Cable ID					YES	ignore
>>Global Cable ID	M		OCTET STRING	Encoded as defined in TS 23.003 [23].	-	
>HFC Node ID New					YES	ignore
>>HFC Node ID	M		OCTET STRING	Indicates the identifier of the HFC node as specified in [37]. Encoded as defined in TS 23.003 [23].	-	
>>TAI	M		9.3.3.11		-	
>Global Cable ID New					YES	ignore
>>Global Cable ID	M		OCTET STRING	Encoded as defined in TS 23.003 [23].	-	
>>TAI	M		9.3.3.11		-	

9.3.1.165 Global eNB ID

This IE is used to globally identify an eNB (see TS 36.401 [38]).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
CHOICE *eNB ID*	M			
>Macro eNB ID				
>>Macro eNB ID	M		BIT STRING (SIZE(20))	Equal to the 20 leftmost bits of the *E-UTRA Cell Identity* IE contained in the *E-UTRA CGI* IE of each cell served by the eNB.
>Home eNB ID				
>>Home eNB ID	M		BIT STRING (SIZE(28))	Equal to the *E-UTRA Cell Identity* IE contained in the *E-UTRA CGI* IE of the cell served by the eNB.
>Short Macro eNB ID				
>>Short Macro eNB ID	M		BIT STRING (SIZE(18))	Equal to the 18 leftmost bits of the *E-UTRA Cell Identity* IE contained in the *E-UTRA CGI* IE of each cell served by the eNB.
>Long Macro eNB ID				
>>Long Macro eNB ID	M		BIT STRING (SIZE(21))	Equal to the 21 leftmost bits of the *E-UTRA Cell Identity* IE contained in the *E-UTRA CGI* IE of each cell served by the eNB.

9.3.1.166 UE History Information from UE

This IE contains information about mobility history report for a UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *UE History Information from UE*	M			
>NR				
>>NR Mobility History Report	M		OCTET STRING	Includes the *mobilityHistoryReport* contained in the *UEInformationResponse* message as defined in TS 38.331 [18].

9.3.1.167 MDT Configuration

This IE defines the MDT configuration parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT Configuration-NR	O		9.3.1.169	
MDT Configuration-EUTRA	O		9.3.1.170	

9.3.1.168 MDT PLMN List

The purpose of the *MDT PLMN List* IE is to provide the list of PLMN allowed for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT PLMN List		*1..<maxnoofMDTPLMNs>*		
>PLMN Identity	M		9.3.3.5	

Range bound	Explanation
maxnoofMDTPLMNs	Maximum no. of PLMNs in the MDT PLMN list. Value is 16.

9.3.1.169 MDT Configuration-NR

This IE defines the MDT configuration parameters of NR.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MDT Activation	M		ENUMERATED (Immediate MDT only, Logged MDT only, Immediate MDT and Trace, …)		-	
CHOICE *Area Scope of MDT*	M				-	
>Cell based						
>>Cell ID List for MDT		*1..<maxnoofCellIDforMDT>*				
>>>NR CGI	M		9.3.1.7		-	
>TA based						
>>TA List for MDT		*1..<maxnoofTAforMDT>*				
>>>TAC	M		9.3.3.10	The TAI is derived using the current serving PLMN.	-	
>PLMN wide			NULL			
>TAI based						
>>TAI List for MDT		*1..<maxnoofTAforMDT>*				
>>>TAI	M				-	
CHOICE *MDT Mode*	M				-	
>Immediate MDT						
>>Measurements to Activate	M		BITSTRING (SIZE(8))	Each position in the bitmap indicates a MDT measurement, as defined in TS 37.320 [41]. First Bit = M1, Second Bit= M2, Third Bit = M4, Fourth Bit = M5, Fifth Bit = M6, Sixth Bit = M7, Seventh Bit = logging of M1 from event triggered measurement reports according to existing RRM configuration, other bits reserved for future use. Value “1” indicates “activate” and value “0” indicates “do not activate”.	-	
>>M1 Configuration	C-ifM1		9.3.1.171		-	
>>M4 Configuration	C-ifM4		9.3.1.172		-	
>>M5 Configuration	C-ifM5		9.3.1.173		-	
>>M6 Configuration	C-ifM6		9.3.1.174		-	
>>M7 Configuration	C-ifM7		9.3.1.175		-	
>>Bluetooth Measurement Configuration	O		9.3.1.177		-	
>>WLAN	O		9.3.1.178		-	

Measurement Configuration						
>>MDT Location Information	O		9.3.1.176		-	
>>Sensor Measurement Configuration	O		9.3.1.179		-	
>Logged MDT						
>>Logging Interval	M		ENUMERATED (320ms, 640ms, 1280ms, 2560ms, 5120ms, 10240ms, 20480ms, 30720ms, 40960ms, 61440ms, infinity, …)	Corresponds to the *LoggingInterval* IE as defined in TS 38.331 [18].	-	
>>Logging Duration	M		ENUMERATED (10, 20, 40, 60, 90,120, …)	Corresponds to the *LoggingDuration* IE as defined in TS 38.331 [18]. Unit: [minute].	-	
>>CHOICE *Report Type*	M				-	
>>>Periodical			NULL			
>>>Event Triggered						
>>>>Event Trigger Logged MDT Configuration	M		9.3.1.180		-	
>>Bluetooth Measurement Configuration	O		9.3.1.177		-	
>>WLAN Measurement Configuration	O		9.3.1.178		-	
>>Sensor Measurement Configuration	O		9.3.1.179		-	
>>Area Scope of Neighbour Cells	O		9.3.1.182		-	
>>Early Measurement	O		ENUMERATED (true, ...)	This IE indicates whether the UE is allowed to log measurements on early measurement related frequencies in logged MDT as specified in TS 38.331 [18].	YES	ignore
Signalling Based MDT PLMN List	O		MDT PLMN List 9.3.1.168		-	

Range bound	Explanation
maxnoofCellIDforMDT	Maximum no. of Cell ID subject for MDT scope. Value is 32.
maxnoofTAforMDT	Maximum no. of TA subject for MDT scope. Value is 8.

Condition	Explanation
C-ifM1	This IE shall be present if the *Measurements to Activate* IE has the first bit set to “1”.
C-ifM4	This IE shall be present if the *Measurements to Activate* IE has the third bit set to “1”.
C-ifM5	This IE shall be present if the *Measurements to Activate* IE has the fourth bit set to “1”.
C-ifM6	This IE shall be present if the Measurements to Activate IE has the fitth bit set to “1”.
C-ifM7	This IE shall be present if the Measurements to Activate IE has the sixth bit set to “1”.

9.3.1.170 MDT Configuration-EUTRA

This IE defines the MDT configuration parameters of EUTRA.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT Activation	M		ENUMERATED (Immediate MDT only, Logged MDT only, Immediate MDT and Trace, …)	
CHOICE *Area Scope of MDT*	M			
>Cell based				
>>Cell ID List for MDT		*1..<maxnoofCellIDforMDT>*		
>>>E-UTRA CGI	M		9.3.1.9	
>TA based				
>>TA List for MDT		*1..<maxnoofTAforMDT>*		
>>>TAC	M		9.3.3.10	The TAI is derived using the current serving PLMN.
>PLMN wide			NULL	
>TAI based				
>>TAI List for MDT		*1..<maxnoofTAforMDT>*		
>>>TAI	M		9.3.3.11	
MDT Mode	M		OCTET STRING	*MDTMode* IE defined in TS 36.413 [16].
Signalling Based MDT PLMN List	O		MDT PLMN List 9.3.1.168	

Range bound	Explanation
maxnoofCellIDforMDT	Maximum no. of Cell ID subject for MDT scope. Value is 32.
maxnoofTAforMDT	Maximum no. of TA subject for MDT scope. Value is 8.

9.3.1.171 M1 Configuration

This IE defines the parameters for M1 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M1 Reporting Trigger	M		ENUMERATED (periodic, A2event-triggered, A2event-triggered periodic, …)		-	
M1 Threshold Event A2	C-ifM1A2trigger				-	
>CHOICE *Threshold Type*	M				-	
>>*RSRP*						
>>>Threshold RSRP	M		INTEGER (0..127)	Corresponds to the *RSRP-Range* IE as defined in TS 38.331 [18].	-	
>>*RSRQ*						
>>>Threshold RSRQ	M		INTEGER (0..127)	Corresponds to the *RSRQ-Range* IE as defined in TS 38.331 [18].	-	
>>*SINR*						
>>>Threshold SINR	M		INTEGER (0..127)	Corresponds to the *SINR-Range* IE as defined in TS 38.331 [18].	-	
M1 Periodic Reporting	C-ifperiodicMDT				-	
>Report Interval	M		ENUMERATED (ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, min1, min6, min12, min30, min60)	Corresponds to the *ReportInterval* IE as defined in TS 38.331 [18]. The value min60 is not used in the specification.	-	
>Report Amount	M		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity)	Number of reports.	-	
>Extended Report Interval	O		ENUMERATED (ms20480, ms40960, ...)	This IE is the extension of the *Report Interval* IE. If this IE is present, the *Report Interval* IE is ignored.	YES	ignore
Include Beam Measurements Indication	O		ENUMERATED (true, …)	To configure whether the UE should include beam level measurements.	YES	ignore
Beam Measurements Report Configuration	C-ifM1BeamMeasInd				YES	ignore
>Beam Measurements Report Quantity		*0..1*		This IE indicates the beam measurement quantity and corresponds to the *MeasReportQuanti*	-	

				ty IE as defined in TS 38.331 [18].		
>>RSRP	M		ENUMERATED (true, …)		-	
>>RSRQ	M		ENUMERATED (true, …)		-	
>>SINR	M		ENUMERATED (true, …)		-	
>MaxNrofRS-IndexesToReport	O		INTEGER (1..64, …)	Indicates the max number of beam measurements to be reported and corresponds to the *maxNrofRS-IndexesToReport* in the *ReportConfigNR* IE as defined in TS 38.331 [18].	-	

Condition	Explanation
C-ifM1A2trigger	This IE shall be present if the *M1 Reporting Trigger* IE is set to “A2event-triggered” or to “A2event-triggered periodic”.
C-ifperiodicMDT	This IE shall be present if the *M1 Reporting Trigger* IE is set to “periodic”, or to “A2event-triggered periodic”.
C-ifM1BeamMeasInd	This IE shall be present if the *Include Beam Measurements Indication* IE is set to “true”.

9.3.1.172 M4 Configuration

This IE defines the parameters for M4 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M4 Collection Period	M		ENUMERATED (ms1024, ms2048, ms5120, ms10240, min1, …)		-	
M4 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, …)		-	
M4 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, …)	Number of reports.	YES	ignore

9.3.1.173 M5 Configuration

This IE defines the parameters for M5 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M5 Collection Period	M		ENUMERATED (ms1024, ms2048, ms5120, ms10240, min1, …)		-	
M5 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, …)		-	
M5 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, …)	Number of reports	YES	ignore

9.3.1.174 M6 Configuration

This IE defines the parameters for M6 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M6 Report Interval	M		ENUMERATED (ms120, ms240, ms480, ms640, ms1024, ms2048, ms5120, ms10240, ms20480, ms40960, min1, min6, min12, min30, …)		-	
M6 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, …)		-	
M6 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, …)	Number of reports	YES	ignore
Excess Packet Delay Threshold Configuration	O		9.3.1.244		YES	ignore

9.3.1.175 M7 Configuration

This IE defines the parameters for M7 measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
M7 Collection Period	M		INTEGER (1..60, …)	Unit: minutes	-	
M7 Links to Log	M		ENUMERATED (uplink, downlink, both-uplink-and-downlink, …)		-	
M7 Report Amount	O		ENUMERATED (1, 2, 4, 8, 16, 32, 64, infinity, …)	Number of reports	YES	ignore

9.3.1.176 MDT Location Information

This IE defines the MDT Location Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT Location Information	M		BITSTRING (SIZE(8))	Each position in the bitmap represents requested location information as defined in TS 37.320 [41]. First Bit = GNSS Other bits are reserved for future use and are ignored if received. Value “1” indicates “activate” and value “0” indicates “do not activate”.

9.3.1.177 Bluetooth Measurement Configuration

This IE defines the parameters for Bluetooth measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Bluetooth Measurement Configuration	M		ENUMERATED (Setup, …)	
Bluetooth Measurement Configuration Name List		*0..1*		This IE is present if the *Bluetooth Measurement Configuration* IE is set to “Setup”.
>Bluetooth Measurement Configuration Name Item		*1..<maxnoofBluetoothName>*		
>>Bluetooth Measurement Configuration Name	M		OCTET STRING (SIZE (1..248))	
BT RSSI	O		ENUMERATED (true, …)	In case of Immediate MDT, it corresponds to M8 measurement as defined in TS 37.320 [41].

Range bound	Explanation
maxnoofBluetoothName	Maximum no. of Bluetooth local name used for Bluetooth measurement collection. Value is 4.

9.3.1.178 WLAN Measurement Configuration

This IE defines the parameters for WLAN measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
WLAN Measurement Configuration	M		ENUMERATED (Setup, …)	
WLAN Measurement Configuration Name List		*0..1*		This IE is present if the *Bluetooth Measurement Configuration* IE is set to “Setup”.
>WLAN Measurement Configuration Name Item		*1..<maxnoofWLANName>*		
>>WLAN Measurement Configuration Name	M		OCTET STRING (SIZE (1..32))	
WLAN RSSI	O		ENUMERATED (true, …)	In case of Immediate MDT, it corresponds to M8 as defined in TS 37.320 [41].
WLAN RTT	O		ENUMERATED (true, …)	In case of Immediate MDT, it corresponds to M9 as defined in TS 37.320 [41].

Range bound	Explanation
maxnoofWLANName	Maximum no. of WLAN SSID used for WLAN measurement collection. Value is 4.

9.3.1.179 Sensor Measurement Configuration

This IE defines the parameters for Sensor measurement collection.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Sensor Measurement Configuration	M		ENUMERATED (Setup, …)	
Sensor Measurement Configuration Name List		*0..1*		
>Sensor Measurement Configuration Name Item		*1..<maxnoofSensorName>*		
>>CHOICE *Sensor Name*	M			
>>>Uncompensated Barometric				
>>>>Uncompensated Barometric Configuration	M		ENUMERATED (true, …)	
>>>UE speed				
>>>>UE Speed Configuration	M		ENUMERATED (true, …)	
>>>UE orientation				
>>>>UE orientation Configuration	M		ENUMERATED (true, …)	

Range bound	Explanation
maxnoofSensorName	Maximum no. of Sensor local name used for Sensor measurement collection. Value is 3

9.3.1.180 Event Trigger Logged MDT Configuration

This IE defines the event trigger logged MDT configuration.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Event trigger type*	M			
>Out-of-coverage				
>>Out-of-Coverage Configuration	M		ENUMERATED (true, ...)	
>L1 Event				
>>CHOICE *L1 Event Threshold*	M			
>>>RSRP				
>>>>Threshold RSRP	M		INTEGER (0..127)	Corresponds to the *RSRP-Range* IE as defined in TS 38.331 [18].
>>>RSRQ				
>>>>Threshold RSRQ	M		INTEGER (0..127)	Corresponds to the *RSRQ-Range* IE as defined in TS 38.331 [18].
>>Hysteresis	M		INTEGER (0..30)	This parameter is used within the entry and leave condition of an event triggered reporting condition.
>>Time to Trigger	M		ENUMERATED (ms0, ms40, ms64, ms80, ms100, ms128, ms160, ms256, ms320, ms480, ms512, ms640, ms1024, ms1280, ms2560, ms5120)	Time during which specific criteria for the event needs to be met in order to trigger a measurement report.

9.3.1.181 NR Frequency Info

This defines the carrier frequency and bands used in a cell for a given direction (UL or DL) in FDD or for both UL and DL directions in TDD.

IE/Group Name	Presence	Range	IE Type and Reference	Semantics Description
NR ARFCN	M		INTEGER (0.. maxNRARFCN)	RF Reference Frequency as defined in TS 38.104 [39], section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A.
NR Frequency Band List		*1*		This IE is not used in this specification and is ignored.
>NR Frequency Band Item		*1..<maxnoofNRCellBands>*		
>>NR Frequency Band	M		INTEGER (1.. 1024, ...)	Primary NR Operating Band as defined in TS 38.104 [39], section 5.4.2.3. The value 1 corresponds to n1, value 2 corresponds to NR operating band n2, etc.

Range bound	Explanation
maxNRARFCN	Maximum value of NRARFCNs. Value is 3279165.
maxnoofNRCellBands	Maximum no. of frequency bands supported for a NR cell. Value is 32.

9.3.1.182 Area Scope of Neighbour Cells

This IE defines the area scope of neighbour cells for logged MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Area Scope of Neighbour Cells Item	M	*1..<maxnoofFreqforMDT>*		
>NR Frequency Info	M		9.3.1.181	
>PCI List for MDT		*0..<maxnoofNeighPCIforMDT>*		
>>NR PCI	M		INTEGER (0..1007, …)	NR Physical Cell ID

Range bound	Explanation
maxnoofFreqforMDT	Maximum no. of Frequency Information subject for MDT scope. Value is 8.
maxnoofNeighPCIforMDT	Maximum no. of Neighbour cells subject for MDT scope. Value is 32.

9.3.1.183 NPN Paging Assistance Information

This IE contains NPN Paging Assistance Information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NPN Paging Assistance Information*	M			
>PNI-NPN Paging Assistance				
>>PNI-NPN Paging Assistance	M		Allowed PNI-NPN List 9.3.3.45	

9.3.1.184 NPN Mobility Information

This IE indicates the access restrictions related to an NPN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NPN Mobility Information*	M			
>SNPN Mobility Information				
>>Serving NID	M		NID 9.3.3.42	
>PNI-NPN Mobility Information				
>>Allowed PNI-NPN List	M		9.3.3.45	

9.3.1.185 Cell CAG Information

This IE provides information about support of closed access groups for a designated cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NG-RAN CGI	M		9.3.1.73	
Cell CAG List	M		9.3.3.47	

9.3.1.186 Target to Source Failure Transparent Container

This IE is used to transparently pass radio related information from the handover target to the handover source through the core network in case of failure of the preparation at the target; it is produced by the target RAN node and is transmitted to the source RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target to Source Failure Transparent Container	M		OCTET STRING	This IE includes a transparent container from the target RAN node to the source RAN node. The octets of the OCTET STRING are encoded according to the specifications of the target system. Note: In the current version of the specification, this IE may carry the *Target NG-RAN Node to Source NG-RAN Node Failure Transparent Container* IE.

9.3.1.187 Target NG-RAN Node to Source NG-RAN Node Failure Transparent Container

This IE is produced by the target NG-RAN node and is transmitted to the source NG-RAN node in case of preparation failure.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Cell CAG Information	O		9.3.1.185		-	
NGAP IE Support Information Response List	O		9.3.1.242		YES	ignore

9.3.1.188 DAPS Request Information

The *DAPS Indicator* IE indicates that the source NG-RAN node requests a DAPS Handover for the concerned DRB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DAPS Indicator	M		ENUMERATED (DAPS HO required, …)	Indicates that DAPS Handover is requested

9.3.1.189 DAPS Response Information

The *DAPS Response Indicator* IE indicates the per DRB response to a requested DAPS Handover.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DAPS Response Indicator	M		ENUMERATED (DAPS HO accepted, DAPS HO not accepted, …)	Indicates that DAPS Handover is accepted or not

9.3.1.190 Early Status Transfer Transparent Container

The *Early Status Transfer Transparent Container* IE is an information element that is produced by the source NG-RAN node and is transmitted to the target NG-RAN node. This IE is used for the NG DAPS handover case.

This IE is transparent to the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Procedure Stage*	M			
>First DL COUNT				
>>DRBs Subject To Early Status Transfer List		*1*		
>>>DRBs Subject To Early Status Transfer Item		*1..<maxnoofDRBs>*		
>>>>DRB ID	M		9.3.1.53	
>>>>CHOICE *First DL COUNT*	M			
>>>>>12 bits				
>>>>>>FIRST DL COUNT Value	M		COUNT Value for PDCP SN Length 12 9.3.1.109	PDCP-SN and Hyper frame number of the first DL SDU that the source NG-RAN node forwards to the target NG-RAN node in case of 12 bit long PDCP-SN
>>>>>18 bits				
>>>>>>FIRST DL COUNT Value	M		COUNT Value for PDCP SN Length 18 9.3.1.110	PDCP-SN and Hyper frame number of the first DL SDU that the source NG-RAN node forwards to the target NG-RAN node in case of 18 bit long PDCP-SN

Range bound	Explanation
maxnoofDRBs	Maximum no. of DRBs allowed towards one UE. Value is 32.

9.3.1.191 Extended Slice Support List

This IE indicates a list of supported slices.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Slice Support Item		*1..<maxnoofExtSliceItems>*		
>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofExtSliceItems	Maximum no. of signalled slice support items. Value is 65535.

9.3.1.192 UE Capability Info Request

This IE indicates the request to provide to the AMF the UE radio capability related information when retrieved from the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Capability Info Request	M		ENUMERATED (requested, …)	

9.3.1.193 Extended RAN Node Name

This IE provides extended human readable name of the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Node Name Visible	O		VisibleString (SIZE(1..150, …))	
RAN Node Name UTF8	O		UTF8String (SIZE(1..150, …))	

9.3.1.194 MICO All PLMN

This IE indicates that the UE is configured with MICO mode by the AMF for the “all PLMN” as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MICO All PLMN	M		ENUMERATED (true, …)	

9.3.1.195 Source Node ID

This IE identifies the source SN for the handover.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Source Node ID*	M			
>Source E-UTRAN Node ID				
>>Source en-gNB ID	M		Global gNB ID 9.3.1.6	This IE is used for handover from EN-DC to SA. The source en-gNB ID is the identity of the source SN.

9.3.1.196 E-UTRAN Composite Available Capacity Group

This IE indicates the overall available resource level in the cell in downlink and uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Composite Available Capacity Downlink	M		E-UTRAN Composite Available Capacity 9.3.1.197	For the downlink
Composite Available Capacity Uplink	M		E-UTRAN Composite Available Capacity 9.3.1.197	For the uplink

9.3.1.197 E-UTRAN Composite Available Capacity

This IE indicates the overall available resource level in the cell in either downlink or uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-UTRAN Cell Capacity Class Value	O		9.3.1.198	
E-UTRAN Capacity Value	M		9.3.1.199	‘0’ indicates no resource is available, Measured on a linear scale.

9.3.1.198 E-UTRAN Cell Capacity Class Value

This IE indicates the value that classifies the cell capacity with regards to the other cells. This IE only indicates resources that are configured for traffic purposes.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell Capacity Class Value	M		INTEGER (1..100, ...)	Value 1 indicates the minimum cell capacity, and 100 indicates the maximum cell capacity. There should be a linear relation between cell capacity and Cell Capacity Class Value.

9.3.1.199 E-UTRAN Capacity Value

This IE indicates the amount of resources that are available relative to the total E-UTRAN resources. The capacity value should be measured and reported so that the minimum E-UTRAN resource usage of existing services is reserved according to implementation. This IE can be weighted according to the ratio of cell capacity class values, if available.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Capacity Value	M		INTEGER (0..100)	Value 0 indicates no available capacity, and 100 indicates maximum available capacity. Capacity Value should be measured on a linear scale.

9.3.1.200 E-UTRAN Radio Resource Status

This IE indicates the usage of the PRBs for all traffic in downlink and uplink (TS 36.314 [49], TS 23.203 [50]) and the usage of PDCCH CCEs for downlink and uplink scheduling.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL GBR PRB Usage	M		INTEGER (0..100)	
UL GBR PRB Usage	M		INTEGER (0..100)	
DL Non-GBR PRB Usage	M		INTEGER (0..100)	
UL Non-GBR PRB Usage	M		INTEGER (0..100)	
DL Total PRB Usage	M		INTEGER (0..100)	
UL Total PRB Usage	M		INTEGER (0..100)	
DL Scheduling PDCCH CCE Usage	O		INTEGER (0..100)	
UL Scheduling PDCCH CCE Usage	O		INTEGER (0..100)	

9.3.1.201 Void

9.3.1.202 Void

9.3.1.203 Void

9.3.1.204 Void

9.3.1.205 NR Radio Resource Status

This IE indicates the usage of the PRBs per cell for MIMO for all traffic in Downlink and Uplink.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell DL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
UL GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell UL GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
DL Non-GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell DL Non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
UL Non-GBR PRB Usage for MIMO	M		INTEGER (0..100)	Per cell UL Non-GBR PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
DL Total PRB Usage for MIMO	M		INTEGER (0..100)	Per cell DL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].
UL Total PRB usage for MIMO	M		INTEGER (0..100)	Per cell UL Total PRB usage for MIMO in percentage of the cell total PRB number as defined in TS 38.314 [48].

9.3.1.206 MBS Session ID

This IE uniquely identifies an MBS session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TMGI	M		OCTET STRING (SIZE(6))	Encoded as defined in TS 23.003 [23].
NID	O		9.3.3.42	

9.3.1.207 MBS Area Session ID

This IE indicates the Area Session ID for MBS session with location dependent context.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Area Session ID	M		INTEGER (0..65535, …)	

9.3.1.208 MBS Service Area

This IE contains the MBS service area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Session Type*	M			
>location independent				
>>MBS Service Area Information	M		9.3.1.209	
>location dependent				
>>MBS Service Area Information List		*1*		
>>>MBS Service Area Information Item		*1..<maxnoofMBSServiceAreaInformation>*		
>>>>MBS Area Session ID	M		9.3.1.207	
>>>>MBS Service Area Information	M		9.3.1.209	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information elements in the *MBS Service Area Information Location Dependent List* IE. Value is 256.

9.3.1.209 MBS Service Area information

This IE contains MBS service area information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Service Area Cell List		*0..<maxnoofCellsforMBS>*		
>NR CGI	M		9.3.1.7	
MBS Service Area TAI List		*0..<maxnoofTAIforMBS>*		
>TAI	M		9.3.3.11	

Range bound	Explanation
maxnoofCellsforMBS	Maximum no. of cells allowed within one MBS Service Area. Value is 8192.
maxnoofTAIforMBS	Maximum no. of TAs allowed within one MBS Service Area. Value is 1024.

9.3.1.210 MBS Support Indicator

This IE indicates whether MBS is supported for the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Support Indicator	M		ENUMERATED (true, …)	

9.3.1.211 MBS Session Setup Request List

This IE provides information related to MBS sessions joined by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Setup Request List		*1*		
>MBS Session Setup Request Item		*1..<maxnoofMBSSessions>*		
>>MBS Session ID	M		9.3.1.206	
>>MBS Area Session ID	O		9.3.1.207	
>>Associated MBS QoS Flow Setup Request List		*0..1*		
>>>Associated MBS QoS Flow Setup Request Item		*1..<maxnoofMBSQoSflows>*		
>>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51	
>>>>Associated Unicast QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51	

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.

9.3.1.212 MBS Session Setup or Modify Request List

This IE provides information related to MBS sessions joined by the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Setup or Modify Request List		*1*		
>MBS Session Setup or Modify Request Item		*1..<maxnoofMBSSessions>*		
>>MBS Session ID	M		9.3.1.206	
>>MBS Area Session ID	O		9.3.1.207	
>>Associated MBS QoS Flow Setup or Modify Request List		*0..1*		
>>>Associated MBS QoS Flow Setup or Modify Request Item		*1..<maxnoofMBSQoSflows>*		
>>>>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51	
>>>>Associated Unicast QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51	
>>MBS QoS Flow To Release List	O		QoS Flow List with Cause 9.3.1.13	This IE indicates the MBS QoS Flow Identifiers of the MBS QoS Flows to be released.

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.

9.3.1.213 MBS Session Setup Response List

This IE contains a list of information related to MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Setup Response List		*1*		
>MBS Session Setup Response Item		*1..<maxnoofMBSSessions>*		
>>MBS Session ID	M		9.3.1.206	
>>MBS Area Session ID	O		9.3.1.207	

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.

9.3.1.214 MBS Session Failed to Setup List

This IE contains a list of information related to MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Failed to Setup List		*1*		
>MBS Session Failed to Setup Item		*1..<maxnoofMBSSessions>*		
>>MBS Session ID	M		9.3.1.206	
>>MBS Area Session ID	O		9.3.1.207	
>>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS Sessions allowed within one PDU session. Value is 32.

9.3.1.215 MBS Session To Release List

This IE indicates MBS sessions to be removed.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session To Release List		*1*		
>MBS Session To Release Item		*1..<maxnoofMBSSessions>*		
>>MBS Session ID	M		9.3.1.206	
>>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofMBSSessions	Maximum no. of MBS sessions allowed within one PDU session. Value is 32.

9.3.1.216 Multicast Group Paging Area

This IE contains a list of TAIs corresponding to the multicast group paging area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Area TAI List		*1..<maxnoofTAIforPaging>*		
>TAI	M		9.3.3.11	

Range bound	Explanation
maxnoofTAIforPaging	Maximum no. of TAIs for multicast group paging. Value is 16.

9.3.1.217 MBS Session Status

This IE indicates whether the MBS session is activated or deactivated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session Status	M		ENUMERATED (activated, deactivated, …)	

9.3.1.218 MRB ID

This IE contains the MRB ID as specified in TS 38.401 [2].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MRB ID	M		INTEGER (1..512, ...)	

9.3.1.219 MRB Progress Information

This IE contains the MRB progress information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *PDCP SN Status*	M			
>12bits				
>>PDCP SN Length 12	M		INTEGER (0..4095)	
>18bits				
>>PDCP SN Length 18	M		INTEGER (0..262143)	

9.3.1.220 Time Synchronisation Assistance Information

This IE indicates 5G access stratum time distribution parameters as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time Distribution Indication	M		ENUMERATED (enabled, disabled, …)	
Uu Time Synchronisation Error Budget	C-ifEnabled		INTEGER (1..1000000, …)	Expressed in units of 1ns.

Condition	Explanation
ifEnabled	This IE shall be present if the *Time Distribution Indication* IE is set to “enabled”.

9.3.1.221 Survival Time

This IE indicates the Survival Time of the TSC QoS flow as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Survival Time	M		INTEGER (0.. 1920000, …)	Survival time expressed in units of 1 us.

9.3.1.222 QMC Deactivation

This IE indicates the QMC configurations to be deactivated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoE Reference List		*1..<maxnoofUEAppLayerMeas>*		
>QoE Reference	M		OCTET STRING (SIZE(6))	*QoE Reference*, as defined in clause 5.2 of TS 28.405 [45]. It consists of MCC+MNC+QMC ID, where the MCC and MNC are coming with the QMC activation request from the management system to identify one PLMN containing the management system, and QMC ID is a 3-bytes Octet String.

Range bound	Explanation
maxnoofUEAppLayerMeas	Maximum no. of UE application layer measurement. Value is 16.

9.3.1.223 QMC Configuration Information

This IE contains the configuration information for the QMC functionality.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Application Layer Measurement Information List		*1*		
>UE Application Layer Measurement Information Item		*1..<maxnoofUEAppLayerMeas>*		
>>UE Application Layer Measurement Configuration Information	M		9.3.1.224	

Range bound	Explanation
maxnoofUEAppLayerMeas	Maximum no. of UE application layer measurements. Value is 16.

9.3.1.224 UE Application Layer Measurement Configuration Information

This IE defines configuration information for the QMC functionality.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoE Reference	M		OCTET STRING (SIZE(6))	*QoE Reference*, as defined in clause 5.2 of TS 28.405 [45]. It consists of MCC+MNC+QMC ID, where the MCC and MNC are coming with the QMC activation request from the management system to identify one PLMN containing the management system, and QMC ID is a 3-bytes Octet String.
Service Type	M		ENUMERATED (QMC for DASH streaming, QMC for MTSI, QMC for VR, ...)	This IE indicates the service type of QoE measurements.
CHOICE *Area Scope of QMC*	O			
>Cell based				
>>Cell ID List for QMC		*1..<maxnoofCellIDforQMC>*		
>>>NG-RAN CGI	M		9.3.1.73	This IE can only indicate the NR CGI.
>TA based				
>>TA List for QMC		*1..<maxnoofTAforQMC>*		
>>>TAC	M		9.3.3.10	The TAI is derived using the current serving PLMN.
>TAI based				
>>TAI List for QMC		*1..<maxnoofTAforQMC>*		
>>>TAI	M		9.3.3.11	
>PLMN area based				
>>PLMN List for QMC		*1..<maxnoofPLMNforQMC>*		
>>>PLMN Identity	M		9.3.3.5	
Measurement Collection Entity IP Address	M		Transport Layer Address 9.3.2.4	The IP address of the entity receiving the QoE measurement report.
QoE Measurement Status	O		ENUMERATED (ongoing, …)	Indicates whether the QoE measurement has been started. Present in case of NG-based handover.
Container for Application Layer Measurement Configuration	O		OCTET STRING (SIZE(1.. 8000))	Contains application layer measurement configuration, see Annex L in 26.247 [46], clause 16.5 in TS 26.114 [51] and clause 9 in TS 26.118 [52]. Present in case of initial QoE configuration, and shall be included in *Source to Target Transparent Container* IE for signalling-based QMC during NG-based handover.
Measurement Configuration Application Layer ID	O		INTEGER (0..15, …)	This IE is present only when the message containing it is NG-based handover related. The IE indicates the identity of the application layer measurement configuration and corresponds to the *MeasConfigAppLayerId* IE as defined in TS 38.331 [18].

Slice Support List for QMC		*0..1*		
>Slice Support QMC Item		*1..<maxnoofSNSSAIforQMC>*		
>>S-NSSAI	M		9.3.1.24	
CHOICE *MDT Alignment Information*	O			Indicates the MDT measurements with which alignment is required.
>S-based MDT				
>>NG-RAN Trace ID	M		OCTET STRING (SIZE(8))	This IE is composed of the following: Trace Reference defined in TS 32.422 [11] (leftmost 6 octets, with PLMN information encoded as in 9.3.3.5), and Trace Recording Session Reference defined in TS 32.422 [11] (last 2 octets).
Available RAN Visible QoE Metrics	O		9.3.1.225	Present in case of initial QoE configuration and in case of NG-based handover for signalling-based QoE measurement.

Range bound	Explanation
maxnoofCellIDforQMC	Maximum no. of Cell ID subject for QMC scope. Value is 32.
maxnoofTAforQMC	Maximum no. of TA subject for QMC scope. Value is 8.
maxnoofPLMNforQMC	Maximum no. of PLMNs in the PLMN list for QMC scope. Value is 16.
maxnoofSNSSAIforQMC	Maximum no. of S-NSSAIs in the S-NSSAI list for QMC scope. Value is 16.

9.3.1.225 Available RAN Visible QoE Metrics

This IE defines which RAN visible QoE metrics can be configured by the NG-RAN in the RAN visible QoE measurement.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Application Layer Buffer Level List	O		ENUMERATED (true, ...)	The IE defines whether the Buffer Level can be collected as a RAN visible QoE metric by NG-RAN from UE, for DASH streaming and VR service types.
Playout Delay for Media Startup	O		ENUMERATED (true, ...)	The IE defines whether the Playout delay can be collected as a RAN visible QoE metric by NG-RAN from UE, for DASH streaming and VR service types.

9.3.1.226 Void

9.3.1.227 NR Paging eDRX Information

This IE indicates the NR Paging eDRX parameters as defined in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NR Paging eDRX Cycle	M		ENUMERATED (hfquarter, hfhalf, hf1, hf2, hf4, hf8, hf16, hf32, hf64, hf128, hf256, hf512, hf1024, …)	$T_{eDRX\ CN}$ defined in TS 38.304 [12]. Unit: [number of hyperframes].
NR Paging Time Window	O		ENUMERATED (s1, s2, s3, s4, s5, s6, s7, s8, s9, s10, s11, s12, s13, s14, s15, s16, …, s17, s18, s19, s20, s21, s22, s23, s24, s25, s26, s27, s28, s29, s30, s31, s32)	PTW defined in TS 38.304 [12]. Unit: [1.28 seconds]

9.3.1.228 RedCap Indication

This IE is provided by the NG-RAN node to inform that the UE indicates Reduced Capability.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RedCap Indication	M		ENUMERATED (RedCap, ...)	

9.3.1.229 Target NSSAI Information

This IE contains the Target NSSAI and Index to RAT/Frequency Selection Priority as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target NSSAI	M		9.3.1.230	
Index to RAT/Frequency Selection Priority	M		9.3.1.61	

9.3.1.230 Target NSSAI

This IE contains the Target NSSAI as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target S-NSSAI List		*1*		
>Target S-NSSAI Item		*1..<maxnoofTargetS-NSSAIs>*		
>>S-NSSAI	M		9.3.1.24	

Range bound	Explanation
maxnoofTargetS-NSSAIs	Maximum no. of Target S-NSSAIs. Value is 8.

9.3.1.231 UE Slice Maximum Bit Rate List

This IE contains the UE Slice Maximum Bit Rate List as specified in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE Slice Maximum Bit Rate Item		*1..<maxnoof AllowedS-NSSAIs>*		Applicable across all GBR and Non-GBR QoS flows.
>S-NSSAI	M		9.3.1.24	
>UE Slice Maximum Bit Rate Downlink	M		Bit Rate 9.3.1.4	This IE indicates the downlink UE-Slice-MBR as specified in TS 23.501 [9].
>UE Slice Maximum Bit Rate Uplink	M		Bit Rate 9.3.1.4	This IE indicates the uplink UE-Slice-MBR as specified in TS 23.501 [9].

Range bound	Explanation
maxnoofAllowedS-NSSAIs	Maximum no. of allowed S-NSSAI. Value is 8.

9.3.1.232 PEIPS Assistance Information

This IE provides the information related to CN paging subgrouping for a particular UE, as specified in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CN Subgroup ID	M		INTEGER (0..7, …)	

9.3.1.233 5G ProSe Authorized

This IE provides information on the authorization status of the UE to use the 5G ProSe services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
5G ProSe Direct Discovery	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Direct Discovery
5G ProSe Direct Communication	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Direct Communication
5G ProSe Layer-2 UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 UE-to-Network Relay
5G ProSe Layer-3 UE-to-Network Relay	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-3 UE-to-Network Relay
5G ProSe Layer-2 Remote UE	O		ENUMERATED (authorized, not authorized, ...)	Indicates whether the UE is authorized for 5G ProSe Layer-2 Remote UE

9.3.1.234 5G ProSe PC5 QoS Parameters

This IE provides information on the 5G ProSe PC5 QoS parameters of the UE’s sidelink communication for 5G ProSe services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
5G ProSe PC5 QoS Flow List		*1*		
>5G ProSe PC5 QoS Flow Item		*1..<maxnoofPC5QoSFlows>*		
>>PQI	M		INTEGER (0..255, …)	PQI is a special 5QI as specified in TS 23.501 [9].
>>5G ProSe PC5 Flow Bit Rates		*0..1*		Only applies for GBR QoS flows.
>>>Guaranteed Flow Bit Rate	M		Bit Rate 9.3.1.4	Guaranteed Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>>Maximum Flow Bit Rate	M		Bit Rate 9.3.1.4	Maximum Bit Rate for the PC5 QoS flow. Details in TS 23.501 [9].
>>Range	O		ENUMERATED (m50, m80, m180, m200, m350, m400, m500, m700, m1000, …)	Only applies for groupcast.
5G ProSe PC5 Link Aggregate Bit Rates	O		Bit Rate 9.3.1.4	Only applies for Non-GBR QoS flows.

Range bound	Explanation
maxnoofPC5QoSFlows	Maximum no. of PC5 QoS flows allowed towards one UE. Value is 2048.

9.3.1.235 Last Visited PSCell Information

The Last Visited PSCell Information may contain cell specific information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PSCell ID	O		NG-RAN CGI 9.3.1.73	This IE is present when the SCG resources are configured for the UE.
Time Stay	M		INTEGER (0..40950)	The duration of the time the UE stayed in the cell in 1/10 seconds. If the UE stays in a cell more than 4095s, this IE is set to 40950. Or the duration of the time when no SCG resources are configured for the UE.

9.3.1.236 MBS QoS Flows To Be Setup List

This IE contains a list of QoS flows and their associated QoS parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS QoS Flows Setup Request Item		*1..<maxnoofMBSQoSFlows>*		
>MBS QoS Flow Identifier	M		QoS Flow Identifier 9.3.1.51	
>MBS QoS Flow Level QoS Parameters	M		QoS Flow Level QoS Parameters 9.3.1.12	

Range bound	Explanation
maxnoofMBSQoSflows	Maximum no. of MBS QoS flows allowed within one MBS session. Value is 64.

9.3.1.237 Reporting System

This IE lists the cells on the reporting system.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Reporting System*				
>E-UTRAN				
>>E-UTRAN Cell To Report List		*1*		
>>>E-UTRAN Cell To Report Item		*1..<maxnoofReportedCells>*		
>>>>Cell ID	M		E-UTRA CGI 9.3.1.9	
>NG-RAN				
>>NG-RAN Cell To Report List		*1*		
>>>NG-RAN Cell To Report Item		*1..<maxnoofReportedCells>*		
>>>>Cell ID	M		NG-RAN CGI 9.3.1.73	
>No Reporting			NULL	Resource status not available. Applicable to the *Inter-system Resource Status Reply* IE.

Range bound	Explanation
maxnoofReportedCells	Maximum no. of cells that can be reported. Value is 256.

9.3.1.238 TAI NSAG Support List

This IE contains the slice group mapping for all groups configured at the NG-RAN node per TAI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TAI NSAG Support Item		*1..<maxnoofNSAGs>*		
>NSAG ID	M		INTEGER (0..255, …)	
>NSAG Slice Support List	M		Extended Slice Support List 9.3.1.191	Indicates the list of slices which belong to the NSAG.

Range bound	Explanation
maxnoofNSAGs	Maximum no. of Slice Groups for the TAI. Value is 256.

9.3.1.239 NGAP Protocol IE-Id

This IE uniquely identifies an NGAP Protocol IE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP Protocol IE-Id	M		INTEGER (0..65535)	

9.3.1.240 NGAP Protocol IE Support Information

This IE provides information about support of functions associated to an NGAP Protocol IE-Id.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP Protocol IE Support Information	M		ENUMERATED (supported, not-supported, ...)	

9.3.1.241 NGAP Protocol IE Presence Information

This IE provides information on whether an NGAP Protocol IE-Id was received within the message requesting the information (i.e. HANDOVER REQUEST).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP Protocol IE Presence Information	M		ENUMERATED (present, not present)	

9.3.1.242 NGAP IE Support Information Response List

This IE provides information about support of functions associated to a list of NGAP Protocol IE-Ids.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NGAP IE Support Information Response Item		*1..<maxnoofIE SupportInfo>*		
>NGAP Protocol IE-Id	M		9.3.1.239	
>NGAP Protocol IE Support Information	M		9.3.1.240	
>NGAP Protocol IE Presence Information	M		9.3.1.241	

Range bound	Explanation
maxnoofIESupportInfo	Maximum no. of IE Support Information. Value is 32.

9.3.1.243 MDT PLMN Modification List

This IE provides the modified list of PLMN allowed for MDT.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MDT PLMN Modification List		*0..<maxnoofM DTPLMNs>*		An empty list indicates there is no PLMN allowed for MDT.
>PLMN Identity	M		9.3.3.5	

Range bound	Explanation
maxnoofMDTPLMNs	Maximum no. of PLMNs in the MDT PLMN list. Value is 16.

9.3.1.244 Excess Packet Delay Threshold Configuration

This IE defines the parameters for Excess Packet Delay Threshold configuration to support the calculation of the PDCP Excess Packet Delay in the UL per DRB as specified in TS 38.314 [48].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Excess Packet Delay Threshold Item		*1..<maxnoofThresholdsForExcessPacketDelay>*		
>5QI	M		INTEGER (0..255, …)	Indicates the standardized or pre-configured 5QI as specified in TS 23.501 [9].
>Excess Packet Delay Threshold Value	M		ENUMERATED (ms0.25, ms0.5, ms1, ms2, ms4, ms5, ms10, ms20, ms30, ms40, ms50, ms60, ms70, ms80, ms90, ms100, ms150, ms300, ms500, ...)	

Range bound	Explanation
maxnoofThresholdsForExcessPacketDelay	Maximum no. of thresholds for Excess Packet Delay configuration. Value is 255.

9.3.2 Transport Network Layer Related IEs

9.3.2.1 QoS Flow per TNL Information List

This IE is used to provide a list of additional UP transport layer information for a split PDU session, along with the associated QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow per TNL Information Item		*1..<maxnoofMultiConnectivityMinusOne>*		
>QoS Flow per TNL Information	M		9.3.2.8	

Range bound	Explanation
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.2.2 UP Transport Layer Information

This IE is used to provide the NG user plane transport layer information associated with a PDU session for an NG-RAN node – UPF pair. In this release it corresponds to an IP address and a GTP Tunnel Endpoint Identifier.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *UP Transport Layer Information*	M			
>GTP tunnel				
>>Endpoint IP Address	M		Transport Layer Address 9.3.2.4	
>>GTP-TEID	M		9.3.2.5	

9.3.2.3 E-RAB ID

This IE is the identifier of the LTE E-RAB.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
E-RAB ID	M		INTEGER (0..15, …)	

9.3.2.4 Transport Layer Address

This IE is an IP address.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Transport Layer Address	M		BIT STRING (SIZE(1..160, …))	The Radio Network Layer is not supposed to interpret the address information. It should pass it to the Transport Layer for interpretation. For details, see TS 38.414 [14].

9.3.2.5 GTP-TEID

This IE is the GTP Tunnel Endpoint Identifier to be used for the user plane transport between the NG-RAN node and the UPF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
GTP-TEID	M		OCTET STRING (SIZE(4))	For details and range, see TS 29.281 [15].

9.3.2.6 CP Transport Layer Information

This IE is used to provide the NG control plane transport layer information associated with an NG-RAN node – AMF pair.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *CP Transport Layer Information*					-	
>Endpoint-IP-address						
>>Endpoint IP Address	M		Transport Layer Address 9.3.2.4		-	
>Endpoint-IP-address-and-port					YES	reject
>>Endpoint IP Address	M		Transport Layer Address 9.3.2.4		-	
>>Port Number	M		OCTET STRING (SIZE(2))		-	

9.3.2.7 TNL Association List

This IE contains a list of TNL associations. It is used for example to indicate failed TNL association(s).

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TNL Association Item		*1..<maxnoofTNLAssociations>*		
>TNL Association Address	M		CP Transport Layer Information 9.3.2.6	
>Cause	M		9.3.1.2	

Range bound	Explanation
maxnoofTNLAssociations	Maximum no. of TNL Associations between the NG-RAN node and the AMF. Value is 32.

9.3.2.8 QoS Flow per TNL Information

This IE indicates the NG-U transport layer information and associated list of QoS flows.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UP Transport Layer Information	M		9.3.2.2	
Associated QoS Flow List	M		9.3.1.99	

9.3.2.9 TNL Association Usage

This IE indicates the usage of the TNL association.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TNL Association Usage	O		ENUMERATED (ue, non-ue, both, …)	Indicates whether the TNL association is only used for UE-associated signalling, or non-UE-associated signalling, or both.

9.3.2.10 TNL Address Weight Factor

This IE indicates the weight factor of the TNL address.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TNL Address Weight Factor	M		INTEGER (0..255)	Value 0 indicates the TNL address is not permitted for the initial NGAP message. If the value for each TNL address is the same, it indicates the deployments that rely solely on 5GC-based load balancing.

9.3.2.11 UP Transport Layer Information Pair List

This IE is used to provide a list of uplink UP transport layer information and associated downlink UP transport layer information for a split PDU session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UP Transport Layer Information Pair Item		*1..<maxnoofMultiConnectivityMinusOne>*		
>UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	
>DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.2.12 UP Transport Layer Information List

This IE is used to provide a list of additional UP transport layer information for a split PDU session.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UP Transport Layer Information Item		*1..<maxnoofMultiConnectivityMinusOne>*			-	
>NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2		-	
>Common Network Instance	O		9.3.1.120		YES	ignore

Range bound	Explanation
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.2.13 QoS Flow List with Data Forwarding

This IE is used to provide a list of QoS flows with indication if forwarding is accepted.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoS Flow Item with Data Forwarding		*1..<maxnoofQoSFlows>*			-	
>QoS Flow Identifier	M		9.3.1.51		-	
>Data Forwarding Accepted	O		9.3.1.62	This IE is included for the QoS flows in the PDU session to be forwarded over the PDU session forwarding tunnel. It may be included for the QoS flows in the PDU session to be forwarded over the DRB forwarding tunnel(s).	-	
>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.2.14 URI

This IE is an URI.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
URI	M		VisibleString	String representing URI (Uniform Resource Identifier)

9.3.2.15 MBS Session TNL Information 5GC

This IE provides 5GC TNL information for location dependent and location independent MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Session Type*	M			
>location independent				
>>Shared NG-U Multicast TNL Information	M		9.3.2.16	
>location dependent				
>>MBS Session TNL Information 5GC List		*1*		
>>>MBS Session TNL Information 5GC Item		*1..<maxnoofMBSServiceAreaInformation>*		
>>>>MBS Area Session ID	M		9.3.1.207	
>>>>Shared NG-U Multicast TNL Information	M		9.3.2.16	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information. Value is 256.

9.3.2.16 Shared NG-U Multicast TNL Information

This IE provides the shared NG user plane transport layer information associated with an MBS session at the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
IP Multicast Address	M		Transport Layer Address 9.3.2.4	
IP Source Address	M		Transport Layer Address 9.3.2.4	
GTP-TEID at 5GC	M		GTP-TEID 9.3.2.5	

9.3.2.17 MBS Session TNL Information NG-RAN

This IE provides NG-RAN TNL information for location dependent and location independent broadcast MBS sessions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Session Type*	M			
>location independent				
>>Shared NG-U Unicast TNL Information	M		UP Transport Layer Information 9.3.2.2	
>location dependent				
>>MBS Session TNL Information NG-RAN List		*1*		
>>>MBS Session TNL Information NG-RAN Item		*1..<maxnoofMBSServiceAreaInformation>*		
>>>>MBS Area Session ID	M		9.3.1.207	
>>>>Shared NG-U Unicast TNL Information	O		UP Transport Layer Information 9.3.2.2	

Range bound	Explanation
maxnoofMBSServiceAreaInformation	Maximum no. of MBS Service Area Information. Value is 256.

9.3.3 NAS Related IEs

9.3.3.1 AMF UE NGAP ID

This IE uniquely identifies the UE association over the NG interface, as described in TS 38.401 [2].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF UE NGAP ID	M		INTEGER (0..2^{40}-1)	

9.3.3.2 RAN UE NGAP ID

This IE uniquely identifies the UE association over the NG interface within the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN UE NGAP ID	M		INTEGER (0..2^{32}-1)	

9.3.3.3 GUAMI

This IE indicates the AMF identity.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
AMF Region ID	M		BIT STRING (SIZE(8))	
AMF Set ID	M		9.3.3.12	
AMF Pointer	M		9.3.3.19	

9.3.3.4 NAS-PDU

This IE contains a 5GC – UE or UE – 5GC message that is transferred without interpretation in the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NAS-PDU	M		OCTET STRING	The content is defined in TS 24.501 [26].

9.3.3.5 PLMN Identity

This IE indicates the PLMN Identity.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		OCTET STRING (SIZE(3))	Digits 0 to 9 encoded 0000 to 1001, 1111 used as filler digit. Two digits per octet: - bits 4 to 1 of octet n encoding digit 2n-1 - bits 8 to 5 of octet n encoding digit 2n PLMN Identity consists of 3 digits from MCC followed by either: - a filler digit plus 2 digits from MNC (in case of 2 digit MNC) or - 3 digits from MNC (in case of 3 digit MNC).

9.3.3.6 SON Configuration Transfer

This IE contains the configuration information, used by e.g., SON functionality, and additionally includes the NG-RAN node identifier of the destination of this configuration information and the NG-RAN node identifier of the source of this information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Target RAN Node ID SON		*1*			-	
>Global RAN Node ID	M		9.3.1.5		-	
>Selected TAI	M		TAI 9.3.3.11		-	
>NR CGI	O		9.3.1.7	This IE is ignored if the *SON Information* IE contains the *SON Information Reply* IE.	YES	ignore
Source RAN Node ID		*1*			-	
>Global RAN Node ID	M		9.3.1.5		-	
>Selected TAI	M		TAI 9.3.3.11		-	
SON Information	M		9.3.3.7		-	
Xn TNL Configuration Info	C-ifSONInformationRequest		9.3.3.9	Source NG-RAN node Xn TNL Configuration Info.	-	

Condition	Explanation
ifSONInformationRequest	This IE shall be present if the *SON Information* IE contains the *SON Information Request* IE set to "Xn TNL Configuration Info"

9.3.3.7 SON Information

This IE identifies the nature of the configuration information transferred, i.e., a request, a reply or a report.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *SON Information*	M				-	
>SON Information Request						
>>SON Information Request	M		ENUMERATED (Xn TNL Configuration Info, …)		-	
>SON Information Reply						
>>SON Information Reply	M		9.3.3.8		-	
>SON Information Report						
>>SON Information Report	M		9.3.3.35		YES	ignore

9.3.3.8 SON Information Reply

This IE contains the configuration information to be replied to the NG-RAN node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Xn TNL Configuration Info	O		9.3.3.9	

9.3.3.9 Xn TNL Configuration Info

This IE is used for signalling Xn TNL Configuration information for automatic Xn SCTP association establishment.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Xn Transport Layer Addresses		*1..<maxnoofXnTLAs>*			-	
>Transport Layer Address	M		9.3.2.4	Transport Layer Addresses for Xn SCTP endpoint.	-	
Xn Extended Transport Layer Addresses		*0..<maxnoofXnExtTLAs>*			-	
>IP-Sec Transport Layer Address	O		Transport Layer Address 9.3.2.4	Transport Layer Addresses for IP-Sec endpoint.	-	
>Xn GTP Transport Layer Addresses		*0..<maxnoofXnGTP-TLAs>*			-	
>>GTP Transport Layer Address	M		Transport Layer Address 9.3.2.4	GTP Transport Layer Addresses for GTP end-points (used for data forwarding over Xn).	-	
>Xn SCTP Transport Layer Addresses		*0..<maxnoofXnTLAs>*			YES	ignore
>>Transport Layer Address SCTP	M		Transport Layer Address 9.3.2.4	Transport Layer Addresses for Xn SCTP endpoint.	-	

Range bound	Explanation
maxnoofXnTLAs	Maximum no. of Xn Transport Layer Addresses for an SCTP end-point. Value is 2.
maxnoofXnExtTLAs	Maximum no. of Xn Extended Transport Layer Addresses in the message. Value is 16.
maxnoofXnGTP-TLAs	Maximum no. of Xn GTP Transport Layer Addresses for a GTP end-point in the message. Value is 16.

9.3.3.10 TAC

This IE is used to uniquely identify a Tracking Area Code.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
TAC	M		OCTET STRING (SIZE(3))	

9.3.3.11 TAI

This IE is used to uniquely identify a Tracking Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
TAC	M		9.3.3.10	

9.3.3.12 AMF Set ID

This IE is used to uniquely identify an AMF Set within the AMF Region.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Set ID	M		BIT STRING (SIZE(10))	

9.3.3.13 Routing ID

This IE is used to identify an LMF within the 5GC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Routing ID	M		OCTET STRING	The maximum length is 16 octets, referring to the length of a Universally Unique Identifier (UUID) version 4 as specified in section 4.4 in IETF RFC 4122 [53]

9.3.3.14 NRPPa-PDU

This IE contains an NG-RAN node – LMF or LMF – NG-RAN node message that is transferred without interpretation in the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NRPPa-PDU	M		OCTET STRING	

9.3.3.15 RAN Paging Priority

This IE contains the service priority as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Paging Priority	M		INTEGER (1..256)	Values ordered in decreasing order of priority, i.e. with 1 as the highest priority and 256 as the lowest priority

9.3.3.16 EPS TAC

This IE is used to uniquely identify an EPS Tracking Area Code.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
EPS TAC	M		OCTET STRING (SIZE(2))	

9.3.3.17 EPS TAI

This IE is used to uniquely identify an EPS Tracking Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
PLMN Identity	M		9.3.3.5	
EPS TAC	M		9.3.3.16	

9.3.3.18 UE Paging Identity

This IE represents the Identity with which the UE is paged.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *UE Paging Identity*	M			
>5G-S-TMSI				
>>5G-S-TMSI	M		9.3.3.20	

9.3.3.19 AMF Pointer

This IE is used to identify one or more AMF(s) within the AMF Set.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Pointer	M		BIT STRING (SIZE(6))	

9.3.3.20 5G-S-TMSI

This IE is used for security reasons, to hide the identity of a subscriber.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Set ID	M		9.3.3.12	
AMF Pointer	M		9.3.3.19	
5G-TMSI	M		OCTET STRING (SIZE(4))	5G-TMSI is unique within the AMF that allocated it.

9.3.3.21 AMF Name

This IE is used to uniquely identify the AMF (see TS 38.300 [8]). It may also be used as a human readable name of the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Name	M		PrintableString (SIZE(1..150, …))	

9.3.3.22 Paging Origin

This IE indicates whether Paging is originated due to the PDU sessions from the non-3GPP access.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Paging Origin	M		ENUMERATED (non-3GPP, …)	

9.3.3.23 UE Identity Index Value

This IE is used by the NG-RAN node to calculate the Paging Frame as specified in TS 38.304 [12] and TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *UE Identity Index Value*				
>Index Length 10				
>>Index Length 10	M		BIT STRING (SIZE(10))	Coded as specified in TS 38.304 [12] and TS 36.304 [29].

9.3.3.24 Periodic Registration Update Timer

This IE is used to assist NG-RAN to generate corresponding timer for periodic RNA update for RRC_INACTIVE UEs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Periodic Registration Update Timer	M		BIT STRING (SIZE(8))	Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit for the Periodic Registration Update Timer as follows: Bits **8 7 6** 0 0 0 value is incremented in multiples of 10 minutes 0 0 1 value is incremented in multiples of 1 hour 0 1 0 value is incremented in multiples of 10 hours 0 1 1 value is incremented in multiples of 2 seconds 1 0 0 value is incremented in multiples of 30 seconds 1 0 1 value is incremented in multiples of 1 minute 1 1 1 value indicates that the timer is deactivated. 1 1 0 value is incremented in multiples of 1 hour in this version of the protocol.

9.3.3.25 UE-associated Logical NG-connection List

This IE contains a list of UE-associated logical NG-connections.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE-associated Logical NG-connection Item		*1..<maxnoofNGConnectionsToReset>*		
>AMF UE NGAP ID	O		9.3.3.1	
>RAN UE NGAP ID	O		9.3.3.2	

Range bound	Explanation
maxnoofNGConnectionsToReset	Maximum no. of UE-associated logical NG-connections allowed to reset in one message. Value is 65536.

9.3.3.26 NAS Security Parameters from NG-RAN

This IE provides security related parameters for inter-system handover from NG-RAN to E-UTRAN or from NG-RAN to UTRAN via the eNB to the UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NAS Security Parameters from NG-RAN	M		OCTET STRING	Refers to the *N1 mode to S1 mode NAS transparent container* IE, the details of the IE definition and the encoding are specified in TS 24.501 [26].

9.3.3.27 Source to Target AMF Information Reroute

This IE is used to transparently pass information provided by NSSF from the source AMF to the target AMF through the NG-RAN node; it is produced by the source core network node and is transmitted to the target core network node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Configured NSSAI	O		OCTET STRING (SIZE(128))	The maximum number of S-NSSAI in Configured NSSAI is 16.This IE contains optional mapping S-NSSAI. When present, this IE shall be transmitted transparent from the source Core network node to the target Core network node. The octets of the OCTET STRING are encoded according to description in TS 29.531 [30]
Rejected NSSAI in PLMN	O		OCTET STRING (SIZE(32))	This IE contain the rejected NSSAI(s) in the PLMN. When present, this IE shall be transmitted transparent from the source Core network node to the target Core network node. The octets of the OCTET STRING are encoded according to description in TS 29.531 [30].
Rejected NSSAI in TA	O		OCTET STRING (SIZE(32))	This IE contain the rejected NSSAI(s) in the TA. When present, this IE shall be transmitted transparent from the source Core network node to the target Core network node. The octets of the OCTET STRING are encoded according to description in TS 29.531 [30].

9.3.3.28 RIM Information Transfer

This IE contains information used by the RIM functionality, and additionally includes the NG-RAN node identifier of the destination of the RIM information and the NG-RAN node identifier of the source of this information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target RAN Node ID RIM		*1*		
>Global RAN Node ID	M		9.3.1.5	
>Selected TAI	M		TAI 9.3.3.11	
Source RAN Node ID		*1*		
>Global RAN Node ID	M		9.3.1.5	
>Selected TAI	M		TAI 9.3.3.11	
RIM Information	M		9.3.3.29	

9.3.3.29 RIM Information

This IE contains the RIM information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Target gNB Set ID	M		gNB Set ID 9.3.1.122	The victim gNB Set ID.
RIM-RS Detection	M		ENUMERATED (RS detected, RS disappeared, …)	

9.3.3.30 LAI

This IE is used to uniquely identify a Location Area.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
LAI				
>PLMN Identity	M		9.3.3.5	
>LAC	M		OCTET STRING (SIZE(2))	0000 and FFFE not allowed.

9.3.3.31 Extended Connected Time

This IE indicates the minimum time the RAN should keep the UE in RRC_CONNECTED state regardless of inactivity, as defined in TS 23.501 [9].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended Connected Time	M		INTEGER (0..255)	Minimum time the RAN should keep the UE in RRC_CONNECTED state. Unit is second. Value of “0” indicates that the AMF is aware of pending data traffic, but no specific time value is requested.

9.3.3.32 End Indication

This IE indicates that there are no further NAS PDUs to be transmitted for this UE.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
End Indication	M		ENUMERATED (no further data, further data exists, …)	

9.3.3.33 Inter-system SON Configuration Transfer

This IE contains the configuration information, used by e.g., SON functionality, transmitted between an NG-RAN node and an eNB and additionally includes the node identifier of the destination of this configuration information and the node identifier of the source of this information.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Transfer Type*	M			
>from E-UTRAN to NG-RAN				
>>Source eNB-ID		*1*		
>>>Global eNB ID	M		9.3.1.165	
>>>Selected EPS TAI	M		EPS TAI 9.3.3.17	
>>Target NG-RAN node ID		*1*		
>>>Global RAN Node ID	M		9.3.1.5	
>>>Selected TAI	M		TAI 9.3.3.11	
>from NG-RAN to E-UTRAN				
>>Source NG-RAN Node ID		*1*		
>>>Global RAN Node ID	M		9.3.1.5	
>>>Selected TAI	M		TAI 9.3.3.11	
>>Target eNB-ID		*1*		
>>>Global eNB ID	M		9.3.1.165	
>>>Selected EPS TAI	M		EPS TAI 9.3.3.17	
Inter-system SON Information	M		9.3.3.34	

9.3.3.34 Inter-system SON Information

This IE identifies the nature of the configuration information transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *Inter-system SON Information*	M				-	
>Inter-system SON Information Report						
>>Inter-system SON Information Report	M		9.3.3.36		-	
>Inter-system SON Information Request						
>>Inter-system SON Information Request	M		9.3.3.54		YES	ignore
>Inter-system SON Information Reply						
>>Inter-system SON Information Reply	M		9.3.3.55		YES	ignore

9.3.3.35 SON Information Report

This IE contains the configuration information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *SON Information Report*	M				-	
>Failure Indication Information						
>>Failure Indication	M		9.3.3.37		-	
>HO Report Information						
>>HO Report	M		9.3.3.39		-	
>Successful HO Report Information						
>>Successful HO Report List		*1*			YES	ignore
>>>Successful HO Report Item		*1..<maxnoofSuccessfulHOReports>*			-	
>>>>Successful HO Report Container	M		OCTET STRING	Includes the *SuccessHO-Report* IE as defined in TS 38.331 [18].	-	

Range bound	Explanation
maxnoofSuccessfulHOReports	Maximum no. of Successful HO Reports. Value is 64.

9.3.3.36 Inter-system SON Information Report

This IE contains the configuration information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
CHOICE *SON Information Report*	M				-	
>HO Report Information						
>>Inter-system HO Report	M		9.3.3.40		-	
>Failure Indication Information						
>>Inter-system Failure Indication	M		9.3.3.38		-	
>Energy Savings Indication						
>>Inter-system Cell State Indication	M		9.3.3.57		YES	ignore
>Resource Status Report						
>>Inter-system Resource Status Report	M		9.3.3.60		YES	ignore

9.3.3.37 Failure Indication

This IE contains the failure indication to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE RLF Report Container	O		9.3.3.41	

9.3.3.38 Inter-system Failure Indication

This IE contains the failure indication to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE RLF Report Container	O		9.3.3.41	Only contains the LTE RLF report in this version of the specification.

9.3.3.39 HO Report

This IE contains the HO report to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Handover Report Type	M		ENUMERATED (HO too early, HO to wrong cell, Inter-system ping-pong, …)		-	
Handover Cause	M		Cause 9.3.1.2	Indicates handover cause employed for handover from source cell	-	
Source Cell CGI	M		NG-RAN CGI 9.3.1.73	NG-RAN CGI of the source cell for handover procedure	-	
Target Cell CGI	M		NG-RAN CGI 9.3.1.73	NG-RAN CGI of the target cell for handover procedure. If the Handover Report Type is set to “Inter-system ping-pong”, it contains the target cell of the inter system handover from the other system to NG-RAN node cell	-	
Re-establishment Cell CGI	C-ifHandoverReportType HoToWrongCell		NG-RAN CGI 9.3.1.73	NG-RAN CGI of the cell where UE attempted re-establishment or where the UE successfully re-connected after the failure	-	
Source Cell C-RNTI	O		BIT STRING (SIZE (16))	C-RNTI allocated at the source NG-RAN node	-	
Target Cell in E-UTRAN	C-ifHandoverReportType Intersystempingpong		E-UTRA CGI 9.3.1.9	E-UTRA CGI of the E-UTRAN target cell for handover procedure	-	
Mobility Information	O		BIT STRING (SIZE (16))	This IE is not used in the specification. If received, the IE is ignored.	-	
UE RLF Report Container	O		9.3.3.41	The UE RLF Report Container IE received in the FAILURE INDICATION message.	-	
Extended Mobility Information	O		BIT STRING (SIZE (32))	Corresponds to the *Mobility Information* IE provided in the HANDOVER REQUEST message from the source NG-RAN	YES	ignore

				node		

Condition	Explanation
ifHandoverReportTypeHoToWrongCell	This IE shall be present if the *Handover Report Type* IE is set to the value "HO to wrong cell"
ifHandoverReportTypeIntersystempingpong	This IE shall be present if the *Handover Report Type* IE is set to the value "Inter-system ping-pong"

9.3.3.40 Inter-system HO Report

This IE contains the inter-system HO report to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Handover Report Type*	M			
>Too early Inter-system HO				
>>Source Cell ID	M		E-UTRA CGI 9.3.1.9	CGI of the source cell for the HO.
>>Failure Cell ID	M		NG-RAN CGI 9.3.1.73	CGI of the target cell for the HO.
>>UE RLF Report Container	O		9.3.3.41	
>Inter-system Unnecessary HO				
>>Source Cell CGI	M		NG-RAN CGI 9.3.1.73	Source NR cell in NG-RAN
>>Target Cell CGI	M		E-UTRA CGI 9.3.1.9	Target cell in E-UTRAN
>>Early IRAT HO	M		ENUMERATED (true, false, ...)	Is set to “true” if the measurement period expired due to an inter-RAT handover towards NR executed within the configured measurement duration and otherwise set to “false”
>>Candidate Cell List		*1*		
>>>Candidate Cell Item		*1..<maxnoofCandidateCells>*		
>>>>CHOICE *Candidate Cell Type*	M			
>>>>>Candidate CGI				
>>>>>>Candidate Cell ID	M		NR CGI 9.3.1.7	This IE contains an NR CGI.
>>>>>Candidate PCI				
>>>>>>Candidate PCI	M		INTEGER (0..1007, …)	This IE includes the NR Physical Cell Identifier of detected cells not included in the *Candidate Cell List* IE and for which an NR CGI could not be derived.
>>>>>>Candidate NR ARFCN	M		INTEGER (0.. maxNRARFCN)	RF Reference Frequency as defined in TS 38.104 [39], section 5.4.2.1. The frequency provided in this IE identifies the absolute frequency position of the reference resource block (Common RB 0) of the carrier. Its lowest subcarrier is also known as Point A.

Range bound	Explanation
maxnoofCandidateCells	Maximum no. of candidate cells. Value is 32
maxNRARFCN	Maximum value of NRARFCNs. Value is 3279165.

9.3.3.41 UE RLF Report Container

This IE contains the RLF Report to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *RLF type*	M			
>NR				
>>NR UE RLF Report Container	M		OCTET STRING	Includes the *nr-RLF-Report* contained in the *UEInformationResponse* message as defined in TS 38.331 [18].
>LTE				
>>LTE UE RLF Report Container	M		OCTET STRING	Includes the *rLF-Report-r9* contained in the *UEInformationResponse* message as defined in TS 36.331 [21]

9.3.3.42 NID

This IE is used to identify (together with a PLMN identifier) a Stand-alone Non-Public Network.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
NID	M		BIT STRING (SIZE(44))	Defined in TS 23.003 [23].

9.3.3.43 CAG ID

This IE is used to identify (together with a PLMN identifier) a Public Network Integrated NPN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CAG ID	M		BIT STRING (SIZE(32))	Defined in TS 23.003 [23].

9.3.3.44 NPN Support

For SNPN, this IE identifies a supported SNPN together with the associated PLMN ID.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NPN Support*	M			
>SNPN				
>>NID	M		9.3.3.42	

9.3.3.45 Allowed PNI-NPN List

This IE contains information on allowed UE mobility in PNI-NPN including allowed PNI-NPNs and whether the UE is allowed to access non-CAG cells for each PLMN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Allowed PNI-NPN Item		*1..<maxnoofEPLMNs+1>*		
>PLMN Identity	M		9.3.3.5	
>PNI-NPN Restricted	M		ENUMERATED (restricted, not-restricted, …)	If set to “restricted”, indicates that the UE is not allowed to access non-CAG cells for this PLMN.
>Allowed CAG List per PLMN		*1..<maxnoofAllowedCAGsperPLMN>*		
>>CAG ID	M		9.3.3.43	

Range bound	Explanation
maxnoofEPLMNs+1	Maximum no. of equivalent PLMNs plus one serving PLMN. Value is 16.
maxnoofAllowedCAGsperPLMN	Maximum number of CAGs per PLMN in UE’s Allowed PNI-NPN list. Value is 256.

9.3.3.46 NPN Access Information

This IE contains information to perform access control for NPN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *NPN Access Information*	M			
>PNI-NPN Access Information				
>>Cell CAG List	M		9.3.3.47	

9.3.3.47 Cell CAG List

This IE indicates the list of CAG IDs supported by a cell.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cell CAG List		*1..<maxnoofCAGsperCell>*		
>CAG ID	M		9.3.3.43	

Range bound	Explanation
maxnoofCAGsperCell	Maximum no. of CAGs per cell. Value is 64. Max is 12 in this release.

9.3.3.48 UL CP Security Information

This IE contains NAS level security information to enable UE authentication by the AMF as described in TS 33.401 [27].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UL NAS MAC	M		BIT STRING (SIZE(16))	Defined in TS 33.401 [27].
UL NAS Count	M		BIT STRING (SIZE(5))	Defined in TS 33.401 [27].

9.3.3.49 DL CP Security Information

This IE contains NAS level security information to be forwarded to the UE as described in TS 33.401 [27].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
DL NAS MAC	M		BIT STRING (SIZE(16))	Defined in TS 33.401 [27].

9.3.3.50 Configured TAC Indication

This IE indicates that in all NR cells served by the gNB, the TAC with which this IE is associated, is only configured but not broadcast.

NOTE: This IE is defined in accordance to the possibility foreseen in TS 38.331 [18] to not broadcast the TAC if the NR cell only supports PSCell/SCell functionality.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Configured TAC Indication	M		ENUMERATED (true, ...)	

9.3.3.51 Extended AMF Name

This IE provides extended human readable name of the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
AMF Name Visible	O		VisibleString (SIZE(1..150, …))	
AMF Name UTF8	O		UTF8String (SIZE(1..150, …))	

9.3.3.52 Extended UE Identity Index Value

This IE is used by the NG-RAN node to calculate the Paging Frame and Paging Occasion as specified in TS 36.304 [29], the Paging Frame and Paging Occasion for eDRX and the UE_ID based subgroup ID as specified in TS 38.304 [12].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Extended UE Identity Index Value	M		BIT STRING (SIZE(16))	

9.3.3.53 NR NTN TAI Information

This IE contains the TAI information for NR NTN.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Serving PLMN	M		PLMN Identity 9.3.3.5	Indicates the UE's serving PLMN.
TAC List in NR NTN		*1..<maxnoofTACsinNTN>*		Includes all TAC(s) broadcast in the cell, for the UE's serving PLMN.
>TAC	M		9.3.3.10	
UE Location Derived TAC in NR NTN	O		TAC 9.3.3.10	This IE contains TAC information derived from the actual UE location, if available.

Range bound	Explanation
maxnoofTACsinNTN	Maximum no. of TACs broadcast per cell. Value is 12.

9.3.3.54 Inter-system SON Information Request

This IE contains the request information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Inter-system SON Information Request*	M			
>NG-RAN Cell Activation				
>>Inter-system Cell Activation Request	M		9.3.3.56	
>Resource Status				
>>Inter-system Resource Status Request	M		9.3.3.59	

9.3.3.55 Inter-system SON Information Reply

This IE contains the reply information to be transferred.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Inter-system SON Information Reply*	M			
>NG-RAN Cell Activation				
>>Inter-system Cell Activation Reply	M		9.3.3.58	
>Resource Status				
>>Inter-system Resource Status Reply	M		9.3.3.61	

9.3.3.56 Inter-system Cell Activation Request

This IE contains request information for inter-system Cell Activation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Activation ID	M		INTEGER (0..16384, …)	Allocated by the eNB.
Cells to Activate List		*1..<maxnoofCellsinNGRANNode>*		
>NG-RAN CGI	M		9.3.1.73	

Range bound	Explanation
maxnoofCellsinNGRANNode	Maximum no. of cells that can be served by a NG-RAN node. Value is 16384.

9.3.3.57 Inter-system Cell State Indication

This IE contains notification information for inter-system Cell Activation and Deactivation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Notification Cell List		*1*		
>Notification Cell Item		*1..<maxnoofCellsinNGRANNode>*		
>>NG-RAN CGI	M		9.3.1.73	
>>Notify Flag	M		ENUMERATED (Activated, Deactivated, ...)	

Range bound	Explanation
maxnoofCellsinNGRANNode	Maximum no. of cells that can be served by a NG-RAN node. Value is 16384.

9.3.3.58 Inter-system Cell Activation Reply

This IE contains reply information for inter-system Cell Activation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Activated Cells List		*1..<maxnoofCellsinNGRANNode>*		
>NG-RAN CGI	M		9.3.1.73	
Activation ID	M		INTEGER (0..16384, …)	Allocated by the eNB.

Range bound	Explanation
maxnoofCellsinNGRANNode	Maximum no. of cells that can be served by a NG-RAN node. Value is 16384.

9.3.3.59 Inter-system Resource Status Request

This IE contains information on the requested Inter-system Load Reporting reporting.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reporting System	M		9.3.1.237	
Report Characteristics	M		BITSTRING (SIZE(32))	Each position in the bitmap indicates measurement object the receiving node is requested to report. First Bit = Number of Active UEs, Second Bit = RRC connections, Third Bit = PRB. Other bits reserved for future use.
CHOICE *report type*	M			
>Event based reporting				
>>Inter-system Resource Threshold Low	M		INTEGER (0..100)	Inter-system Resource Status reporting is enabled when Composite Available Capacity is above this threshold or is disabled when Composite Available Capacity is below this threshold. The reporting node sends a report when the cell Composite Available Capacity becomes greater than or equal to the threshold.
>>Inter-system Resource Threshold High	M		INTEGER (0..100)	Inter-system Resource Status reporting is enabled when Composite Available Capacity is below this threshold or is disabled when Composite Available Capacity is above this threshold. The reporting node sends a report when the cell Composite Available Capacity becomes smaller than or equal to the threshold.
>>Number of Measurement Reporting Levels	M		ENUMERATED (2, 3, 4, 5, 10, ..., 0)	The reporting node divides the cell load scale into the indicated number of reporting levels, evenly distributed on a linear scale between the Inter-system Resource Threshold Low and the Inter-system Resource Threshold High. The reporting node sends a report each time the cell load changes from one reporting level to another. If the value is “0”, the IE is ignored and only the Inter-system Resource Threshold Low and the Inter-system Resource Threshold High are used as the reporting levels.
>Periodic Reporting				
>>Reporting Periodicity	M		ENUMERATED (stop, single, 1000ms, 2000ms, 5000ms, 10000ms, …)	Periodicity that can be used for reporting. If the value is “stop”, the reporting node is asked to stop the periodic reporting. If the value is “single” there is only one report.

Range bound	Explanation
maxnoofReportedCells	Maximum no. of cells that can be reported. Value is 256.

9.3.3.60 Inter-system Resource Status Report

This IE contains the Inter-system load report.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Reporting System*	M			
>E-UTRAN				
>>E-UTRAN Cell Report List		*1*		
>>>E-UTRAN Cell Report Item		*1..<maxnoofReportedCells>*		
>>>>Cell ID	M		E-UTRA CGI 9.3.1.9	
>>>>E-UTRAN Composite Available Capacity Group	M		9.3.1.196	Corresponds to the *Composite Available Capacity Group* IE as defined in TS 36.423 [40].
>>>>Number of Active UEs	O		INTEGER (0..16777215, …)	Corresponds to the *Number of Active UEs* IE as defined in TS 36.314 [49].
>>>>RRC Connections	O		INTEGER (1..65536, …)	Corresponds to the *RRC Connections* IE as defined in TS 38.423 [24].
>>>>E-UTRAN Radio Resource Status	O		9.3.1.200	
>NG-RAN				
>>NG-RAN Cell Report List		*1*		
>>>NG-RAN Cell Report Item		*1..<maxnoofReportedCells>*		
>>>>Cell ID	M		NG-RAN CGI 9.3.1.73	
>>>>NR Composite Available Capacity Group	M		E-UTRAN Composite Available Capacity Group 9.3.1.196	Corresponds to the *Composite Available Capacity Group* IE as defined in TS 36.423 [40].
>>>>Number of Active UEs	O		INTEGER (0..16777215, …)	Corresponds to the *Number of Active UEs* IE as defined in TS 38.423 [24].
>>>>RRC Connections	O		INTEGER (1..65536, …)	Corresponds to the *RRC Connections* IE as defined in TS 38.423 [24].
>>>>NR Radio Resource Status	O		9.3.1.205	

Range bound	Explanation
maxnoofReportedCells	Maximum no. of cells that can be reported. Value is 256.

9.3.3.61 Inter-system Resource Status Reply

This IE indicates for which cell(s) the inter-system load reporting was successfully initiated.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reporting System	M		9.3.1.237	

9.3.3.62 Hashed UE Identity Index Value

This IE is the 13 Most Significant Bits (MSBs) of the Hashed ID defined in TS 38.304 [12] or TS 36.304 [29].

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Hashed UE Identity Index Value	M		BIT STRING (SIZE(13, …))	

9.3.4 SMF Related IEs

9.3.4.1 PDU Session Resource Setup Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PDU Session Aggregate Maximum Bit Rate	O		9.3.1.102	This IE shall be present when at least one Non-GBR QoS flow is being setup and is ignored otherwise.	YES	reject
UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs.	YES	reject
Additional UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s), for delivery of UL PDUs for split PDU session.	YES	reject
Data Forwarding Not Possible	O		9.3.1.63	This IE may be present in case of HANDOVER REQUEST message and is ignored otherwise.	YES	reject
PDU Session Type	M		9.3.1.52		YES	reject
Security Indication	O		9.3.1.27		YES	reject
Network Instance	O		9.3.1.113	This IE is ignored if the *Common Network Instance* IE is included.	YES	reject
QoS Flow Setup Request List		*1*			YES	reject
>QoS Flow Setup Request Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>QoS Flow Level QoS Parameters	M		9.3.1.12		-	
>>E-RAB ID	O		9.3.2.3		-	
>>TSC Traffic Characteristics	O		9.3.1.130	This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
>>Redundant QoS Flow Indicator	O		9.3.1.134	This IE indicates whether this QoS flow is requested for the redundant transmission.	YES	ignore
Common Network Instance	O		9.3.1.120		YES	ignore
Direct Forwarding Path Availability	O		9.3.1.64	This IE may be present in case of inter-system handover and intra-system handover.	YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission.	YES	ignore

Additional Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s), for delivery of redundant UL PDUs for split PDU session.	YES	ignore
Redundant Common Network Instance	O		Common Network Instance 9.3.1.120		YES	ignore
Redundant PDU Session Information	O		9.3.1.136		YES	ignore
MBS Session Setup Request List	O		9.3.1.211		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.2 PDU Session Resource Setup Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL QoS Flow per TNL Information	M		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer for delivery of DL PDUs, together with associated QoS flows.	-	
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows and corresponding to the *Additional UL NG-U UP TNL Information* IE in the *PDU Session Resource Setup Request Transfer* IE.	-	
Security Result	O		9.3.1.59		-	
QoS Flow Failed to Setup List	O		QoS Flow List with Cause 9.3.1.13		-	
Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer(s) for delivery of DL PDUs of the indicated Redundant QoS Flow(s) and corresponding to the *Redundant UL NG-U UP TNL Information* IE in the *PDU Session Resource Setup Request Transfer* IE.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of redundant DL PDUs for split PDU session, together with associated QoS flows and corresponding to the *Additional Redundant UL NG-U UP TNL Information* IE in the *PDU Session Resource Setup*	YES	ignore

				Request Transfer IE.		
Used RSN Information	O		Redundant PDU Session Information 9.3.1.136		YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore
MBS Session Setup Response List	O		9.3.1.213		YES	ignore
MBS Session Failed to Setup List	O		9.3.1.214		YES	ignore

9.3.4.3 PDU Session Resource Modify Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
PDU Session Aggregate Maximum Bit Rate	O		9.3.1.102		YES	reject
UL NG-U UP TNL Modify List		*0..1*			YES	reject
>UL NG-U UP TNL Modify Item		*1..<maxnoofMultiConnectivity>*		This IE(s) are included only for modification of an existing tunnel.	-	
>>UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs.	-	
>>DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the NG-RAN node.	-	
>>Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission.	YES	ignore
>>Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the NG-RAN node for the redundant transmission.	YES	ignore
Network Instance	O		9.3.1.113	This IE is ignored if the *Common Network Instance* IE is included.	YES	reject
QoS Flow Add or Modify Request List		*0..1*			YES	reject
>QoS Flow Add or Modify Request Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>QoS Flow Level QoS Parameters	O		9.3.1.12		-	
>>E-RAB ID	O		9.3.2.3		-	
>>TSC Traffic Characteristics	O		9.3.1.130	This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
>>Redundant QoS Flow Indicator	O		9.3.1.134	This IE indicates whether this QoS flow is requested for the redundant transmission.	YES	ignore
QoS Flow to Release List	O		QoS Flow List with Cause 9.3.1.13		YES	reject
Additional UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s) proposed for delivery of UL PDUs for split PDU session.	YES	reject
Common Network Instance	O		9.3.1.120		YES	ignore

Additional Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	UPF endpoint of the additional NG-U transport bearer(s) proposed for delivery of redundant UL PDUs for split PDU session.	YES	ignore
Redundant Common Network Instance	O		Common Network Instance 9.3.1.120		YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission of the Redundant QoS Flow(s).	YES	ignore
Security Indication	O		9.3.1.27		YES	ignore
MBS Session Setup or Modify Request List	O		9.3.1.212		YES	ignore
MBS Session To Release List	O		9.3.1.215		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.
maxnoofMultiConnectivity	Maximum no. of connectivity allowed for a UE. Value is 4. The current version of the specification supports up to 2 connectivity.

9.3.4.4 PDU Session Resource Modify Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.	-	
UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the 5GC node.	-	
QoS Flow Add or Modify Response List		*0..1*			-	
>QoS Flow Add or Modify Response Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set	YES	Ignore
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows.	-	
QoS Flow Failed to Add or Modify List	O		QoS Flow List with Cause 9.3.1.13		-	
Additional NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer corresponding to the modified UPF endpoint received in the *PDU Session Resource Modify Request Transfer* IE in case of PDU session split.	YES	ignore
Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs for the redundant transmission.	YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	Identifies the NG-U transport bearer at the 5GC node for the redundant transmission.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of redundant DL PDUs for split PDU	YES	ignore

				session, together with associated QoS flows.		
Additional Redundant NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer for delivery of redundant DL PDUs corresponding to the modified UPF endpoint(s) received in the *UL NG-U UP TNL Modify List* IE of the *PDU Session Resource Modify Request Transfer* IE in case of PDU session split.	YES	ignore
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore
MBS Session Setup or Modify Response List	O		MBS Session Setup Response List 9.3.1.213		YES	ignore
MBS Session Failed to Setup or Modify List	O		MBS Session Failed to Setup List 9.3.1.214		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.5 PDU Session Resource Notify Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoS Flow Notify List		*0..1*			-	
>QoS Flow Notify Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Notification Cause	M		ENUMERATED (fullfilled, not fulfilled, …)		-	
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Notify Index 9.3.1.153	Index to the currently fulfilled alternative QoS parameters set. Value 0 indicates that NG-RAN cannot even fulfil the lowest alternative parameters set.	YES	Ignore
QoS Flow Released List	O		QoS Flow List with Cause 9.3.1.13		-	
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
QoS Flow Feedback List		*0..1*			YES	ignore
>QoS Flow Feedback Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Update Feedback	O		BIT STRING { CN PDB DL(0), CN PDB UL(1)} (SIZE(8, …))	Each position in the bitmap represents a QoS parameter. If a bit is set to "1", the respective parameter was not updated. If a bit is set to "0", the respective parameter was successfully updated. Bits 2-7 reserved for future use.	-	
>>CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Indicates when the packet delay budget downlink was not updated in path switch that NG-RAN can offer this value	-	
>>CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Indicates when the packet delay budget uplink was not updated in path switch that NG-RAN can offer this value	-	

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.6 PDU Session Resource Modify Indication Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL QoS Flow per TNL Information	M		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer for delivery of DL PDUs, together with associated QoS flows.	-	
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows	-	
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore
Security Result	O		9.3.1.59	Current UP security status	YES	ignore
Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information 9.3.2.8	NG-RAN node endpoint of the NG-U transport bearer for delivery of DL PDUs for the redundant transmission, together with associated QoS flows.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of Redundant DL PDUs for split PDU session, together with associated QoS flows.	YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore

9.3.4.7 PDU Session Resource Modify Confirm Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
QoS Flow Modify Confirm List		*1*			-	
>QoS Flow Modify Confirm Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
UL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer corresponding to the *DL QoS Flow per TNL Information* IE received in the *PDU Session Resource Modify Indication Transfer* IE.	-	
Additional NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer indicated in the *PDU Session Resource Modify Indication Transfer* IE and the corresponding UPF endpoint for split PDU session.	-	
QoS Flow Failed to Modify List	O		QoS Flow List with Cause 9.3.1.13		-	
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer identified by the above redundant DL NG-U UP TNL Information IE for the redundant transmission.	YES	ignore
Additional Redundant NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer for the redundant transmission indicated in the PDU Session Resource Modify Indication Transfer IE and the corresponding UPF endpoint for split PDU session.	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.8 Path Switch Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.	-	
DL NG-U TNL Information Reused	O		ENUMERATED (true, …)	Indicates that DL NG-U TNL Information has been reused.	-	
User Plane Security Information	O		9.3.1.60		-	
QoS Flow Accepted List		*1*		QoS flows associated with the *DL NG-U UP TNL Information* IE.	-	
>QoS Flow Accepted Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Current QoS Parameters Set Index	O		Alternative QoS Parameters Set Index 9.3.1.152	Index to the currently fulfilled alternative QoS parameters set.	YES	ignore
Additional DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of DL PDUs for split PDU session, together with associated QoS flows.	YES	ignore
Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of redundant DL PDUs.	YES	ignore
Redundant DL NG-U TNL Information Reused	O		ENUMERATED (true, …)	Indicates that Redundant DL NG-U TNL Information has been reused.	YES	ignore
Additional Redundant DL QoS Flow per TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint of the additional NG-U transport bearer(s) for delivery of Redundant DL PDUs for split PDU session, together with associated QoS flows.	YES	ignore
Used RSN Information	O		Redundant PDU Session Information 9.3.1.136		YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.9 Path Switch Request Acknowledge Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer corresponding to the DL NG-U UP TNL Information IE received in the Path Switch Request Transfer IE.	-	
Security Indication	O		9.3.1.27		-	
Additional NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer indicated in the Path Switch Request Transfer IE and the corresponding UPF endpoint for split PDU session.	YES	ignore
Redundant UL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	UPF endpoint of the NG-U transport bearer, for delivery of UL PDUs for the redundant transmission.	YES	ignore
Additional Redundant NG-U UP TNL Information	O		UP Transport Layer Information Pair List 9.3.2.11	NG-RAN node endpoint of the NG-U transport bearer for the redundant transmission indicated in the Path Switch Request Transfer IE and the corresponding UPF endpoint for split PDU session.	YES	ignore
QoS Flow Parameters List		*0..1*			YES	ignore
>QoS Flow Parameters Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
>>Alternative QoS Parameters Set List	O		9.3.1.151	Indicates alternative sets of QoS parameters for the QoS flow.	-	
>>CN Packet Delay Budget Downlink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9]. This IE may be present in case of GBR QoS flows and is ignored otherwise.	YES	ignore
>>CN Packet Delay Budget Uplink	O		Extended Packet Delay Budget 9.3.1.135	Core Network Packet Delay Budget is specified in TS 23.501 [9].	YES	ignore

				This IE may be present in case of GBR QoS flows and is ignored otherwise.		
>>Burst Arrival Time Downlink	O		Burst Arrival Time 9.3.1.133	Indicates the downlink Burst Arrival Time of the TSC QoS flow	YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.10 Handover Command Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded DL PDUs per PDU session tunnel.	-	
QoS Flow to be Forwarded List		*0..1*			-	
>QoS Flow to be Forwarded Item		*1..<maxnoofQoSFlows>*			-	
>>QoS Flow Identifier	M		9.3.1.51		-	
Data Forwarding Response DRB List	O		9.3.1.77		-	
Additional DL Forwarding UP TNL Information	O		QoS Flow per TNL Information List 9.3.2.1	NG-RAN node endpoint to deliver forwarded DL PDUs for split PDU session tunnel, together with associated QoS flows to be forwarded.	YES	ignore
UL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded UL PDUs	YES	reject
Additional UL Forwarding UP TNL Information	O		UP Transport Layer Information List 9.3.2.12	NG-RAN node endpoint to deliver forwarded UL PDUs for split PDU session tunnel.	YES	reject
Data Forwarding Response E-RAB List	O		9.3.1.121		YES	ignore
QoS Flow Failed to Setup List	O		QoS Flow List with Cause 9.3.1.13		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.

9.3.4.11 Handover Request Acknowledge Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.	-	
DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded DL PDUs.	-	
Security Result	O		9.3.1.59		-	
QoS Flow Setup Response List	M		QoS Flow List with Data Forwarding 9.3.2.13	QoS flows associated with the *DL NG-U UP TNL Information* IE.	-	
QoS Flow Failed to Setup List	O		QoS Flow List with Cause 9.3.1.13		-	
Data Forwarding Response DRB List	O		9.3.1.77		-	
Additional DL UP TNL Information for HO List		*0..1*			YES	ignore
>Additional DL UP TNL Information for HO Item		*1..<maxnoofMultiConnectivityMinusOne>*		Additional DL UP TNL Information for split PDU session, in the same order as the UPF endpoint of the additional NG-U transport bearer(s) received in the *Handover Request Transfer* IE of the Handover Request message.	-	
>>Additional DL NG-U UP TNL Information	M		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the additional NG-U transport bearer for delivery of DL PDUs.	-	
>>Additional QoS Flow Setup Response List	M		QoS Flow List with Data Forwarding 9.3.2.13	QoS flows associated with the *Additional DL NG-U UP TNL Information* IE.	-	
>>Additional DL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint to deliver forwarded DL PDUs.	-	
>>Additional Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the additional NG-U transport bearer for delivery of redundant DL PDUs.	YES	ignore
UL Forwarding UP TNL Information	O		UP Transport Layer Information 9.3.2.2	To deliver forwarded UL PDUs	YES	reject
Additional UL Forwarding UP TNL Information	O		UP Transport Layer Information List	NG-RAN node endpoint to deliver forwarded UL	YES	reject

			9.3.2.12	PDUs for split PDU session.		
Data Forwarding Response E-RAB List	O		9.3.1.121		YES	ignore
Redundant DL NG-U UP TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs for the redundant transmission.	YES	ignore
Used RSN Information	O		Redundant PDU Session Information 9.3.1.136		YES	ignore
Global RAN Node ID of Secondary NG-RAN Node	O		Global RAN Node ID 9.3.1.5		YES	ignore
MBS Support Indicator	O		9.3.1.210		YES	ignore

Range bound	Explanation
maxnoofQoSFlows	Maximum no. of QoS flows allowed within one PDU session. Value is 64.
maxnoofMultiConnectivityMinusOne	Maximum no. of connectivity allowed for a UE minus one. Value is 3. The current version of the specification supports 1.

9.3.4.12 PDU Session Resource Release Command Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.13 PDU Session Resource Notify Released Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Cause	M		9.3.1.2		-	
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore

9.3.4.14 Handover Required Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Direct Forwarding Path Availability	O		9.3.1.64	

9.3.4.15 Path Switch Request Setup Failed Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.16 PDU Session Resource Setup Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.4.17 PDU Session Resource Modify Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.4.18 Handover Preparation Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.19 Handover Resource Allocation Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.4.20 Path Switch Request Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.21 PDU Session Resource Release Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Secondary RAT Usage Information	O		9.3.1.114		YES	ignore

9.3.4.22 PDU Session Resource Modify Indication Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	

9.3.4.23 Secondary RAT Data Usage Report Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Secondary RAT Usage Information	O		9.3.1.114	

9.3.4.24 UE Context Resume Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Failed to Resume List	O		QoS Flow List with Cause 9.3.1.13	

9.3.4.25 UE Context Resume Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
QoS Flow Failed to Resume List	O		QoS Flow List with Cause 9.3.1.13	

9.3.4.26 UE Context Suspend Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Suspend Indicator	O		ENUMERATED (true, …)	

9.3.5 MB-SMF Related IEs

9.3.5.1 Void

9.3.5.2 Void

9.3.5.3 MBS Session Setup or Modification Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Session TNL Information 5GC	O		9.3.2.15		YES	reject
MBS QoS Flows To Be Setup or Modified List	M		MBS QoS Flows To Be Setup List 9.3.1.236		YES	reject
MBS Session FSA ID List		*0..<maxnoofMBSFSAs>*			YES	ignore
>MBS Frequency Selection Area Identity	M		OCTET STRING (SIZE(3))		-	

Range bound	Explanation
maxnoofMBSQoSFlows	Maximum no. of QoS Flows allowed within one MBS session. Value is 64.
maxnoofMBSFSAs	Maximum no. of FSA IDs for one MBS session. Value is 64.

9.3.5.4 Void

9.3.5.5 MBS Session Setup or Modification Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session TNL Information NG-RAN	O		9.3.2.17	

9.3.5.6 MBS Session Setup or Modification Failure Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.5.7 MBS Distribution Setup Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Area Session ID	O		9.3.1.207	
Shared NG-U Unicast TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.

9.3.5.8 MBS Distribution Setup Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Area Session ID	O		9.3.1.207	
Shared NG-U Multicast TNL Information	O		9.3.2.16	
MBS QoS Flows To Be Setup List	M		9.3.1.236	
MBS Session Status	M		9.3.1.217	
MBS Service Area	O		9.3.1.208	

Range bound	Explanation
maxnoofMBSQoSFlows	Maximum no. of QoS Flows allowed within one MBS session. Value is 64.

9.3.5.9 MBS Distribution Setup Unsuccessful Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Area Session ID	O		9.3.1.207	
Cause	M		9.3.1.2	
Criticality Diagnostics	O		9.3.1.3	

9.3.5.10 MBS Distribution Release Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	
MBS Area Session ID	O		9.3.1.207	
Shared NG-U Unicast TNL Information	O		UP Transport Layer Information 9.3.2.2	NG-RAN node endpoint of the NG-U transport bearer, for delivery of DL PDUs.
Cause	M		9.3.1.2	

9.3.5.11 Multicast Session Activation Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	

9.3.5.12 Multicast Session Deactivation Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session ID	M		9.3.1.206	

9.3.5.13 Multicast Session Update Request Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
MBS Session ID	M		9.3.1.206		YES	reject
MBS Service Area	O		9.3.1.208		YES	reject
MBS QoS Flows To Be Setup or Modified List	O		MBS QoS Flows To Be Setup List 9.3.1.236		YES	reject
MBS QoS Flow To Release List	O		QoS Flow List with Cause 9.3.1.13	This IE indicates the MBS QoS flow Identifiers of the MBS QoS flows to be released.	YES	reject
MBS Session TNL Information 5GC	O		9.3.2.15		YES	reject

Range bound	Explanation
maxnoofMBSQoSFlows	Maximum no. of QoS Flows allowed within one MBS session. Value is 64.

9.3.5.14 MBS Session Release Response Transfer

This IE is transparent to the AMF.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
MBS Session TNL Information NG-RAN	O		9.3.2.17	